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Soft gluon wave function and evolution operator with quark interactions in the CGC at next-to-leading order

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Abstract: We extend the construction of the soft gluon light-cone wave function of a fast-moving hadron and of the associated unitary evolution operator Ω to QCD with N_f massless quark flavours, up to $\mathcal{O}(g^2)$, within the Color Glass Condensate (CGC) framework. Working in light-cone gauge, we perform the Born–Oppenheimer separation between fast valence and soft modes and implement the eikonal approximation; the valence colour charge density then acquires a fundamental representation component carried by the valence quarks, while still obeying the same non-commutative algebra. The evolution operator is written as a fully normal ordered series in soft gluon, quark and antiquark creation and annihilation operators, whose coefficients are functionals of the non-commuting valence charge density ρ. Besides the coefficients that create and annihilate soft quark–antiquark pairs, a complete construction requires the coefficients describing gluon emission and absorption by soft quark and antiquark lines and the eikonal rescattering of a soft quark on the valence charge; we include all of them. The coefficients are fixed by unitarity and by explicit diagrammatic calculations within light-cone perturbation theory (LCPT), and every second order coefficient is shown to reduce to products of first order ones weighted by ratios of energy denominators, plus instantaneous terms. As an application we diagonalize the soft QCD Hamiltonian through orders g and g^2: all off-diagonal mixing between Fock sectors cancels, the quark and gluon self-energies are scaleless, and the only surviving correction is the coherent background field energy proportional to ρ^2, now with ρ including the valence quark charge.

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1 Introduction

The dynamics of gluon saturation [1, 2, 3, 4] at small-x can be systematically studied within the framework of the Color Glass Condensate (CGC) effective theory [5, 6, 7, 8, 9, 10]. In this regime the rapidity evolution of the gluon distributions is governed by the nonlinear BK and JIMWLK equations [11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22]. While these equations describe the evolution of Wilson line correlators, physical observables such as single and double inclusive gluon production [23, 24] and, more generally, particle production [25, 26, 27, 28, 29, 30] are obtained from matrix elements of operators evaluated between boosted hadronic states. An explicit and controlled construction of the light-cone wave function (LCWF) of a fast-moving hadron is therefore of both theoretical and phenomenological interest.

From the light-cone perspective, boosting a hadron by a rapidity interval opens an additional longitudinal phase space in which new soft modes are generated. The effect of the boost is represented by a unitary operator Ω acting on the valence state, which describes the emission, absorption and recombination of the soft quanta. At leading order Ω generates the Weizsäcker–Williams gluon field [31, 32]; beyond leading order it develops multi-particle and colour correlated structures responsible for fluctuations and correlations [33]. In a companion paper [34], hereafter referred to as Paper I, we constructed the soft gluon wave function and the evolution operator Ω explicitly up to $\mathcal{O}(g^2)$ in pure Yang–Mills theory ($N_f = 0$), fixing all coefficients of the normal ordered expansion of Ω by unitarity and by direct light-cone perturbation theory (LCPT) [35, 36, 37], and used it to diagonalize the soft Yang–Mills Hamiltonian through order g^2.

The purpose of the present work is to include the quark degrees of freedom and thereby obtain the evolution operator of full QCD to the same accuracy. Quarks enter the problem in two distinct ways. First, the valence sector of a hadron contains quarks, and in the Born–Oppenheimer picture [38, 39, 40] their only effect on the soft modes is through their colour charge: the valence charge density acquires a fundamental representation component, $\rho = \rho_g + \rho_q$, which obeys the same $SU(N_c)$ algebra as in the pure gauge theory. Second, and more importantly, soft gluons can fluctuate into soft quark–antiquark pairs already at order g, soft quarks radiate and absorb soft gluons, and soft quarks rescatter eikonally on the valence charge. These processes generate new operator structures in Ω. They are the source of the N_f dependence of NLO small-x evolution [41, 42, 43, 22, 20], of the quark contributions to forward particle production and of the sea quark component of the projectile wave function, and they must be under control before the full QCD single inclusive gluon spectrum at NLO [44] can be computed. Light-cone wave functions involving quarks have been computed in several related contexts (see e.g. Refs. [45, 46, 47, 48]); here we organize them into the unitary operator Ω in the presence of the non-commuting valence charge.

We follow the strategy of Paper I closely. After the Born–Oppenheimer separation and the eikonal approximation, we parameterize Ω as a fully normal ordered series in soft gluon, quark and antiquark creation and annihilation operators with operator valued coefficients that are functionals of ρ. We act with Ω on elementary Fock states — the vacuum, one, two and three-gluon states, a single quark (antiquark), a quark–antiquark pair and a quark–gluon state — and equate the result, sector by sector, with the corresponding LCWF components computed in LCPT. This one-to-one matching fixes the coefficients associated with real transitions, while the unitarity condition $\Omega^\dagger\Omega = 1$, imposed order by order in the coupling, determines the remaining ones and provides nontrivial checks of the

coefficients that preserve the particle content. Two results of the construction deserve to be emphasized at the outset. First, the ansatz that contains only the pair creation and annihilation structures is not complete: at order g the evolution operator must also contain the coefficients $\mathfrak{F}$ and $\mathfrak{G}$ describing the emission of a soft gluon by a soft quark and by a soft antiquark, and at order g^2 the coefficients $\mathfrak{P}$ and $\bar{\mathfrak{P}}$ describing the eikonal rescattering of a soft quark (antiquark) on the valence charge. Without them Ω is neither unitary nor able to diagonalize the Hamiltonian at order g. Second, every second order coefficient can be written as a sum of products of first order coefficients weighted by ratios of energy denominators, together with the instantaneous terms; this reduction formula makes all the unitarity relations transparent and allows a compact presentation of the coefficients with many external legs.

As an application we use the evolution operator to diagonalize the soft QCD Hamiltonian up to orders g and g^2. At order g the Hamiltonian is fully diagonalized once the coefficients A, C of Paper I are supplemented by $\mathfrak{A}_3$, $\mathfrak{F}$ and $\mathfrak{G}$. At order g^2 all quark induced off-diagonal terms are removed, the quark loop contribution to the gluon self-energy and the quark self-energy are scaleless, and the only surviving correction is the coherent background field energy proportional to ρ^2. The sole modification relative to the pure Yang–Mills result is that ρ now contains the valence quark charge.

Our article is structured as follows. In section 2 we introduce the light-cone QCD Hamiltonian with quarks and discuss the Born–Oppenheimer separation and the eikonal approximation. In section 3 we review the light-cone wave function formalism and LCPT and derive the reduction formula for second order amplitudes. In section 4 we construct the perturbative evolution operator with quarks, derive the unitarity relations and its action on the relevant Fock states, and compute the corresponding LCWFs. In section 5 we determine all the coefficients. In section 6 we diagonalize the soft QCD Hamiltonian at orders g and g^2. Section 7 contains the concluding remarks. Technical details are collected in the appendices.

2 Background and the setup

2.1 Light-cone coordinates and conventions

We use the light-cone conventions of Paper I [34, 49, 50]: $x^\pm = (x^0 \pm x^3)/\sqrt{2}$, quantization on surfaces of fixed x^+, longitudinal momentum $p^+ > 0$ for physical particles, $d^3p \equiv dp^+ d^2\boldsymbol{p}$ and $\delta^{(3)}(k-p) \equiv \delta(k^+ - p^+)\delta^{(2)}(\boldsymbol{k}-\boldsymbol{p})$. For an on-shell massless particle the light-cone energy is $p^- = \boldsymbol{p}^2/2p^+$, which we abbreviate as

$$\varepsilon_p \equiv \frac{\boldsymbol{p}^2}{2p^+}. \tag{2.1}$$

We also use the shorthand $\int_p \equiv \int d^3p/(2\pi)^3$. We work in light-cone gauge $A^+ = 0$, in which only the transverse gluon fields A_i^a, $i = 1, 2$, are dynamical; their creation and annihilation operators satisfy

$$\big[a_i^a(k), a_j^{\dagger b}(p)\big] = (2\pi)^3 \delta^{ab}\delta_{ij}\, \delta^{(3)}(k-p). \tag{2.2}$$

The quark field is decomposed as $\psi = \psi_+ + \psi_-$ with $\psi_\pm = \Lambda_\pm \psi$, $\Lambda_\pm = \frac{1}{2}\gamma^\mp\gamma^\pm$. Only the "good" component ψ_+ is dynamical; ψ_- is constrained and is eliminated through its equation of motion (appendix A). In the two-component representation [51, 36] ψ_+ is described

by a two-component field ξ, which we expand as

$$\xi^{\beta}(x) = 2^{-1/4}\int_0^\infty \frac{dk^+}{2\pi}\int\frac{d^2k}{(2\pi)^2}\sum_{\lambda=\pm}\Big[b_\lambda^{\beta}(k)\,\chi_\lambda\,e^{-ik\cdot x} + d_\lambda^{\dagger\beta}(k)\,\chi_\lambda\,e^{ik\cdot x}\Big], \tag{2.3}$$

where β is a colour index in the fundamental representation, χ_λ are the eigenvectors of σ^3 with $\sum_\lambda \chi_\lambda\chi_\lambda^\dagger = \mathbf{1}$, and a flavour index is suppressed.[1] The quark and antiquark operators obey

$$\big\{b_\lambda^{\beta}(l), b_{\lambda'}^{\dagger\beta'}(l')\big\} = \big\{d_\lambda^{\beta}(l), d_{\lambda'}^{\dagger\beta'}(l')\big\} = (2\pi)^3\delta^{\beta\beta'}\delta_{\lambda\lambda'}\,\delta^{(3)}(l-l'), \tag{2.4}$$

all other anticommutators vanishing; fermion operators commute with gluon operators.

All the quark–gluon vertices of light-cone QCD follow from one structure. For a gluon of momentum k and transverse polarization i attached to a fermion line, let P_L and P_R be the momenta of the two fermion legs, P_L being the one at the head of the fermion number arrow and P_R the one at its tail (antiquark momenta are physical, i.e. positive, momenta). We define

$$\Gamma^i(k;P_L,P_R) \equiv \frac{2k^i}{k^+} - \frac{(\boldsymbol{\sigma}\cdot\boldsymbol{P}_L)\,\sigma^i}{P_L^+} - \frac{\sigma^i\,(\boldsymbol{\sigma}\cdot\boldsymbol{P}_R)}{P_R^+}, \qquad \Phi^i_{\lambda\lambda'}(k;P_L,P_R) \equiv \chi_\lambda^\dagger\,\Gamma^i(k;P_L,P_R)\,\chi_{\lambda'}. \tag{2.5}$$

The three elementary processes then carry the assignments

process	P_L	P_R
$g(k)\to q(l)\,\bar q(m)$	l (quark)	m (antiquark)
$q(l')\to q(l)\,g(k)$	l (outgoing quark)	l' (incoming quark)
$\bar q(m')\to \bar q(m)\,g(k)$	m' (incoming antiquark)	m (outgoing antiquark)

and the conjugate (absorption) processes follow from $\big[\Gamma^i(k;P_L,P_R)\big]^\dagger = \Gamma^i(k;P_R,P_L)$. Since $2k^i - (\boldsymbol{\sigma}\cdot\boldsymbol{k})\sigma^i - \sigma^i(\boldsymbol{\sigma}\cdot\boldsymbol{k}) = 0$, the vertex depends only on the relative transverse momentum of the splitting. For $g(k)\to q(l)\bar q(m)$, with $z = l^+/k^+$ and $\boldsymbol{\kappa} = \boldsymbol{l} - z\boldsymbol{k}$,

$$\Gamma^i(k;l,m) = \frac{(2z-1)\kappa^i + i\epsilon^{ij}\kappa^j\sigma^3}{k^+z(1-z)}, \qquad \varepsilon_k - \varepsilon_l - \varepsilon_m = -\frac{\boldsymbol{\kappa}^2}{2z(1-z)k^+}, \tag{2.6}$$

and for $q(l')\to q(l)g(p)$, with $z = p^+/l'^+$ and $\boldsymbol{\kappa} = \boldsymbol{p} - z\boldsymbol{l'}$,

$$\Gamma^j(p;l,l') = \frac{(2-z)\kappa^j - iz\,\epsilon^{jk}\kappa^k\sigma^3}{l'^+z(1-z)}, \qquad \varepsilon_{l'} - \varepsilon_l - \varepsilon_p = -\frac{\boldsymbol{\kappa}^2}{2z(1-z)l'^+}. \tag{2.7}$$

Finally, the three-gluon vertex of Paper I is written as

$$\mathcal{V}^{ijl}(k;p,q) = \Big(p^i - q^i + \tfrac{q^+-p^+}{k^+}k^i\Big)\delta^{jl} + \Big(k^j + q^j - \tfrac{k^++q^+}{p^+}p^j\Big)\delta^{il} + \Big(\tfrac{k^++p^+}{q^+}q^l - p^l - k^l\Big)\delta^{ij}. \tag{2.8}$$

2.2 Light-cone QCD Hamiltonian with quarks

The derivation of the QCD Hamiltonian in light-cone gauge is standard [36, 51, 52, 20]; it is summarized in appendix A. After the constrained fields A^- and ψ_- are eliminated, the Hamiltonian reads

$$H_{\mathrm{LC\,QCD}} = H_{\mathrm{LC\,YM}} + H_q, \tag{2.9}$$

[1] With this labelling the physical helicity of an antiquark created by $d_\lambda^\dagger$ is $-\lambda$. It has the advantage that every quark–gluon vertex reduces to a single spinor bilinear of the form (2.5).

where $H_{\mathrm{LC\,YM}}$ is the light-cone Yang–Mills Hamiltonian of Paper I and, for massless quarks,

$$H_q = \int dx^- d^2x \Big[\frac{1}{2}\bar\psi\gamma^+ \frac{-\boldsymbol{\partial}^2}{i\partial^+}\psi + g\,\bar\psi\gamma^i t^a \psi\, A_i^a + \frac{g^2}{2}\,\bar\psi\gamma^i t^a A_i^a\,\frac{\gamma^+}{2i\partial^+}\,\gamma^j t^b A_j^b\,\psi \\ + g^2\, J_q^{+a}\frac{1}{(i\partial^+)^2}J_g^{+a} + \frac{g^2}{2}\,J_q^{+a}\frac{1}{(i\partial^+)^2}J_q^{+a}\Big], \tag{2.10}$$

with the quark and gluon light-cone colour currents

$$J_q^{+a} = \bar\psi\gamma^+ t^a\psi, \qquad J_g^{+a} = f^{abc}A_i^b\partial^+ A_i^c. \tag{2.11}$$

The first term of (2.10) is the free quark Hamiltonian; the second is the quark–gluon vertex; the third is the instantaneous quark exchange (it arises from eliminating ψ_-); the last two are the instantaneous gluon exchange terms involving the quark current (they arise, together with the J_gJ_g term contained in $H_{\mathrm{LC\,YM}}$, from eliminating A^-). The instantaneous gluon exchange is most usefully regarded as a single current–current interaction,

$$H_{\mathrm{inst}}^{(A)} = \frac{g^2}{2}\int dx^- d^2x\; \mathcal{J}^a \frac{1}{(i\partial^+)^2}\mathcal{J}^a, \qquad \mathcal{J}^a = J_{\mathrm{val}}^{+a} + J_g^{+a} + J_q^{+a}, \tag{2.12}$$

because after the Born–Oppenheimer separation discussed below the classification of its contributions becomes automatic: one expands $\mathcal{J}$ and reads off the cross terms. The free Hamiltonian of the soft sector is

$$H_0 = \int_k \varepsilon_k\, a_i^{\dagger a}(k)a_i^a(k) + \int_l \varepsilon_l \sum_{\lambda,\beta}\Big[b_\lambda^{\dagger\beta}(l)b_\lambda^\beta(l) + d_\lambda^{\dagger\beta}(l)d_\lambda^\beta(l)\Big], \tag{2.13}$$

whose eigenstates are the free Fock states of soft gluons, quarks and antiquarks.

2.3 Born–Oppenheimer formalism and the eikonal approximation

As in Paper I, the modes with longitudinal momentum $k^+ \geq \vee$ are the valence degrees of freedom and those with $\Lambda < k^+ < \vee$ the soft ones [7, 8]. With quarks, both the gluon and the quark fields are decomposed into valence and soft components,

$$A_i^a = \mathcal{A}_i^a + A_i^a, \qquad \xi = \Xi + \xi, \tag{2.14}$$

where calligraphic (capital) symbols denote the valence modes. Substituting into the Hamiltonian generates soft–soft, soft–valence and valence–valence interactions. Since the valence modes carry parametrically larger longitudinal momenta, contributions suppressed by powers of $k^+_{\mathrm{soft}}/k^+_{\mathrm{valence}}$ are neglected (eikonal approximation), as are terms in which inverse longitudinal derivatives act on valence fields [20, 34]. In particular, a soft quark cannot absorb a valence gluon (nor a soft gluon a valence quark) without leaving the soft sector, and the transverse couplings $\bar\psi\gamma^i\psi A_i$ of the valence quarks are power suppressed. The only coupling of the valence sector to the soft modes that survives is through the plus component of the valence colour current, i.e. through the total valence colour charge density

$$\rho^a(-p) = \rho_g^a(-p) + \rho_q^a(-p), \tag{2.15}$$

with the gluonic part of Paper I and a fundamental representation part carried by the valence quarks and antiquarks,

$$\rho_g^a(-p) = -if^{abc}\int_\vee^\infty \frac{dk^+}{2\pi}\int\frac{d^2k}{(2\pi)^2}\,\bar a_j^{\dagger b}(k)\,\bar a_j^c(k+p), \tag{2.16}$$

$$\rho_q^a(-p) = \int_\vee^\infty \frac{dk^+}{2\pi} \int \frac{d^2k}{(2\pi)^2} \sum_\lambda \Big[\bar{b}_\lambda^{\dagger\alpha}(k)\, t^a_{\alpha\beta}\, \bar{b}_\lambda^{\beta}(k+p) - \bar{d}_\lambda^{\dagger\alpha}(k+p)\, t^a_{\beta\alpha}\, \bar{d}_\lambda^{\beta}(k) \Big], \tag{2.17}$$

barred operators denoting valence modes and a sum over flavours being implied. Both pieces satisfy the same algebra, so that

$$\big[\rho^a(k), \rho^b(p)\big] = i f^{abc} \rho^c(k+p) \tag{2.18}$$

holds for the total charge density. Every result of Paper I therefore carries over verbatim with the enlarged ρ; what changes is the operator content of ρ and, more importantly, the appearance of soft quark degrees of freedom.

After performing the mode decomposition and applying the eikonal approximation, the soft interaction Hamiltonian of QCD can be organized as

$$H_{\rm int} = \underbrace{H_g + H_{ggg} + H_{gggg} + H_{gg-{\rm inst}} + H_{gggg-{\rm inst}}}_{\text{Paper I}} + H_{gq\bar{q}} + H_{ggq\bar{q}} + H_{q-{\rm inst}} + H_V, \tag{2.19}$$

where the first five terms are those of Paper I with ρ given by (2.15), and H_V contains valence–valence interactions only. In particular

$$H_g = \int_k \frac{g\, k^i}{\sqrt{2}\,|k^+|^{3/2}} \Big[a_i^{\dagger a}(k) \rho^a(-k) + a_i^a(k) \rho^a(k) \Big]. \tag{2.20}$$

The quark–gluon vertex of the soft sector, written in terms of the vertex function (2.5), contains three structures and their Hermitian conjugates,

$$\begin{aligned} H_{gq\bar{q}} = & \int_{k,l,m} \Big[V^{a\alpha\beta}_{3\, i\lambda_2\lambda_1}(k,m,l)\, b^{\dagger\beta}_{\lambda_1}(l) d^{\dagger\alpha}_{\lambda_2}(m) a^a_i(k) + \text{h.c.} \Big] \\ & + \int_{p,l,l'} \Big[V^{\beta'\beta b}_{F\,\lambda'\lambda j}(l',l,p)\, a^{\dagger b}_j(p) b^{\dagger\beta}_\lambda(l) b^{\beta'}_{\lambda'}(l') + \text{h.c.} \Big] \\ & + \int_{p,m,m'} \Big[V^{\alpha'\alpha b}_{G\,\lambda'\lambda j}(m',m,p)\, a^{\dagger b}_j(p) d^{\dagger\alpha}_\lambda(m) d^{\alpha'}_{\lambda'}(m') + \text{h.c.} \Big], \end{aligned} \tag{2.21}$$

with

$$V^{a\alpha\beta}_{3\, i\lambda_2\lambda_1}(k,m,l) = \frac{(2\pi)^3 g\, t^a_{\beta\alpha}}{2\sqrt{2}\sqrt{k^+}}\, \delta^{(3)}(k-l-m)\, \Phi^i_{\lambda_1\lambda_2}(k;l,m), \tag{2.22}$$

$$V^{\beta'\beta b}_{F\,\lambda'\lambda j}(l',l,p) = \frac{(2\pi)^3 g\, t^b_{\beta\beta'}}{2\sqrt{2}\sqrt{p^+}}\, \delta^{(3)}(l'-l-p)\, \Phi^j_{\lambda\lambda'}(p;l,l'), \tag{2.23}$$

$$V^{\alpha'\alpha b}_{G\,\lambda'\lambda j}(m',m,p) = -\frac{(2\pi)^3 g\, t^b_{\alpha'\alpha}}{2\sqrt{2}\sqrt{p^+}}\, \delta^{(3)}(m'-m-p)\, \Phi^j_{\lambda'\lambda}(p;m',m). \tag{2.24}$$

The first line of (2.21) creates (annihilates) a soft quark–antiquark pair from (into) a soft gluon; the second and third describe the emission and absorption of a soft gluon by a soft quark and by a soft antiquark. The minus sign and the transposed colour matrix in (2.24) reflect the antiquark charge. The instantaneous quark exchange of (2.10) restricted to soft fields is denoted by $H_{ggq\bar{q}}$. Expanding the current $\mathcal{J}$ of (2.12), the cross terms $J_{\rm val}\cdot J_g$ and $J_g\cdot J_g$ are the $H_{gg-{\rm inst}}$ and $H_{gggg-{\rm inst}}$ of Paper I, while the quark current gives

$$H_{q-{\rm inst}} = g^2 \int dx^- d^2x\, \Big[\rho^a\, \frac{1}{(i\partial^+)^2}\, J_q^{+a} + J_g^{+a} \frac{1}{(i\partial^+)^2} J_q^{+a} + \frac{1}{2}\, J_q^{+a} \frac{1}{(i\partial^+)^2} J_q^{+a} \Big], \tag{2.25}$$

with all currents built from soft fields except ρ. The first term describes the instantaneous interaction of soft quarks with the valence charge (eikonal colour rotation and pair creation off ρ), the second the instantaneous interaction between the soft gluon and quark currents, and the third the instantaneous interaction among soft quarks. The matrix elements of all these operators between soft Fock states are collected in appendix D. We count $\rho = \mathcal{O}(1)$ in the dilute regime, as in Paper I, so that H_g, H_{ggg} and $H_{gq\bar{q}}$ are of order g and H_{gggg}, the instantaneous terms and $H_{ggq\bar{q}}$ of order g^2.

3 Light-cone wave functions and light-cone perturbation theory

3.1 Fock space expansion with quarks

The soft Fock space is now generated by the gluon, quark and antiquark creation operators acting on the soft vacuum $|0\rangle \equiv |0\rangle_{\rm soft} \otimes |v\rangle_{\rm val}$ (appendix B). A dressed state is expanded as

$$|\Psi\rangle = \psi_0|0\rangle + \int d^3k\, \psi_g\, |g\rangle + \int d^3l\, d^3m\, \psi_{q\bar{q}}\, |q\bar{q}\rangle + \int d^3k\, d^3p\, \psi_{gg}\, |gg\rangle + \int d^3l\, d^3m\, d^3r\, \psi_{q\bar{q}g}\, |q\bar{q}\, g\rangle + \ldots, \tag{3.1}$$

where the sums over colours, polarizations, helicities and flavours are implicit. As in Paper I, the expansion is in the Fock basis of the soft subspace only: we do not expand the valence part, and the coefficients ψ are functionals of the valence charge density, i.e. operators on $\mathcal{H}_{\rm val}$. Since the ρ's do not commute, Eq. (2.18), their ordering is kept throughout. Relative to the pure Yang–Mills case, the soft vacuum acquires a quark–antiquark component at order g^2 (a gluon emitted by ρ splits into a pair, or the pair is created instantaneously off ρ), a soft gluon acquires a quark–antiquark component already at order g, and a soft quark acquires a quark–gluon component at order g.

3.2 Light-cone perturbation theory

Light-cone perturbation theory is the Hamiltonian formulation of old fashioned perturbation theory in light-cone quantization [35, 36, 52]. The interaction Hamiltonian acts on free Fock states and generates transitions between components with different particle content; the light-cone energy of a multiparticle state is the sum of ε over its constituents, and each intermediate state contributes an energy denominator. The rules for fermions are the standard ones: each quark–gluon vertex contributes the matrix element of $H_{gq\bar{q}}$ (appendix D), an instantaneous fermion line carrying plus momentum P^+ contributes $1/(2P^+)$ through $H_{ggq\bar{q}}$, an instantaneous gluon line contributes $1/(P^+)^2$ through $H^{(A)}_{\rm inst}$, and the fermion signs follow from the anticommutators (2.4). The derivation for non-commuting matrix elements is identical to that of Paper I (appendix E).

As in Paper I, the denominators are defined by the $i\epsilon$ prescription $E_i - E_f \to E_i - E_f + in\epsilon$ at n-th order, which follows from the adiabatic definition of Ω as the Møller operator [37, 53]. With $H_{fi} \equiv \langle f|\, H_{\rm int}\, |i\rangle$, the first and second order amplitudes are

$$\mu_1(f \leftarrow i) = \frac{H_{fi}}{E_i - E_f + i\epsilon},$$

$$\mu_2(f \leftarrow i) = \sum_m \frac{H_{fm} H_{mi}}{(E_i - E_f + 2i\epsilon)(E_i - E_m + i\epsilon)} + \frac{H^{(2)}_{fi}}{E_i - E_f + i\epsilon}, \tag{3.2}$$

where $H^{(2)}$ denotes the order g^2 (instantaneous and four-point) part of the interaction, and the ordering $H_{fm}H_{mi}$ is kept for the non-commuting coefficients. A simple rewriting of (3.2) will be used throughout this paper. Since $H_{mi} = (E_i - E_m + i\epsilon)\,\mu_1(m \leftarrow i)$ and $H_{fm} = (E_m - E_f + i\epsilon)\,\mu_1(f \leftarrow m)$,

$$\boxed{\mu_2(f \leftarrow i) = \sum_m \mu_1(f \leftarrow m)\,\mu_1(m \leftarrow i)\,\frac{E_m - E_f + i\epsilon}{E_i - E_f + 2i\epsilon} + \frac{H^{(2)}_{fi}}{E_i - E_f + i\epsilon}} \tag{3.3}$$

We refer to (3.3) as the reduction formula: every second order amplitude is a sum, over intermediate states, of products of first order amplitudes weighted by the fraction of the total energy difference that is released in the second step, plus the instantaneous terms. Two immediate consequences are worth stating.

(i) *Independent subprocesses.* If the transition $i \to f$ consists of two subprocesses X and Y acting on disjoint sets of particles, with energy releases Δ_X and Δ_Y, there are two intermediate states and (3.3) gives

$$\mu_2 = \frac{YX\,\Delta_Y + XY\,\Delta_X}{\Delta_X + \Delta_Y} \xrightarrow{[X,Y]=0} X\,Y. \tag{3.4}$$

For commuting first order amplitudes the overall energy denominator cancels between the two time orderings and the amplitude factorizes; this is the familiar partial fraction identity $1/(D_f D_X) + 1/(D_f D_Y) = 1/(D_X D_Y)$ with $D_f = D_X + D_Y$. Since ρ appears only in the Weizsäcker–Williams amplitude, all the quark structures of this paper that involve two independent subprocesses factorize exactly.

(ii) *Poles.* The overall denominator $E_i - E_f$ survives in (3.3) only for transitions that can be on the energy shell, namely those that preserve the number of soft particles of each kind ($2 \to 2$ scattering of soft partons, or the rescattering of a soft parton on ρ). These are understood with the $i\epsilon$ prescription and their residue is the on-shell scattering amplitude.

With the prescription (3.2) the unitarity relations hold identically. In particular, for every intermediate state m [34],

$$\mu_2(f \leftarrow i) + \big[\mu_2(i \leftarrow f)\big]^\dagger = -\big[\mu_1(m \leftarrow f)\big]^\dagger \mu_1(m \leftarrow i), \tag{3.5}$$

while the instantaneous terms in (3.2) are anti-Hermitian and drop out of the left-hand side. We use (3.5) to verify the unitarity relations of the coefficients that are fixed by LCPT and preserve the particle content.

3.3 Soft wave function and the evolution operator

As in Paper I, the dressed eigenstates of the full soft Hamiltonian are related to the free Fock states by the unitary evolution operator,

$$|\Psi_n\rangle = \Omega\,|n\rangle\,, \qquad \Omega^\dagger\Omega = 1, \tag{3.6}$$

with $\Omega = U_\epsilon(-\infty, 0)$ the adiabatic (Møller) operator [53]. At leading order in the coupling and the highest order in ρ it reduces to the coherent operator

$$\mathcal{C} = \exp\Big\{ \int_p \Big[A_j^b(p) a_j^{\dagger b}(p) - A_j^{\dagger b}(p) a_j^b(p) \Big] \Big\}, \tag{3.7}$$

which creates the Weizsäcker–Williams field of the (now enlarged) valence charge. The same operator diagonalizes the Hamiltonian: acting with H on (3.6) one finds $\Omega^\dagger H \Omega \,|n\rangle = E_n \,|n\rangle$, so that

$$H_{\rm diag} \equiv \Omega^\dagger H \Omega = \sum_n E_n \,|n\rangle \,\langle n| \,. \tag{3.8}$$

In QCD with quarks the Hamiltonian couples the gluonic Fock sectors to sectors with any number of soft quark–antiquark pairs, and the diagonalization can only be carried out perturbatively.

4 Perturbative expansion of the evolution operator (Ω) with quarks

In the previous section the evolution operator Ω was defined implicitly through Eq. (3.6). Beyond leading order no closed form for Ω is available and it must be constructed perturbatively. Our strategy is the one of Paper I. We first write down the most general ansatz for Ω as a fully normal ordered series in the soft gluon, quark and antiquark creation and annihilation operators, $a^\dagger, a, b^\dagger, b, d^\dagger, d$, with unknown operator valued coefficients, retaining only those operator structures that can contribute up to $\mathcal{O}(g^2)$. The unknown coefficients are then fixed by two complementary sets of conditions:

- the unitarity requirement $\Omega^\dagger \Omega = 1$, imposed order by order in the coupling;
- the requirement that Ω, acting on the free Fock states, reproduces the dressed states computed directly in LCPT, i.e. $\Omega \,|n\rangle = |\Psi_n\rangle$ for the vacuum, one, two and three-gluon states, a single quark (antiquark), a quark–antiquark pair and a quark–gluon state.

Each coefficient, containing a definite number of annihilation and creation operators, is probed only by the transition between the corresponding incoming and outgoing Fock sectors; acting on these states therefore fixes the coefficients one by one. Since ρ satisfies (2.18), the coefficients are non-commuting operators on $\mathcal{H}_{\rm val}$ and their ordering carries physical significance.

The ansatz

We write

$$\Omega = \Omega_{\rm YM} + \Omega_q, \tag{4.1}$$

where $\Omega_{\rm YM}$ is the pure gauge evolution operator of Paper I, Eq. (4.1) there, with coefficients $N, A, B_{1,2,3}, C, C_2, D_{1,2,3}, E_{1,2,3}$, now evaluated with the total charge density (2.15), and Ω_q contains every normal ordered structure with at least one soft quark or antiquark operator. The latter is organized by order and by the type of process,

$$\Omega_q = \Omega_q^{(1)} + \Omega_{q\bar{q}}^{(2)} + \Omega_F^{(2)} + \Omega_\times^{(2)}. \tag{4.2}$$

At order g,

$$\begin{aligned}\Omega_q^{(1)} = \int_{p,l,m} \Big[& \mathfrak{A}_{3\,j\lambda_2\lambda_1}^{b\alpha\beta}(p,m,l)\, b_{\lambda_1}^{\beta\dagger}(l) d_{\lambda_2}^{\alpha\dagger}(m) a_j^b(p) \\ & \qquad + \mathfrak{A}_{4\,\lambda_1\lambda_2 j}^{\beta\alpha b}(l,m,p)\, a_j^{\dagger b}(p) d_{\lambda_2}^{\alpha}(m) b_{\lambda_1}^{\beta}(l) \Big] \\ + \int_{p,l,l'} \Big[& \mathfrak{F}_{1\,\lambda'\lambda j}^{\beta'\beta b}(l',l,p)\, a_j^{\dagger b}(p) b_{\lambda}^{\beta\dagger}(l) b_{\lambda'}^{\beta'}(l') \\ & \qquad + \mathfrak{F}_{2\,j\lambda\lambda'}^{b\beta\beta'}(p,l,l')\, b_{\lambda'}^{\beta'\dagger}(l') b_{\lambda}^{\beta}(l) a_j^b(p) \Big] \\ + \int_{p,m,m'} \Big[& \mathfrak{G}_{1\,\lambda'\lambda j}^{\alpha'\alpha b}(m',m,p)\, a_j^{\dagger b}(p) d_{\lambda}^{\alpha\dagger}(m) d_{\lambda'}^{\alpha'}(m') \\ & \qquad + \mathfrak{G}_{2\,j\lambda\lambda'}^{b\alpha\alpha'}(p,m,m')\, d_{\lambda'}^{\alpha'\dagger}(m') d_{\lambda}^{\alpha}(m) a_j^b(p) \Big]. \end{aligned} \tag{4.3}$$

$\mathfrak{A}_3$ splits a soft gluon into a soft pair and $\mathfrak{A}_4$ merges a soft pair into a soft gluon; $\mathfrak{F}_1$, $\mathfrak{G}_1$ ($\mathfrak{F}_2$, $\mathfrak{G}_2$) describe the emission (absorption) of a soft gluon by a soft quark and by a soft antiquark. At order g^2, the structures that create or annihilate soft pairs are

$$\begin{aligned}\Omega_{q\bar{q}}^{(2)} = \int_{l,m} & \Big[\mathfrak{A}_{1\,\lambda_2\lambda_1}^{\alpha\beta}(m,l)\, b_{\lambda_1}^{\beta\dagger}(l) d_{\lambda_2}^{\alpha\dagger}(m) + \mathfrak{A}_{2\,\lambda_1\lambda_2}^{\beta\alpha}(l,m)\, d_{\lambda_2}^{\alpha}(m) b_{\lambda_1}^{\beta}(l) \Big] \\ + \int_{p,q,l,m} & \Big[\mathfrak{B}_{1\,kj\lambda_2\lambda_1}^{cb\alpha\beta}(q,p,m,l)\, b_{\lambda_1}^{\beta\dagger}(l) d_{\lambda_2}^{\alpha\dagger}(m) a_j^b(p) a_k^c(q) \\ & \quad + \mathfrak{B}_{2\,\lambda_1\lambda_2 kj}^{\beta\alpha cb}(l,m,q,p)\, a_j^{\dagger b}(p) a_k^{\dagger c}(q) d_{\lambda_2}^{\alpha}(m) b_{\lambda_1}^{\beta}(l) \\ & \quad + \mathfrak{B}_{3\,\lambda_2\lambda_1 jk}^{\alpha\beta bc}(m,l,p,q)\, a_k^{\dagger c}(q) a_j^b(p) b_{\lambda_1}^{\beta\dagger}(l) d_{\lambda_2}^{\alpha\dagger}(m) \\ & \quad + \mathfrak{B}_{4\,kj\lambda_1\lambda_2}^{cb\beta\alpha}(q,p,l,m)\, d_{\lambda_2}^{\alpha}(m) b_{\lambda_1}^{\beta}(l) a_j^{\dagger b}(p) a_k^c(q) \Big] \\ + \int_{p,q,r,s,l,m} & \Big[\mathfrak{C}_{1\,\lambda_1\lambda_2 jkon}^{\beta\alpha bcde}(l,m,p,q,r,s)\, a_n^{\dagger e}(s) a_o^{\dagger d}(r) a_k^{\dagger c}(q) a_j^b(p) d_{\lambda_2}^{\alpha}(m) b_{\lambda_1}^{\beta}(l) \\ & \quad + \mathfrak{C}_{2\,\lambda_1\lambda_2 jkon}^{\beta\alpha bcde}(l,m,p,q,r,s)\, a_n^{\dagger e}(s) a_o^{\dagger d}(r) a_k^c(q) a_j^b(p) d_{\lambda_2}^{\alpha}(m) b_{\lambda_1}^{\beta}(l) \\ & \quad + \mathfrak{C}_{3\,jkon\lambda_2\lambda_1}^{bcde\alpha\beta}(p,q,r,s,m,l)\, b_{\lambda_1}^{\beta\dagger}(l) d_{\lambda_2}^{\alpha\dagger}(m) a_n^{\dagger e}(s) a_o^d(r) a_k^c(q) a_j^b(p) \\ & \quad + \mathfrak{C}_{4\,jkon\lambda_2\lambda_1}^{bcde\alpha\beta}(p,q,r,s,m,l)\, b_{\lambda_1}^{\beta\dagger}(l) d_{\lambda_2}^{\alpha\dagger}(m) a_n^{\dagger e}(s) a_o^{\dagger d}(r) a_k^c(q) a_j^b(p) \Big] \\ + \int_{p,q,l,m,l',m'} & \Big[\mathfrak{D}_{1\,kj\lambda_2'\lambda_1'\lambda_2\lambda_1}^{cb\alpha'\beta'\alpha\beta}(q,p,m',l',m,l) \\ & \qquad \times b_{\lambda_1}^{\beta\dagger}(l) d_{\lambda_2}^{\alpha\dagger}(m) b_{\lambda_1'}^{\beta'\dagger}(l') d_{\lambda_2'}^{\alpha'\dagger}(m') a_j^b(p) a_k^c(q) \\ & \quad + \mathfrak{D}_{2\,\lambda_1\lambda_2\lambda_1'\lambda_2' kj}^{\beta\alpha\beta'\alpha' cb}(l,m,l',m',q,p) \\ & \qquad \times a_j^{\dagger b}(p) a_k^{\dagger c}(q) d_{\lambda_2'}^{\alpha'}(m') b_{\lambda_1'}^{\beta'}(l') d_{\lambda_2}^{\alpha}(m) b_{\lambda_1}^{\beta}(l) \Big] \\ + \int_{l,m,l',m'} & \mathfrak{D}_{3\,\lambda_1'\lambda_2'\lambda_2\lambda_1}^{\beta'\alpha'\alpha\beta}(l',m',m,l)\, b_{\lambda_1}^{\beta\dagger}(l) d_{\lambda_2}^{\alpha\dagger}(m) d_{\lambda_2'}^{\alpha'}(m') b_{\lambda_1'}^{\beta'}(l'). \end{aligned} \tag{4.4}$$

The labels of each coefficient are listed in the order in which the operators appear when the string is read from right to left. Every pair is annihilated in the order $d\,b$ and created in the order $b^\dagger d^\dagger$; with this convention all the matrix elements of Ω between states containing one soft pair, defined in appendix B, carry a positive sign. The structures that describe soft quark (antiquark) lines are

$$\Omega_F^{(2)} = \int_{l,l'} \mathfrak{P}_{\lambda'\lambda}^{\beta'\beta}(l',l)\, b_{\lambda}^{\beta\dagger}(l) b_{\lambda'}^{\beta'}(l')$$

$$
\begin{aligned}
&+\int_{q,r,l,l'}\Big[\mathfrak{Q}_{1\,\lambda'\lambda kn}^{\beta'\beta cd}(l',l,q,r)\,a_n^{\dagger d}(r)a_k^{\dagger c}(q)b_\lambda^{\beta\dagger}(l)b_{\lambda'}^{\beta'}(l')\\
&\quad+\mathfrak{Q}_{2\,\lambda' k\lambda n}^{\beta' c\beta d}(l',q,l,r)\,a_n^{\dagger d}(r)b_\lambda^{\beta\dagger}(l)a_k^{c}(q)b_{\lambda'}^{\beta'}(l')\\
&\quad+\mathfrak{Q}_{3\,nk\lambda\lambda'}^{dc\beta\beta'}(r,q,l,l')\,b_{\lambda'}^{\beta'\dagger}(l')b_\lambda^{\beta}(l)a_k^{c}(q)a_n^{d}(r)\Big]\\
&+\int_{l,l',l_2,m}\Big[\mathfrak{K}_{1\,\lambda'\lambda\lambda_2\lambda_m}^{\beta'\beta\beta_2\alpha}(l',l,l_2,m)\,b_\lambda^{\beta\dagger}(l)b_{\lambda_2}^{\beta_2\dagger}(l_2)d_{\lambda_m}^{\alpha\dagger}(m)b_{\lambda'}^{\beta'}(l')\\
&\quad+\mathfrak{K}_{2\,\lambda_m\lambda_2\lambda\lambda'}^{\alpha\beta_2\beta\beta'}(m,l_2,l,l')\,b_{\lambda'}^{\beta'\dagger}(l')b_\lambda^{\beta}(l)b_{\lambda_2}^{\beta_2}(l_2)d_{\lambda_m}^{\alpha}(m)\Big]+\big(q\to\bar q\big),
\end{aligned}
\tag{4.5}
$$

where $(q\to\bar q)$ denotes the analogous antiquark structures, with coefficients $\bar{\mathfrak{P}},\bar{\mathfrak{Q}}_{1,2,3},\bar{\mathfrak{K}}_{1,2}$, obtained by $b\leftrightarrow d$. $\mathfrak{P}$ ($\bar{\mathfrak{P}}$) describes the eikonal rescattering of a soft quark (antiquark) on the valence charge; $\mathfrak{Q}_1$ the emission of two soft gluons by a soft quark, $\mathfrak{Q}_2$ quark–gluon scattering and $\mathfrak{Q}_3$ the absorption of two gluons; $\mathfrak{K}_1$ the production of a soft pair off a soft quark and $\mathfrak{K}_2$ the reverse process. Finally, $\Omega_\times^{(2)}$ collects the remaining products of two first order structures that act on disjoint sets of particles,

$$
\Omega_\times^{(2)}=\frac{1}{2}:\Omega^{(1)}\Omega^{(1)}:\Big|_{\text{no contraction, not in (4.4),(4.5)}},\qquad \Omega^{(1)}\equiv\Omega_{\rm YM}^{(1)}+\Omega_q^{(1)},
\tag{4.6}
$$

where $\Omega_{\rm YM}^{(1)}$ contains the A and C terms of Paper I and : : denotes normal ordering. These six-operator structures (e.g. $\mathfrak{A}_3\,\mathfrak{F}_1$ or $\mathfrak{F}_1\,C$) are fixed completely by (3.4), as shown in section 5; we list them separately only to keep (4.4) and (4.5) readable. The structures $\mathfrak{C}_{1,\dots,4}$ and $\mathfrak{D}_{1,2}$ in (4.4) are of the same type, but we keep them explicitly since they act on the states considered below. Normal ordered structures with more operators than those displayed contribute beyond $\mathcal{O}(g^2)$ and are omitted. The coefficients $\mathfrak{A}_3$, $\mathfrak{A}_4$, $\mathfrak{F}_{1,2}$, $\mathfrak{G}_{1,2}$ are of order g; all others are of order g^2. The identity component of Ω is the normalization N of Paper I; since soft pairs are not created out of the vacuum at order g, N receives no quark contribution at $\mathcal{O}(g^2)$.

Unitarity constraints

Expanding $\Omega=1+g\Omega^{(1)}+g^2\Omega^{(2)}+\mathcal{O}(g^3)$, the unitarity condition gives, as in Paper I,

$$
\Omega^{\dagger(1)}=-\Omega^{(1)},\qquad \Omega^{(2)}+\Omega^{\dagger(2)}=-\Omega^{\dagger(1)}\Omega^{(1)}=\Omega^{(1)}\Omega^{(1)}.
\tag{4.7}
$$

In the quark sector the first condition gives

$$
\begin{aligned}
\mathfrak{A}_{4\,\lambda_1\lambda_2 j}^{\beta\alpha b}(l,m,p)&=-\big[\mathfrak{A}_{3\,j\lambda_2\lambda_1}^{b\alpha\beta}(p,m,l)\big]^\dagger,\\
\mathfrak{F}_{2\,j\lambda\lambda'}^{b\beta\beta'}(p,l,l')&=-\big[\mathfrak{F}_{1\,\lambda'\lambda j}^{\beta'\beta b}(l',l,p)\big]^\dagger,\qquad \mathfrak{G}_{2\,j\lambda\lambda'}^{b\alpha\alpha'}(p,m,m')=-\big[\mathfrak{G}_{1\,\lambda'\lambda j}^{\alpha'\alpha b}(m',m,p)\big]^\dagger.
\end{aligned}
\tag{4.8}
$$

From here onward we write $\mathfrak{F}\equiv\mathfrak{F}_1$, $\mathfrak{G}\equiv\mathfrak{G}_1$. The second condition, projected onto the independent normal ordered structures, gives the following relations. To lighten the notation we suppress the colour, polarization and helicity labels: each coefficient carries the labels of its momentum arguments, in the order of Eqs. (4.3)–(4.5), and repeated internal momenta are integrated with $\int$.

$$
\mathfrak{A}_2(l,m)=-\mathfrak{A}_1^\dagger(m,l)+\int_p A^\dagger(p)\,\mathfrak{A}_3^\dagger(p,m,l),
\tag{4.9}
$$

$$
\mathfrak{B}_2(l,m,q,p)=-\mathfrak{B}_1^\dagger(q,p,m,l)-\mathfrak{A}_3^\dagger(p,m,l)A(q)-A(p)\mathfrak{A}_3^\dagger(q,m,l)
$$

$$
\begin{aligned}
&- \int_u C(u,q,p)\,\mathfrak{A}_3^\dagger(u,m,l) \\
&- \int_{l_1} \mathfrak{A}_3^\dagger(q,m,l_1)\,\mathfrak{F}(l,l_1,p) - \int_{m_1} \mathfrak{A}_3^\dagger(q,m_1,l)\,\mathfrak{G}(m,m_1,p), \qquad (4.10)
\end{aligned}
$$

$$
\begin{aligned}
\mathfrak{B}_4(q,p,l,m) = &-\mathfrak{B}_3^\dagger(m,l,p,q) + \mathfrak{A}_3^\dagger(p,m,l)A^\dagger(q) + A^\dagger(q)\mathfrak{A}_3^\dagger(p,m,l) \\
&+ 2\int_u C^\dagger(p,q,u)\,\mathfrak{A}_3^\dagger(u,m,l) \\
&+ \int_{l_1} \mathfrak{A}_3^\dagger(p,m,l_1)\,\mathfrak{F}^\dagger(l_1,l,q) + \int_{m_1} \mathfrak{A}_3^\dagger(p,m_1,l)\,\mathfrak{G}^\dagger(m_1,m,q), \qquad (4.11)
\end{aligned}
$$

$$
\mathfrak{C}_1(l,m,p,q,r,s) = -\mathfrak{C}_3^\dagger(p,q,r,s,m,l) - \mathcal{S}_3\Big\{\mathfrak{A}_3^\dagger(s,m,l),\, C(p,q,r)\Big\}, \qquad (4.12)
$$

$$
\mathfrak{C}_2(l,m,p,q,r,s) = -\mathfrak{C}_4^\dagger(p,q,r,s,m,l) + \mathcal{S}_2\Big\{\mathfrak{A}_3^\dagger(s,m,l),\, C^\dagger(r,q,p)\Big\}, \qquad (4.13)
$$

$$
\begin{aligned}
\mathfrak{D}_2(l,m,l',m',q,p) = &-\mathfrak{D}_1^\dagger(q,p,m',l',m,l) \\
&+ \mathfrak{A}_3^\dagger(p,m,l)\mathfrak{A}_3^\dagger(q,m',l') + \mathfrak{A}_3^\dagger(q,m,l)\mathfrak{A}_3^\dagger(p,m',l'), \qquad (4.14)
\end{aligned}
$$

$$
\begin{aligned}
\mathfrak{Q}_3(r,q,l,l') = &-\mathfrak{Q}_1^\dagger(l',l,q,r) + A^\dagger(q)\mathfrak{F}^\dagger(l',l,r) \\
&+ \mathfrak{F}^\dagger(l',l,r)A^\dagger(q) \\
&+ \int_{l_1} \mathfrak{F}^\dagger(l_1,l,r)\,\mathfrak{F}^\dagger(l',l_1,q) + \int_s \mathfrak{F}^\dagger(l',l,s)\,C^\dagger(s,q,r) + (q \leftrightarrow r), \qquad (4.15)
\end{aligned}
$$

$$
\mathfrak{K}_2(m,l_2,l,l') = -\mathfrak{K}_1^\dagger(l',l,l_2,m) + \Big[\int_k \mathfrak{F}^\dagger(l',l,k)\,\mathfrak{A}_3^\dagger(k,m,l_2) - \big(l \leftrightarrow l_2\big)\Big]. \qquad (4.16)
$$

Here $\mathcal{S}_3$ symmetrizes over the three gluon labels created by $\mathfrak{C}_3^\dagger$ and $\mathcal{S}_2$ over the two created by $\mathfrak{C}_4^\dagger$, $\{X,Y\} \equiv XY + YX$, A and C are the Weizsäcker–Williams and three-gluon coefficients of Paper I, and the antisymmetrization in (4.16) acts on all the labels of the two identical quarks. The antiquark relations follow by $\mathfrak{F} \to \mathfrak{G}$, $b \to d$. The structures $\mathfrak{B}_2$, $\mathfrak{B}_4$ and $\mathfrak{Q}_3$ receive contributions from the emission and absorption coefficients $\mathfrak{F}$ and $\mathfrak{G}$: they correspond to the t- and u-channel quark exchange in $gg \to q\bar{q}$ and to the sequential emission of two gluons by a quark, and they are absent if the ansatz is restricted to the pair structures.

The coefficients that preserve the particle content, $\mathfrak{D}_3$, $\mathfrak{P}$, $\bar{\mathfrak{P}}$ and $\mathfrak{Q}_2$, are fixed completely by matching to the LCWFs below. Their unitarity relations are

$$
\begin{aligned}
\mathfrak{D}_3(l',m',m,l) + \mathfrak{D}_3^\dagger(l,m,m',l') = &- \int_k \mathfrak{A}_3(k,m,l)\,\mathfrak{A}_3^\dagger(k,m',l') \\
&- \int_p \Big[\mathfrak{G}^\dagger(m,m',p)\,\mathfrak{F}(l',l,p) + \mathfrak{F}^\dagger(l,l',p)\,\mathfrak{G}(m',m,p)\Big], \qquad (4.17)
\end{aligned}
$$

$$
\begin{aligned}
\mathfrak{P}(l',l) + \mathfrak{P}^\dagger(l,l') = &- \int_p \Big[A^\dagger(p)\,\mathfrak{F}(l',l,p) + \mathfrak{F}^\dagger(l,l',p)\,A(p)\Big] \\
&- \int_{p,l_1} \mathfrak{F}^\dagger(l,l_1,p)\,\mathfrak{F}(l',l_1,p), \qquad (4.18)
\end{aligned}
$$

$$
\mathfrak{Q}_2(f \leftarrow i) + \mathfrak{Q}_2^\dagger(i \leftarrow f) = -(2\pi)^6 \sum_m \big[\mu_1(m \leftarrow f)\big]^\dagger \mu_1(m \leftarrow i), \qquad (4.19)
$$

and they are therefore nontrivial checks of the calculation. In (4.19) the intermediate states are q, qgg, $qq\bar{q}$ and, for the ρ dependent part, qgg with one gluon exchanged with the valence; each term is the product of two of the first order coefficients $A, C, \mathfrak{A}_3, \mathfrak{F}$. All of them are verified intermediate state by intermediate state with the identity (3.5). The relation for $\mathfrak{D}_3$ requires the t-channel exchange of a gluon between the quark and the antiquark, and the relation for $\mathfrak{P}$ requires the exchange of a gluon between the soft quark and the valence charge; both involve precisely the coefficients $\mathfrak{F}$ and $\mathfrak{G}$. Finally, the pure gauge relation for the coefficient B_3 of Paper I (the $a^\dagger a$ structure) acquires the quark loop term $-\int_{l,m} \mathfrak{A}_3^\dagger(k',m,l)\mathfrak{A}_3(k,m,l)$ on its right-hand side; it affects only the momentum diagonal part of B_3, discussed in section 4.2.

Action of Ω on Fock states

Since Ω is normal ordered, each structure contributes to the action on a given Fock state in a simple way: every annihilation operator either contracts with an incoming particle or annihilates the vacuum, while the creation operators produce the particles of the outgoing component. If a structure contains n annihilation and M creation operators and acts on an n-particle state, it produces

$$\Omega\,|n\rangle \supset \frac{1}{(2\pi)^{3(n+M)/2}} \int d^3(M \text{ momenta})\; K\;\; |M\rangle\,, \qquad \text{i.e.} \qquad K = (2\pi)^{3(n+M)/2}\,\psi, \tag{4.20}$$

where K is the coefficient and ψ the coefficient of the outgoing state in the LCWF (3.1). We list only the terms that involve the quark sector and carry new information through $\mathcal{O}(g^2)$; the purely gluonic terms are those of Paper I.

Vacuum.

$$\Omega|0\rangle \supset \frac{1}{(2\pi)^3} \int d^3l\, d^3m\; \mathfrak{A}_{1\,\lambda_2\lambda_1}^{\alpha\beta}(m,l)\; \Big|q_{\lambda_1}^{\beta}(l)\,\bar{q}_{\lambda_2}^{\alpha}(m)\Big\rangle\,. \tag{4.21}$$

One gluon.

$$\begin{aligned}
\Omega\,|g_i^a(k)\rangle \supset\; & \frac{1}{(2\pi)^{9/2}} \int d^3l\, d^3m\; \mathfrak{A}_{3\,i\lambda_2\lambda_1}^{a\alpha\beta}(k,m,l)\; \Big|q_{\lambda_1}^{\beta}(l)\bar{q}_{\lambda_2}^{\alpha}(m)\Big\rangle \\
& + \frac{1}{(2\pi)^6} \int d^3l\, d^3m\, d^3q\; \mathfrak{B}_{3\,\lambda_2\lambda_1 ik}^{\alpha\beta ac}(m,l,k,q)\; \Big|q_{\lambda_1}^{\beta}(l)\bar{q}_{\lambda_2}^{\alpha}(m)g_k^c(q)\Big\rangle \\
& + \frac{1}{(2\pi)^3} \int d^3l\, d^3m\; \mathfrak{A}_1(m,l)\; |g_i^a(k)\,q\bar{q}\rangle \\
& + \frac{1}{(2\pi)^3} \int d^3p\; B_3(k,p)\Big|_{\text{quark loop}}\, |g_i^a(p)\rangle\,.
\end{aligned} \tag{4.22}$$

Two gluons.

$$\begin{aligned}
\Omega\,\Big|g_i^a(k)g_j^b(p)\Big\rangle \supset\; & \frac{1}{(2\pi)^6} \int d^3l\, d^3m\; \Big|q_{\lambda_1}^{\beta}(l)\bar{q}_{\lambda_2}^{\alpha}(m)\Big\rangle \\
& \qquad \times \Big[\mathfrak{B}_{1\,ji\lambda_2\lambda_1}^{ba\alpha\beta}(p,k,m,l) + \mathfrak{B}_{1\,ij\lambda_2\lambda_1}^{ab\alpha\beta}(k,p,m,l)\Big] \\
& + \frac{1}{(2\pi)^9} \int d^3r\, d^3s\, d^3l\, d^3m\; \Big[\mathfrak{C}_4(k,p,r,s,m,l) + \mathfrak{C}_4(p,k,r,s,m,l)\Big]
\end{aligned}$$

$$
\begin{aligned}
&\times \Big| q\bar{q}\, g_o^d(r) g_n^e(s) \Big\rangle \\
&+ \frac{1}{(2\pi)^9} \int d^3l\, d^3m\, d^3l'\, d^3m' \, \Big[\mathfrak{D}_1(p,k,m',l',m,l) + \mathfrak{D}_1(k,p,m',l',m,l) \Big] \\
&\qquad \times |q\bar{q}\, q\bar{q}\rangle \\
&+ \frac{1}{(2\pi)^{9/2}} \int d^3l\, d^3m \, \Big[\mathfrak{A}_3(k,m,l) \Big| g_j^b(p)\, q\bar{q} \Big\rangle \\
&\qquad\qquad + \mathfrak{A}_3(p,m,l)\, |g_i^a(k)\, q\bar{q}\rangle \Big] + \dots
\end{aligned} \tag{4.23}
$$

where the dots denote the disconnected terms $\mathfrak{A}_1\,|gg\,q\bar{q}\rangle$ and $\mathfrak{B}_3\,|g\,q\bar{q}\,g\rangle$ with a spectator gluon.

Three gluons.

$$
\Omega \Big| g_i^a(k) g_j^b(p) g_k^c(q) \Big\rangle \supset \frac{1}{(2\pi)^9} \int d^3s\, d^3l\, d^3m \sum_{3 \text{ assign.}} \mathfrak{C}_3(k,p,q,s,m,l)\ |q\bar{q}\, g_n^e(s)\rangle + \dots \tag{4.24}
$$

One quark.

$$
\begin{aligned}
\Omega \Big| q_{\lambda'}^{\beta'}(l') \Big\rangle \supset\ & \frac{1}{(2\pi)^3} \int d^3l\, \mathfrak{P}_{\lambda'\lambda}^{\beta'\beta}(l',l) \Big| q_\lambda^\beta(l) \Big\rangle \\
&+ \frac{1}{(2\pi)^{9/2}} \int d^3l\, d^3p\, \mathfrak{F}_{\lambda'\lambda j}^{\beta'\beta b}(l',l,p) \Big| q_\lambda^\beta(l) g_j^b(p) \Big\rangle \\
&+ \frac{1}{(2\pi)^6} \int d^3l\, d^3q\, d^3r\, \mathfrak{Q}_1(l',l,q,r) \Big| q_\lambda^\beta(l) g_k^c(q) g_n^d(r) \Big\rangle \\
&+ \frac{1}{(2\pi)^6} \int d^3l\, d^3l_2\, d^3m\, \mathfrak{K}_1(l',l,l_2,m)\, |q\, q\bar{q}\rangle \\
&+ \frac{1}{(2\pi)^{3/2}} \int d^3p\, A(p) \Big| q_{\lambda'}^{\beta'}(l') g_j^b(p) \Big\rangle + \dots
\end{aligned} \tag{4.25}
$$

and analogously for an antiquark with $\bar{\mathfrak{P}}, \mathfrak{G}, \bar{\mathfrak{Q}}_1, \bar{\mathfrak{K}}_1$.

Quark–antiquark pair.

$$
\begin{aligned}
\Omega \Big| q_{\lambda_1'}^{\beta'}(l') \bar{q}_{\lambda_2'}^{\alpha'}(m') \Big\rangle \supset\ & \frac{1}{(2\pi)^3} \mathfrak{A}_2(l',m') |0\rangle \\
&+ \frac{1}{(2\pi)^{9/2}} \int d^3k\, \mathfrak{A}_4(l',m',k)\, |g_i^a(k)\rangle \\
&+ \frac{1}{(2\pi)^6} \int d^3l\, d^3m\, \mathfrak{D}_3(l',m',m,l) \Big| q_{\lambda_1}^\beta(l) \bar{q}_{\lambda_2}^\alpha(m) \Big\rangle \\
&+ \frac{1}{(2\pi)^3} \int d^3l\, \mathfrak{P}(l',l) \Big| q_{\lambda_1}^\beta(l) \bar{q}_{\lambda_2}^{\alpha'}(m') \Big\rangle \\
&+ \frac{1}{(2\pi)^3} \int d^3m\, \bar{\mathfrak{P}}(m',m) \Big| q_{\lambda_1'}^{\beta'}(l') \bar{q}_{\lambda_2}^\alpha(m) \Big\rangle \\
&+ \frac{1}{(2\pi)^6} \int d^3p\, d^3q\, \mathfrak{B}_2(l',m',q,p) \Big| g_j^b(p) g_k^c(q) \Big\rangle \\
&+ \frac{1}{(2\pi)^{9/2}} \int d^3l\, d^3p\, \mathfrak{F}(l',l,p)\, |q\, \bar{q}\, g\rangle
\end{aligned}
$$

$$+\frac{1}{(2\pi)^{9/2}}\int d^3m\,d^3p\;\mathfrak{G}(m',m,p)\,|q\,\bar{q}\,g\rangle+\dots \tag{4.26}$$

Quark–gluon state.

$$\Omega\left|q^{\beta'}_{\lambda'}(l')g^c_k(q)\right\rangle \supset \frac{1}{(2\pi)^6}\int d^3l\,d^3r\;\mathfrak{Q}_2(l',q,l,r)\left|q^{\beta}_{\lambda}(l)g^d_n(r)\right\rangle + \frac{1}{(2\pi)^{9/2}}\int d^3l\;\mathfrak{F}_2(q,l',l)\left|q^{\beta}_{\lambda}(l)\right\rangle+\dots \tag{4.27}$$

The coefficients of real transitions, $\mathfrak{A}_1$, $\mathfrak{A}_3$, $\mathfrak{F}$, $\mathfrak{G}$, $\mathfrak{B}_1$, $\mathfrak{B}_3$, $\mathfrak{C}_3$, $\mathfrak{C}_4$, $\mathfrak{D}_1$, $\mathfrak{Q}_1$, $\mathfrak{K}_1$, and the particle number preserving coefficients $\mathfrak{D}_3$, $\mathfrak{P}$, $\bar{\mathfrak{P}}$, $\mathfrak{Q}_2$ are determined by matching these expressions to the LCWFs computed in the following subsections. The remaining coefficients, $\mathfrak{A}_2$, $\mathfrak{A}_4$, $\mathfrak{F}_2$, $\mathfrak{G}_2$, $\mathfrak{B}_2$, $\mathfrak{B}_4$, $\mathfrak{C}_1$, $\mathfrak{C}_2$, $\mathfrak{D}_2$, $\mathfrak{Q}_3$, $\mathfrak{K}_2$, follow from the unitarity relations.[2]

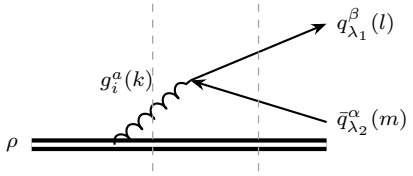
4.1 Calculation of the vacuum incoming state

We now compute the LCWF components that fix the coefficients. Throughout, the outgoing quark is $q^{\beta}_{\lambda_1}(l)$ and the outgoing antiquark $\bar{q}^{\alpha}_{\lambda_2}(m)$, the dashed vertical lines denote the intermediate states, the double line denotes the valence charge, a straight line crossed by a bar denotes an instantaneous exchange, and we abbreviate $E_g(k)=\varepsilon_k$, $E_{q\bar{q}}(l,m)=\varepsilon_l+\varepsilon_m$, $E_{gg}(k,p)=\varepsilon_k+\varepsilon_p$. The matrix elements are collected in appendix D. Each wave function is labelled, as in Paper I, by its outgoing particles and its powers of ρ.

4.1.1 Quark–antiquark outgoing

The order g component of the vacuum is the Weizsäcker–Williams gluon of Paper I. A soft pair appears first at order g^2, through two diagrams.

(a) The valence emits a gluon which splits into a pair.



Applying the LCPT rules,

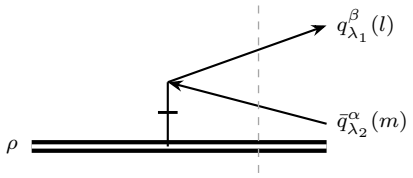
$$\left|\Psi^1_{q\bar{q}\,\rho}\right\rangle=\int d^3k\,d^3l\,d^3m\;\left|q^{\beta}_{\lambda_1}(l)\bar{q}^{\alpha}_{\lambda_2}(m)\right\rangle\;\frac{\left\langle q^{\beta}_{\lambda_1}(l)\bar{q}^{\alpha}_{\lambda_2}(m)\right|H_{g q\bar{q}}\left|g^a_i(k)\right\rangle}{-E_{q\bar{q}}(l,m)}\;\frac{\langle g^a_i(k)|\,H_g|0\rangle}{-E_g(k)}, \tag{4.28}$$

where here and below the sums over colours, polarizations, helicities and flavours are implicit. Using the matrix elements (D.4) and (D.1) and $k=l+m$,

$$\left|\Psi^1_{q\bar{q}\,\rho}\right\rangle=\frac{g^2}{32\pi^3}\int d^3l\,d^3m\;\left|q^{\beta}_{\lambda_1}(l)\bar{q}^{\alpha}_{\lambda_2}(m)\right\rangle\;\frac{t^a_{\beta\alpha}\,\rho^a(-k)\,k^i\,\Phi^i_{\lambda_1\lambda_2}(k;l,m)}{(k^+)^2\,\varepsilon_k\left(\varepsilon_l+\varepsilon_m\right)}. \tag{4.29}$$

[2]For $\mathfrak{A}_4$, $\mathfrak{A}_2$ and $\mathfrak{B}_2$ we have also verified the unitarity relations by a direct LCPT computation of the transitions $q\bar{q}\to g$, $q\bar{q}\to 0$ and $q\bar{q}\to gg$; see section 4.6.

(b) Instantaneous pair creation off the valence charge.



This is the $\rho \cdot J_q$ term of $H_{q-\mathrm{inst}}$, Eq. (2.25):

$$\left|\Psi^2_{q\bar{q}\,\rho}\right\rangle = \int d^3l\, d^3m\, \left|q^\beta_{\lambda_1}(l)\bar{q}^\alpha_{\lambda_2}(m)\right\rangle \frac{\left\langle q^\beta_{\lambda_1}(l)\bar{q}^\alpha_{\lambda_2}(m)\right| H_{q-\mathrm{inst}}|0\rangle}{-E_{q\bar{q}}(l,m)}$$
$$= -\frac{g^2}{8\pi^3}\int d^3l\, d^3m\, |q\bar{q}\rangle \frac{t^a_{\beta\alpha}\,\rho^a(-k)\,\delta_{\lambda_1\lambda_2}}{(k^+)^2\left(\varepsilon_l+\varepsilon_m\right)}, \quad (4.30)$$

with $k \equiv l + m$ and the matrix element (D.7).

4.1.2 Summary of results

Using $k^i\Gamma^i(k;l,m) = 4\varepsilon_k - 2(\varepsilon_l+\varepsilon_m) - \frac{k^+}{l^+m^+}(\boldsymbol{\sigma}\cdot\boldsymbol{l})(\boldsymbol{\sigma}\cdot\boldsymbol{m})$, which follows from (2.5) and $\boldsymbol{k} = \boldsymbol{l} + \boldsymbol{m}$, the term proportional to ε_k in (4.29) cancels the instantaneous contribution (4.30) exactly, and the quark component of the vacuum at order g^2 is

$$|\Psi_{q\bar{q}\,\rho}\rangle = -\frac{g^2}{8\pi^3}\int d^3l\, d^3m\, \left|q^\beta_{\lambda_1}(l)\bar{q}^\alpha_{\lambda_2}(m)\right\rangle\, t^a_{\beta\alpha}\rho^a(-k)\left[\frac{\delta_{\lambda_1\lambda_2}}{k^+\boldsymbol{k}^2} + \frac{\chi^\dagger_{\lambda_1}(\boldsymbol{\sigma}\cdot\boldsymbol{l})(\boldsymbol{\sigma}\cdot\boldsymbol{m})\chi_{\lambda_2}}{2\,l^+m^+\boldsymbol{k}^2\,(\varepsilon_l+\varepsilon_m)}\right]. \quad (4.31)$$

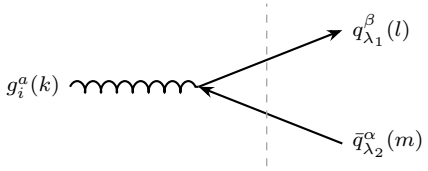
The cancellation of the instantaneous term is required by unitarity: the relation (4.9) contains no instantaneous piece. Since the pair component is of order g^2, its norm is of order g^4, and the normalization N of the vacuum is not modified by quarks at the order we work. The full vacuum state through $\mathcal{O}(g^2)$ is the one of Paper I supplemented by (4.31), with $\rho = \rho_g + \rho_q$ in all the gluonic components.

4.2 Calculation of the single-gluon incoming state

We now examine all quark contributions at orders g and g^2 associated with a single-gluon incoming state $|g^a_i(k)\rangle$. The possible final configurations are a quark–antiquark pair (order g), the one gluon state itself (through the quark loop) and the quark–antiquark–gluon state (order g^2).

4.2.1 Quark–antiquark outgoing

To order g only one diagram contributes: the incoming gluon splits into a pair.



$$|(\Psi_i^a(k))_{q\bar{q}}\rangle = \int d^3l\, d^3m\ \left|q^\beta_{\lambda_1}(l)\bar{q}^\alpha_{\lambda_2}(m)\right\rangle\ \frac{\left\langle q^\beta_{\lambda_1}(l)\bar{q}^\alpha_{\lambda_2}(m)\right| H_{gq\bar{q}}\,|g_i^a(k)\rangle}{E_g(k)-E_{q\bar{q}}(l,m)}$$
$$= \frac{g}{8\pi^{3/2}}\int d^3l\, d^3m\ |q\bar{q}\rangle\ \frac{t^a_{\beta\alpha}\,\delta^{(3)}(k-l-m)\,\Phi^i_{\lambda_1\lambda_2}(k;l,m)}{\sqrt{k^+}\,\left(\varepsilon_k-\varepsilon_l-\varepsilon_m\right)}. \quad (4.32)$$

4.2.2 One gluon outgoing: the quark loop

At order g^2 the incoming gluon can fluctuate into a pair and recombine. Since momentum is conserved along the soft line, this contribution is proportional to $\delta^{(3)}(k-k')$ and has $E_i = E_f$: it is the quark loop contribution to the gluon self-energy, and it enters the coefficient B_3 of Paper I only through its momentum diagonal part. As in Paper I, its effect at $\mathcal{O}(g^2)$ is fixed by the normalization of the dressed state. The norm of the component (4.32) is evaluated with the variables $z = l^+/k^+$, $\boldsymbol{\kappa} = \boldsymbol{l} - z\boldsymbol{k}$ of Eq. (2.6), for which $d^3l = k^+dz\,d^2\kappa$ and, for fixed i after averaging over the direction of $\boldsymbol{\kappa}$,

$$\sum_{\lambda_1\lambda_2}\left|\Phi^i_{\lambda_1\lambda_2}(k;l,m)\right|^2 = \frac{2\,\boldsymbol{\kappa}^2\left[z^2+(1-z)^2\right]}{(k^+)^2z^2(1-z)^2}. \quad (4.33)$$

All powers of $\boldsymbol{\kappa}$ and k^+ cancel against the energy denominator and one finds

$$\left\langle(\Psi^{a'}_{i'}(k'))_{q\bar{q}}\right|(\Psi^a_i(k))_{q\bar{q}}\rangle = \delta^{aa'}\delta_{ii'}\delta^{(3)}(k-k')\,\frac{g^2N_fT_F}{8\pi^3}\int_{\Lambda/k^+}^{1-\Lambda/k^+}dz\left[z^2+(1-z)^2\right]\int\frac{d^2\kappa}{\boldsymbol{\kappa}^2}$$
$$= \delta^{aa'}\delta_{ii'}\delta^{(3)}(k-k')\,\frac{g^2N_fT_F}{8\pi^3}\,\frac{2}{3}\int\frac{d^2\kappa}{\boldsymbol{\kappa}^2}, \quad (4.34)$$

with $T_F = 1/2$; the longitudinal integral is finite and the cutoff Λ may be removed. Combined with the gluon loop of Paper I, Eq. (4.83) there, the order g^2 norm of the two-parton components of a soft gluon is

$$\frac{g^2}{8\pi^3}\left[2N_c\log\frac{k^+}{\Lambda}-\frac{11N_c-4T_FN_f}{6}\right]\int\frac{d^2\kappa}{\boldsymbol{\kappa}^2}, \quad (4.35)$$

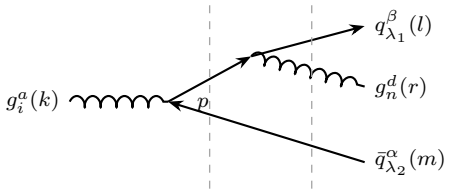
in which the familiar combination $11N_c-4T_FN_f$ of the one-loop β-function appears as the coefficient of the ultraviolet divergence. The transverse integral is scaleless and vanishes in dimensional regularization. Therefore, exactly as in Paper I, the quark loop does not modify the normalization N at $\mathcal{O}(g^2)$, and the momentum diagonal part of B_3 is unchanged.

4.2.3 Quark, antiquark and one gluon outgoing

At order g^2 there are seven connected contributions to the $\left|q^\beta_{\lambda_1}(l)\bar{q}^\alpha_{\lambda_2}(m)g^d_n(r)\right\rangle$ component. We denote the overall energy denominator by

$$D_f \equiv E_i - E_f = \varepsilon_k-\varepsilon_l-\varepsilon_m-\varepsilon_r. \quad (4.36)$$

(a) The gluon splits into a pair, then the quark emits a gluon.

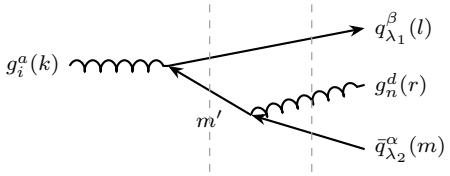


$$\left|(\Psi_i^a(k))^1_{gq\bar{q}}\right\rangle = \int d^3p\, d^3l\, d^3m\, d^3r\ \left|q\bar{q}\, g\right\rangle$$
$$\frac{\left\langle q^\beta_{\lambda_1}(l) g^d_n(r)\right| H_{gq\bar{q}} \left|q^\gamma_\lambda(p)\right\rangle}{D_f}\ \frac{\left\langle q^\gamma_\lambda(p)\bar{q}^\alpha_{\lambda_2}(m)\right| H_{gq\bar{q}}\left|g^a_i(k)\right\rangle}{\varepsilon_k-\varepsilon_p-\varepsilon_m}. \quad (4.37)$$

Using (D.4) and (D.5), closing the intermediate helicity sum with $\sum_\lambda \chi_\lambda\chi^\dagger_\lambda = \mathbf{1}$, and $p = l + r$,

$$\left|(\Psi_i^a(k))^1_{gq\bar{q}}\right\rangle = \frac{g^2}{64\pi^3}\int d^3l\, d^3m\, d^3r\ \left|q\bar{q}\, g\right\rangle$$
$$\times\ (t^d t^a)_{\beta\alpha}\,\frac{\delta^{(3)}(k-l-m-r)}{\sqrt{r^+k^+}}\,\frac{\chi^\dagger_{\lambda_1}\Gamma^n(r;l,p)\,\Gamma^i(k;p,m)\chi_{\lambda_2}}{D_f\left(\varepsilon_k-\varepsilon_p-\varepsilon_m\right)}. \quad (4.38)$$

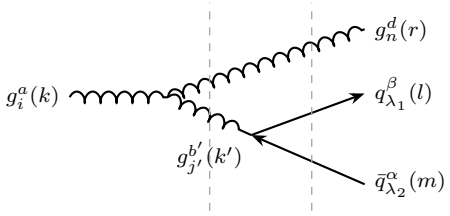
(b) The gluon splits into a pair, then the antiquark emits a gluon.



With $m' = m + r$, the antiquark vertex (D.6) and the same steps,

$$\left|(\Psi_i^a(k))^2_{gq\bar{q}}\right\rangle = -\frac{g^2}{64\pi^3}\int d^3l\, d^3m\, d^3r\ \left|q\bar{q}\, g\right\rangle$$
$$\times\ (t^a t^d)_{\beta\alpha}\,\frac{\delta^{(3)}(k-l-m-r)}{\sqrt{r^+k^+}}\,\frac{\chi^\dagger_{\lambda_1}\Gamma^i(k;l,m')\,\Gamma^n(r;m',m)\chi_{\lambda_2}}{D_f\left(\varepsilon_k-\varepsilon_l-\varepsilon_{m'}\right)}. \quad (4.39)$$

(c) The gluon splits into two gluons, one of which splits into a pair.

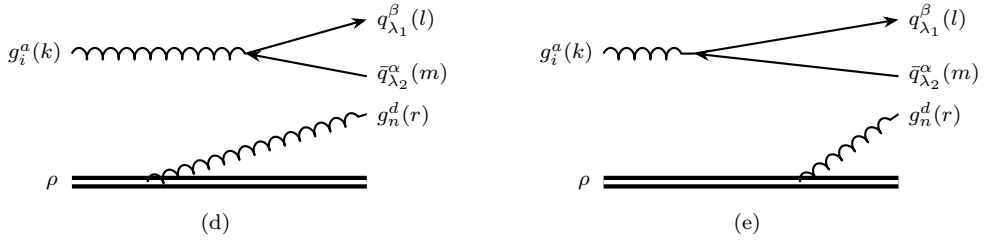


$$\left|(\Psi_i^a(k))^3_{gq\bar{q}}\right\rangle = \frac{1}{2}\int d^3k'\, d^3l\, d^3m\, d^3r\ \left|q\bar{q}\, g\right\rangle\ \frac{\langle q\bar{q}|\, H_{gq\bar{q}}\left|g^{b'}_{j'}(k')\right\rangle}{D_f}\ \frac{\left\langle g^{b'}_{j'}(k') g^d_n(r)\right| H_{ggg}\left|g^a_i(k)\right\rangle}{\varepsilon_k-\varepsilon_{k'}-\varepsilon_r}. \quad (4.40)$$

The factor $\frac{1}{2}$ cancels against the two assignments of the intermediate gluon pair. With $k' = l + m$ and (D.3),

$$\left|(\Psi_i^a(k))^3_{gq\bar{q}}\right\rangle = \frac{ig^2}{64\pi^3}\int d^3l\, d^3m\, d^3r\ \left|q\bar{q}\, g\right\rangle$$
$$\times\ f^{ab'd}t^{b'}_{\beta\alpha}\,\frac{\delta^{(3)}(k-l-m-r)}{k'^+\sqrt{k^+r^+}}\,\frac{\mathcal{V}^{ij'n}(k;k',r)\,\Phi^{j'}_{\lambda_1\lambda_2}(k';l,m)}{D_f\left(\varepsilon_k-\varepsilon_{k'}-\varepsilon_r\right)}. \quad (4.41)$$

(d), (e) The valence emits the gluon r and the incoming gluon splits.



In (d) the valence emits first and the intermediate state is $\big|g_i^a(k)g_n^d(r)\big\rangle$, with $E_i - E_m = -\varepsilon_r$; in (e) the pair is formed first, with $E_i - E_m = \varepsilon_k - \varepsilon_l - \varepsilon_m$:

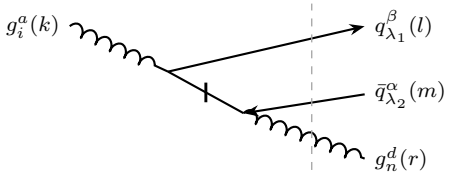
$$\big|(\Psi_i^a(k))^4_{gq\bar{q}\,\rho}\big\rangle = \int d^3l\, d^3m\, d^3r\ \ |q\bar{q}\,g\rangle\ \frac{\big\langle q\bar{q}\,g_n^d(r)\big|\,H_{gq\bar{q}}\,\big|g_i^a(k)g_n^d(r)\big\rangle}{D_f}\ \frac{\big\langle g_i^a(k)g_n^d(r)\big|\,H_g\,|g_i^a(k)\rangle}{-\varepsilon_r}, \tag{4.42}$$

$$\big|(\Psi_i^a(k))^5_{gq\bar{q}\,\rho}\big\rangle = \int d^3l\, d^3m\, d^3r\ \ |q\bar{q}\,g\rangle\ \frac{\big\langle q\bar{q}\,g_n^d(r)\big|\,H_g\,|q\bar{q}\rangle}{D_f}\ \frac{\langle q\bar{q}|\,H_{gq\bar{q}}\,|g_i^a(k)\rangle}{\varepsilon_k - \varepsilon_l - \varepsilon_m}. \tag{4.43}$$

Since $D_f = (\varepsilon_k - \varepsilon_l - \varepsilon_m) + (-\varepsilon_r)$, the two orderings combine by (3.4) and the overall denominator cancels:

$$\big|(\Psi_i^a(k))^{4+5}_{gq\bar{q}\,\rho}\big\rangle = \frac{g^2}{32\pi^3}\int d^3l\, d^3m\, d^3r\ \ |q\bar{q}\,g\rangle\ \frac{t^a_{\beta\alpha}\, r^n \rho^d(-r)\,\Phi^i_{\lambda_1\lambda_2}(k;l,m)\,\delta^{(3)}(k-l-m)}{\sqrt{k^+}(r^+)^{3/2}\,(-\varepsilon_r)\,\big(\varepsilon_k - \varepsilon_l - \varepsilon_m\big)}. \tag{4.44}$$

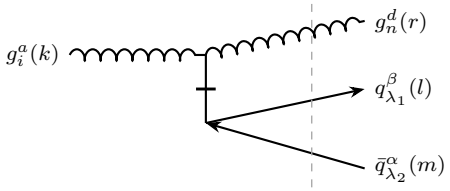
(f) Instantaneous quark exchange.



The two orderings of the gluon attachments give instantaneous fermion lines with plus momenta $l^+ + r^+$ and $-(m^+ + r^+)$; using (D.10),

$$\begin{aligned}\big|(\Psi_i^a(k))^6_{gq\bar{q}}\big\rangle = \frac{g^2}{16\pi^3}\int d^3l\, d^3m\, d^3r\ \ |q\bar{q}\,g\rangle \\ \times\ \frac{\delta^{(3)}(k-l-m-r)}{\sqrt{k^+r^+}\,D_f}\,\chi^\dagger_{\lambda_1}\left[\frac{(t^dt^a)_{\beta\alpha}\sigma^n\sigma^i}{l^+ + r^+} - \frac{(t^at^d)_{\beta\alpha}\sigma^i\sigma^n}{m^+ + r^+}\right]\chi_{\lambda_2}.\end{aligned} \tag{4.45}$$

(g) Instantaneous gluon exchange between the gluon and quark currents.



This is the $J_g \cdot J_q$ term of (2.25); the exchanged gluon carries $P^+ = l^+ + m^+ = k^+ - r^+$ and, with (D.14),

$$\left|(\Psi_i^a(k))^7_{gq\bar{q}}\right\rangle = \frac{ig^2}{2(2\pi)^3}\int d^3l\, d^3m\, d^3r\ \left|q\bar{q}\, g\right\rangle\ \frac{f^{bad}\, t^b_{\beta\alpha}\, \delta_{\lambda_1\lambda_2}\, \delta_{in}\, (k^+ + r^+)\, \delta^{(3)}(k-l-m-r)}{\sqrt{k^+ r^+}\, (l^+ + m^+)^2\, D_f}. \tag{4.46}$$

The remaining topology, in which the valence emits a gluon which splits into the pair while the incoming gluon is a spectator, is disconnected: it is the $\mathfrak{A}_1$ term of (4.22) and is not part of $\mathfrak{B}_3$.

4.2.4 Summary of results

The quark components of the dressed one gluon state through $\mathcal{O}(g^2)$ are

$$|\Psi_i^a(k)\rangle \supset |(\Psi_i^a(k))_{q\bar{q}}\rangle + \sum_{n=1,2,3,6,7} \left|(\Psi_i^a(k))^n_{gq\bar{q}}\right\rangle + \left|(\Psi_i^a(k))^{4+5}_{gq\bar{q}\,\rho}\right\rangle + |g_i^a(k)\rangle \otimes |\Psi_{q\bar{q}\,\rho}\rangle\,, \tag{4.47}$$

where the last term is the vacuum pair component (4.31) with the incoming gluon as a spectator. The diagrams (a)–(c) are the sequential processes $g \to q\bar{q} \to q\bar{q}\, g$ and $g \to gg \to q\bar{q}\, g$, (d)+(e) is the product of the Weizsäcker–Williams emission and the splitting $g \to q\bar{q}$, and (f), (g) are the two instantaneous interactions. The overall denominator D_f cannot vanish except on the collinear boundary of phase space (a set of measure zero), so $\mathfrak{B}_3$ has no on-shell pole. As discussed above, the normalization is unchanged at this order.

4.3 Calculation of the two-gluon incoming states

We now examine the quark contributions at order g^2 associated with an incoming state containing two gluons, $\left|g_i^a(k) g_j^b(p)\right\rangle$. The allowed final configurations are a quark–antiquark pair (virtual corrections to the pair sector and $gg \to q\bar{q}$ scattering), a pair with two gluons, and two pairs. The disconnected configurations, in which one of the incoming gluons is a spectator, reproduce the coefficients $\mathfrak{A}_3$, $\mathfrak{A}_1$ and $\mathfrak{B}_3$ already computed and are not repeated.

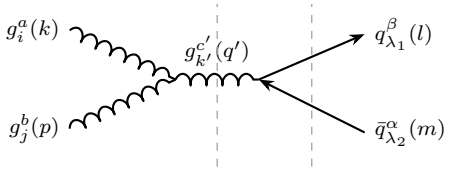
4.3.1 Quark–antiquark outgoing

The overall energy denominator is

$$D_f \equiv \varepsilon_k + \varepsilon_p - \varepsilon_l - \varepsilon_m. \tag{4.48}$$

Six topologies contribute.

(a) s-channel: the two gluons merge and the resulting gluon splits.



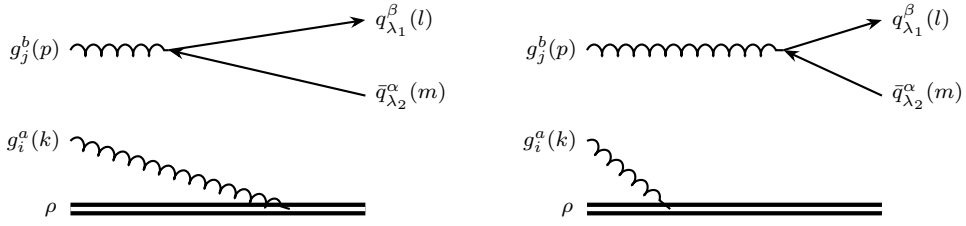
$$\Big|(\Psi^{ab}_{ij}(k,p))^1_{q\bar q}\Big\rangle = \int d^3q'\, d^3l\, d^3m\ |q\bar q\rangle\ \frac{\langle q\bar q|\, H_{gq\bar q}\,\Big|g^{c'}_{k'}(q')\Big\rangle}{D_f}\ \frac{\Big\langle g^{c'}_{k'}(q')\Big|\, H_{ggg}\,\Big|g^a_i(k)g^b_j(p)\Big\rangle}{\varepsilon_k+\varepsilon_p-\varepsilon_{q'}}. \tag{4.49}$$

The $2\to1$ gluon vertex is the Hermitian conjugate of (D.3); with $q'=k+p$,

$$\Big|(\Psi^{ab}_{ij}(k,p))^1_{q\bar q}\Big\rangle = -\frac{ig^2}{64\pi^3}\int d^3l\, d^3m \\ \times|q\bar q\rangle\ f^{abc'}t^{c'}_{\beta\alpha}\,\frac{\delta^{(3)}(k+p-l-m)}{q'^+\sqrt{k^+p^+}}\,\frac{\mathcal{V}^{k'ij}(q';k,p)\ \Phi^{k'}_{\lambda_1\lambda_2}(q';l,m)}{\big(\varepsilon_k+\varepsilon_p-\varepsilon_{q'}\big)\,D_f}. \tag{4.50}$$

This contribution is symmetric under $(a,i,k)\leftrightarrow(b,j,p)$.

(b) One gluon is absorbed by the valence charge, the other splits.



In (b1) the intermediate state is $|q\bar q\, g^a_i(k)\rangle$, with $E_i-E_m=\varepsilon_p-\varepsilon_l-\varepsilon_m$; in (b2) it is $\Big|g^b_j(p)\Big\rangle$, with $E_i-E_m=\varepsilon_k$:

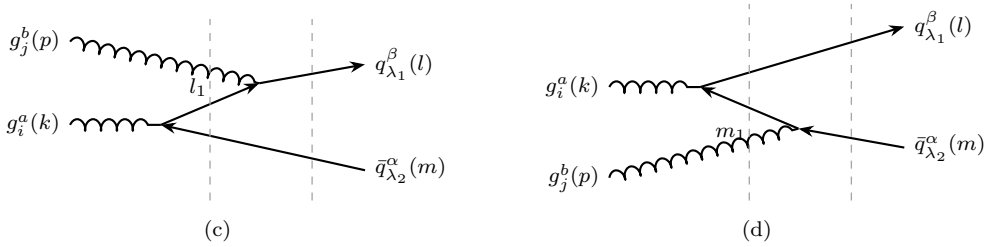
$$\Big|(\Psi^{ab}_{ij}(k,p))^2_{q\bar q\,\rho}\Big\rangle = \int d^3l\, d^3m\ |q\bar q\rangle\ \frac{\langle q\bar q|\, H_g\,|q\bar q\, g^a_i(k)\rangle}{D_f}\ \frac{\langle q\bar q\, g^a_i(k)|\, H_{gq\bar q}\,\Big|g^a_i(k)g^b_j(p)\Big\rangle}{\varepsilon_p-\varepsilon_l-\varepsilon_m}, \tag{4.51}$$

$$\Big|(\Psi^{ab}_{ij}(k,p))^3_{q\bar q\,\rho}\Big\rangle = \int d^3l\, d^3m\ |q\bar q\rangle\ \frac{\langle q\bar q|\, H_{gq\bar q}\,\Big|g^b_j(p)\Big\rangle}{D_f}\ \frac{\Big\langle g^b_j(p)\Big|\, H_g\,\Big|g^a_i(k)g^b_j(p)\Big\rangle}{\varepsilon_k}. \tag{4.52}$$

Since $D_f=(\varepsilon_p-\varepsilon_l-\varepsilon_m)+\varepsilon_k$, the two orderings combine by (3.4),

$$\Big|(\Psi^{ab}_{ij}(k,p))^{2+3}_{q\bar q\,\rho}\Big\rangle = \frac{g^2}{32\pi^3}\int d^3l\, d^3m\ |q\bar q\rangle\ \frac{\rho^a(k)\,t^b_{\beta\alpha}\,k^i\,\Phi^j_{\lambda_1\lambda_2}(p;l,m)\,\delta^{(3)}(p-l-m)}{(k^+)^{3/2}\sqrt{p^+}\ \varepsilon_k\,\big(\varepsilon_p-\varepsilon_l-\varepsilon_m\big)} \\ +\big((a,i,k)\leftrightarrow(b,j,p)\big). \tag{4.53}$$

(c) t-channel quark exchange: one gluon splits and the quark absorbs the other.



The gluon k splits into $q^\gamma_\lambda(l_1)\bar q^\alpha_{\lambda_2}(m)$ and the quark absorbs the gluon p, so that $l_1 = l - p$ and the process requires $l^+ > p^+$. The absorption vertex is the Hermitian conjugate of (D.5), $\big[\Phi^j_{\lambda'\lambda}(p;l_1,l)\big]^* = \Phi^j_{\lambda\lambda'}(p;l,l_1)$:

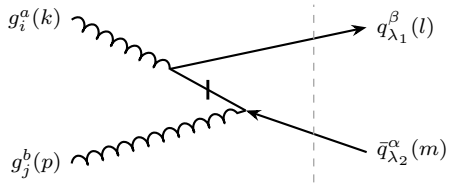
$$\begin{aligned}\Big|(\Psi^{ab}_{ij}(k,p))^4_{q\bar q}\Big\rangle &= \int d^3l_1\, d^3l\, d^3m\ |q\bar q\rangle\ \frac{\Big\langle q^\beta_{\lambda_1}(l)\Big|\, H_{gq\bar q}\,\Big|q^\gamma_\lambda(l_1)g^b_j(p)\Big\rangle}{D_f}\\ &\qquad\times\frac{\big\langle q^\gamma_\lambda(l_1)\bar q^\alpha_{\lambda_2}(m)\big|\, H_{gq\bar q}\,|g^a_i(k)\rangle}{\varepsilon_k-\varepsilon_{l_1}-\varepsilon_m}\\ &= \frac{g^2}{64\pi^3}\int d^3l\, d^3m\ |q\bar q\rangle\ (t^bt^a)_{\beta\alpha}\,\frac{\delta^{(3)}(k+p-l-m)\,\theta(l^+-p^+)}{\sqrt{k^+p^+}}\\ &\qquad\times\frac{\chi^\dagger_{\lambda_1}\Gamma^j(p;l,l_1)\,\Gamma^i(k;l_1,m)\chi_{\lambda_2}}{\big(\varepsilon_k-\varepsilon_{l_1}-\varepsilon_m\big)\,D_f}\\ &\quad+\big((a,i,k)\leftrightarrow(b,j,p)\big).\end{aligned}\tag{4.54}$$

(d) u-channel quark exchange: one gluon splits and the antiquark absorbs the other. Now $m_1 = m - p$, the process requires $m^+ > p^+$, and the antiquark vertex (D.6) supplies a minus sign:

$$\begin{aligned}\Big|(\Psi^{ab}_{ij}(k,p))^5_{q\bar q}\Big\rangle &= -\frac{g^2}{64\pi^3}\int d^3l\, d^3m\ |q\bar q\rangle\\ &\quad\times(t^at^b)_{\beta\alpha}\,\frac{\delta^{(3)}(k+p-l-m)\,\theta(m^+-p^+)}{\sqrt{k^+p^+}}\,\frac{\chi^\dagger_{\lambda_1}\Gamma^i(k;l,m_1)\,\Gamma^j(p;m_1,m)\chi_{\lambda_2}}{\big(\varepsilon_k-\varepsilon_l-\varepsilon_{m_1}\big)\,D_f}\\ &\qquad+\big((a,i,k)\leftrightarrow(b,j,p)\big).\end{aligned}\tag{4.55}$$

The diagrams (c) and (d) are the light-cone time ordered t- and u-channel quark exchange graphs of $gg\to q\bar q$. In the language of the evolution operator they are the products $\mathfrak{F}_2\,\mathfrak{A}_3$ and $\mathfrak{G}_2\,\mathfrak{A}_3$ in (3.3), and they are the origin of the $\mathfrak{F}$ and $\mathfrak{G}$ terms of the unitarity relation (4.10).

(e) Instantaneous quark exchange.

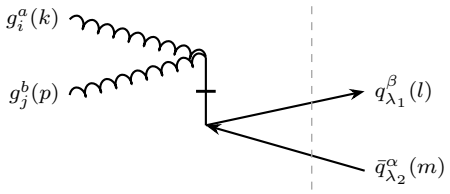


With (D.11),

$$\begin{aligned}\Big|(\Psi^{ab}_{ij}(k,p))^6_{q\bar q}\Big\rangle &= \frac{g^2}{16\pi^3}\int d^3l\, d^3m\ |q\bar q\rangle\\ &\qquad\times\frac{\delta^{(3)}(k+p-l-m)}{\sqrt{k^+p^+}\,D_f}\,\chi^\dagger_{\lambda_1}\left[\frac{(t^at^b)_{\beta\alpha}\sigma^i\sigma^j}{l^+-k^+}+\frac{(t^bt^a)_{\beta\alpha}\sigma^j\sigma^i}{l^+-p^+}\right]\chi_{\lambda_2}.\end{aligned}\tag{4.56}$$

The second term is the instantaneous part of the t-channel exchange (c) (the fermion line carries $l_1^+ = l^+ - p^+$) and the first that of the u-channel exchange (d) (it carries $l^+ - k^+ = -m_1^+$); unlike (c) and (d), the instantaneous term has support for either sign of these momenta.

(f) Instantaneous gluon exchange between the gluon and quark currents.

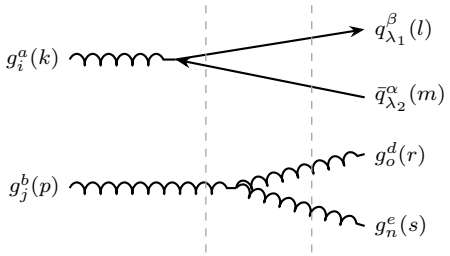


The two incoming gluons are annihilated by J_g and the pair is created by J_q; the exchanged gluon carries $P^+ = k^+ + p^+$ and, with (D.15),

$$\Big|(\Psi^{ab}_{ij}(k,p))^{7}_{q\bar{q}}\Big\rangle = -\frac{ig^2}{2(2\pi)^3}\int d^3l\, d^3m\ |q\bar{q}\rangle\ \frac{f^{cab}\, t^c_{\beta\alpha}\, \delta_{\lambda_1\lambda_2}\, \delta_{ij}\, (p^+ - k^+)\, \delta^{(3)}(k+p-l-m)}{\sqrt{k^+p^+}\,(k^+ + p^+)^2\, D_f}. \tag{4.57}$$

4.3.2 Two gluons, quark and antiquark outgoing

One incoming gluon splits into the pair and the other into the two outgoing gluons $g^d_o(r)g^e_n(s)$; both time orderings contribute.



With $D_1 = \varepsilon_k - \varepsilon_l - \varepsilon_m$, $D_2 = \varepsilon_p - \varepsilon_r - \varepsilon_s$ and $D_f = D_1 + D_2$, (3.4) gives

$$\begin{aligned}\Big|(\Psi^{ab}_{ij}(k,p))_{gg q\bar{q}}\Big\rangle = \frac{ig^2}{64\pi^3}\int d^3l\, d^3m\, d^3r\, d^3s\ \Big|q\bar{q}\, g^d_o(r) g^e_n(s)\Big\rangle \\ \times\, t^a_{\beta\alpha} f^{bde}\,\frac{\delta^{(3)}(k-l-m)\delta^{(3)}(p-r-s)}{\sqrt{k^+p^+r^+s^+}}\,\frac{\Phi^i_{\lambda_1\lambda_2}(k;l,m)\ \mathcal{V}^{jon}(p;r,s)}{D_1\, D_2} \\ +\,\big((a,i,k)\leftrightarrow(b,j,p)\big),\end{aligned} \tag{4.58}$$

where the factor $\frac{1}{2}$ of the two identical outgoing gluons has cancelled against their two assignments, as in Paper I.

4.3.3 Two quark–antiquark pairs outgoing

Each gluon splits independently and both orderings again combine by (3.4):

$$\begin{aligned}\Big|(\Psi^{bc}_{jk}(p,q))_{q\bar{q}q\bar{q}}\Big\rangle = \frac{g^2}{64\pi^3}\int d^3l\, d^3m\, d^3l'\, d^3m'\ |q\bar{q}\, q\bar{q}\rangle\ t^b_{\beta\alpha} t^c_{\beta'\alpha'}\,\frac{\delta^{(3)}(p-l-m)\delta^{(3)}(q-l'-m')}{\sqrt{p^+q^+}} \\ \times\,\frac{\Phi^j_{\lambda_1\lambda_2}(p;l,m)\ \Phi^k_{\lambda'_1\lambda'_2}(q;l',m')}{\big(\varepsilon_p - \varepsilon_l - \varepsilon_m\big)\big(\varepsilon_q - \varepsilon_{l'} - \varepsilon_{m'}\big)}.\end{aligned} \tag{4.59}$$

4.3.4 Summary of results

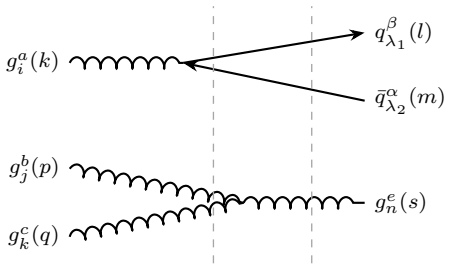
The connected quark components of the two-gluon state at order g^2 are

$$\left|\Psi^{ab}_{ij}(k,p)\right\rangle \supset \sum_{n=1,4,5,6,7} \left|(\Psi^{ab}_{ij}(k,p))^{n}_{q\bar{q}}\right\rangle + \left|(\Psi^{ab}_{ij}(k,p))^{2+3}_{q\bar{q}\,\rho}\right\rangle + \left|(\Psi^{ab}_{ij}(k,p))_{gg q\bar{q}}\right\rangle + \left|(\Psi^{ab}_{ij}(k,p))_{q\bar{q}q\bar{q}}\right\rangle . \quad (4.60)$$

The ρ dependent component and the components with four outgoing partons factorize into products of first order amplitudes. The ρ independent part of the $q\bar{q}$ component is the $gg \to q\bar{q}$ scattering amplitude; its overall denominator D_f vanishes on the energy shell and is understood with the prescription of section 3.2. The new connected components do not affect the normalization at this order.

4.4 Calculation of the three-gluon incoming states

As in Paper I, we restrict the calculation to the components needed for Ω at $\mathcal{O}(g^2)$. For an incoming state $\left|g^a_i(k) g^b_j(p) g^c_k(q)\right\rangle$ the only new connected quark component is $|q\bar{q}\, g^e_n(s)\rangle$: one gluon splits into the pair while the other two merge.



With $D_A = \varepsilon_k - \varepsilon_l - \varepsilon_m$ and $D_B = \varepsilon_p + \varepsilon_q - \varepsilon_s$, $D_A + D_B = D_f$, and (3.4) gives

$$\left|(\Psi^{abc}_{ijk}(k,p,q))_{gq\bar{q}}\right\rangle = -\frac{ig^2}{64\pi^3}\int d^3l\, d^3m\, d^3s\ |q\bar{q}\, g^e_n(s)\rangle\ t^a_{\beta\alpha} f^{ebc}\, \frac{\delta^{(3)}(k-l-m)\delta^{(3)}(s-p-q)}{\sqrt{k^+ s^+ p^+ q^+}} \times \frac{\Phi^i_{\lambda_1\lambda_2}(k;l,m)\ \mathcal{V}^{njk}(s;p,q)}{D_A\, D_B}, \quad (4.61)$$

summed over the three inequivalent choices of the gluon that splits. Since $f^{ebc}\mathcal{V}^{njk}(s;p,q)$ is symmetric under $(b,j,p) \leftrightarrow (c,k,q)$, a sum over all six permutations would double count.

4.4.1 Summary of results

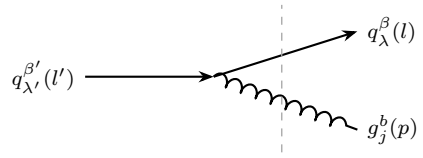
The only new connected quark component of the three-gluon state at order g^2 is (4.61); it factorizes into the product of the pair splitting amplitude and the three-gluon merging amplitude of Paper I.

4.5 Calculation of the single-quark incoming state

The pure Yang–Mills construction has no analogue of this sector. A soft quark $\left|q^{\beta'}_{\lambda'}(l')\right\rangle$ emits and absorbs soft gluons and rescatters on the valence charge; these processes fix the coefficients $\mathfrak{F}$, $\mathfrak{P}$, $\mathfrak{Q}_1$ and $\mathfrak{K}_1$. The results for an incoming antiquark are given at the end of this subsection. The outgoing quark is $q^\beta_\lambda(l)$.

4.5.1 Quark and one gluon outgoing

To order g the quark emits a soft gluon.



$$\left|(\Psi^{\beta'}_{\lambda'}(l'))_{qg}\right\rangle = \int d^3l\, d^3p\, \left|q^{\beta}_{\lambda}(l)g^b_j(p)\right\rangle \frac{\left\langle q^{\beta}_{\lambda}(l)g^b_j(p)\right| H_{gq\bar{q}} \left|q^{\beta'}_{\lambda'}(l')\right\rangle}{\varepsilon_{l'} - \varepsilon_l - \varepsilon_p}$$
$$= \frac{g}{8\pi^{3/2}} \int d^3l\, d^3p\, |qg\rangle \frac{t^b_{\beta\beta'}\, \delta^{(3)}(l'-l-p)\, \Phi^j_{\lambda\lambda'}(p;l,l')}{\sqrt{p^+}\left(\varepsilon_{l'} - \varepsilon_l - \varepsilon_p\right)}, \quad (4.62)$$

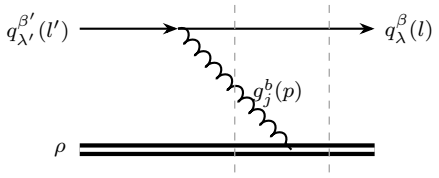
with the matrix element (D.5). The Weizsäcker–Williams gluon emitted by ρ with the quark as a spectator is the disconnected A term of (4.25).

4.5.2 One quark outgoing

At order g^2 the soft quark exchanges a gluon with the valence charge. Three diagrams contribute to the transition $l' \to l$ with momentum transfer to the valence; we write

$$\Delta \equiv E_i - E_f = \varepsilon_{l'} - \varepsilon_l. \quad (4.63)$$

(a) The quark emits a gluon which is absorbed by the valence.



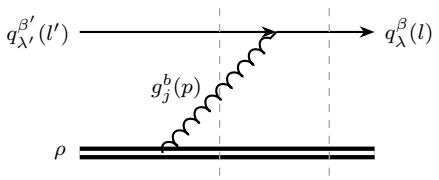
Here $p = l' - l$ and the process requires $l'^+ > l^+$:

$$\left|(\Psi^{\beta'}_{\lambda'}(l'))^1_{q\,\rho}\right\rangle = \int d^3l\, d^3p\, \left|q^{\beta}_{\lambda}(l)\right\rangle \frac{\left\langle q^{\beta}_{\lambda}(l)\right| H_g \left|q^{\beta}_{\lambda}(l)g^b_j(p)\right\rangle}{\Delta} \frac{\left\langle q^{\beta}_{\lambda}(l)g^b_j(p)\right| H_{gq\bar{q}} \left|q^{\beta'}_{\lambda'}(l')\right\rangle}{\Delta - \varepsilon_p}. \quad (4.64)$$

Using (D.2) and (D.5),

$$\left|(\Psi^{\beta'}_{\lambda'}(l'))^1_{q\,\rho}\right\rangle = \frac{g^2}{32\pi^3} \int d^3l\, \left|q^{\beta}_{\lambda}(l)\right\rangle \frac{t^b_{\beta\beta'}\, \rho^b(p)\, p^j \Phi^j_{\lambda\lambda'}(p;l,l')}{(p^+)^2\,(\Delta - \varepsilon_p)\,\Delta}\, \theta(l'^+ - l^+). \quad (4.65)$$

(b) The valence emits a gluon which is absorbed by the quark.

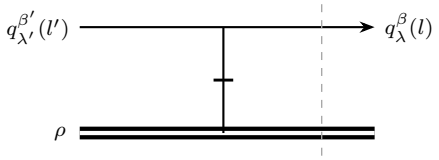


Now $p = l - l'$, the process requires $l^+ > l'^+$ and the intermediate state $\left| q^{\beta'}_{\lambda'}(l') g^b_j(p) \right\rangle$ has $E_i - E_m = -\varepsilon_p$:

$$\left| (\Psi^{\beta'}_{\lambda'}(l'))^2_{q\,\rho} \right\rangle = -\frac{g^2}{32\pi^3} \int d^3l \, \left| q^{\beta}_{\lambda}(l) \right\rangle \, \frac{t^b_{\beta\beta'}\, \rho^b(-p)\, p^j \Phi^j_{\lambda\lambda'}(p; l, l')}{(p^+)^2\, \varepsilon_p\, \Delta}\, \theta(l^+ - l'^+), \tag{4.66}$$

where the absorption vertex is $\left[\Phi^j_{\lambda'\lambda}(p; l', l)\right]^* = \Phi^j_{\lambda\lambda'}(p; l, l')$.

(c) Instantaneous gluon exchange with the valence.



This is the eikonal colour rotation of the soft quark by the $\rho \cdot J_q$ term of (2.25); with (D.8),

$$\left| (\Psi^{\beta'}_{\lambda'}(l'))^3_{q\,\rho} \right\rangle = \int d^3l \, \left| q^{\beta}_{\lambda}(l) \right\rangle \, \frac{\left\langle q^{\beta}_{\lambda}(l) \right| H_{q-\mathrm{inst}} \left| q^{\beta'}_{\lambda'}(l') \right\rangle}{\Delta}$$
$$= \frac{g^2}{8\pi^3} \int d^3l \, \left| q^{\beta}_{\lambda}(l) \right\rangle \, \frac{t^b_{\beta\beta'}\, \rho^b(l' - l)\, \delta_{\lambda\lambda'}}{(l'^+ - l^+)^2\, \Delta}. \tag{4.67}$$

(d) Self-energy. The quark may also emit and reabsorb a soft gluon. This contribution is proportional to $\delta^{(3)}(l - l')$, has $E_i = E_f$, and enters only the normalization, discussed below.

Simplification. For $p = l' - l$ the vertex satisfies

$$p^j \Gamma^j(p; l, l') = 4\varepsilon_p - 2\Delta - \frac{p^+}{l^+ l'^+} (\boldsymbol{\sigma}\cdot\boldsymbol{l})(\boldsymbol{\sigma}\cdot\boldsymbol{l}'), \tag{4.68}$$

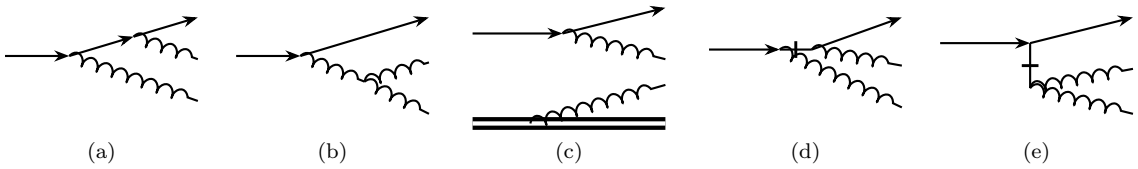
and for $p = l - l'$, $p^j \Gamma^j(p; l, l') = 4\varepsilon_p + 2\Delta - \frac{p^+}{l^+ l'^+} (\boldsymbol{\sigma}\cdot\boldsymbol{l})(\boldsymbol{\sigma}\cdot\boldsymbol{l}')$. Writing $4\varepsilon_p - 2\Delta = -4(\Delta - \varepsilon_p) + 2\Delta$ in (4.65), and $4\varepsilon_p + 2\Delta$ unchanged in (4.66), the terms proportional to $\delta_{\lambda\lambda'}/\Delta$ cancel the instantaneous contribution (4.67) in both kinematic regions, exactly as in the vacuum sector. The sum is

$$\left| (\Psi^{\beta'}_{\lambda'}(l'))_{q\,\rho} \right\rangle = \frac{g^2}{32\pi^3} \int d^3l \, \left| q^{\beta}_{\lambda}(l) \right\rangle \, \frac{t^b_{\beta\beta'} \rho^b(l' - l)}{(p^+)^2}\, \chi^\dagger_\lambda \Bigg\{$$
$$\theta(l'^+ - l^+) \left[\frac{2}{\Delta - \varepsilon_p} - \frac{p^+\, (\boldsymbol{\sigma}\cdot\boldsymbol{l})(\boldsymbol{\sigma}\cdot\boldsymbol{l}')}{l^+ l'^+\, (\Delta - \varepsilon_p)\, \Delta} \right]$$
$$- \theta(l^+ - l'^+) \left[\frac{2}{\varepsilon_p} - \frac{p^+\, (\boldsymbol{\sigma}\cdot\boldsymbol{l})(\boldsymbol{\sigma}\cdot\boldsymbol{l}')}{l^+ l'^+\, \varepsilon_p\, \Delta} \right] \Bigg\} \chi_{\lambda'}, \tag{4.69}$$

where $p = \pm(l' - l)$ is the (positive plus momentum) gluon exchanged in the two regions. The overall denominator $\Delta = \varepsilon_{l'} - \varepsilon_l$ survives in the helicity dependent term: the coefficient $\mathfrak{P}$ has an on-shell pole, exactly as the coefficient B_3 of Paper I, and its residue is the amplitude for the scattering of a soft quark on the valence charge.

4.5.3 Quark and two gluons outgoing

At order g^2 the transition $q(l') \to q(l)\, g^c_k(q) g^d_n(r)$ receives five contributions:



(a) (b) (c) (d) (e)

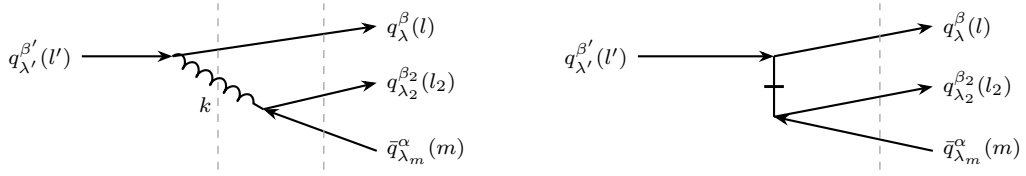
(a) two sequential emissions by the quark, in either order; (b) emission of a gluon which splits into two through H_{ggg}; (c) emission of one gluon by the quark and of the other by the valence charge; (d) instantaneous quark exchange; (e) instantaneous gluon exchange between the quark and gluon currents. With $D_f = \varepsilon_{l'} - \varepsilon_l - \varepsilon_q - \varepsilon_r$,

$$\begin{aligned}
\left|(\Psi^{\beta'}_{\lambda'}(l'))_{qgg}\right\rangle = \frac{1}{2}\int d^3l\, d^3q\, d^3r\ \left|q^\beta_\lambda(l) g^c_k(q) g^d_n(r)\right\rangle \\
\times \Bigg\{ \int d^3l_1\, \frac{\left\langle q^\beta_\lambda(l) g^c_k(q)\right| H_{gq\bar q} \left|q(l_1)\right\rangle}{D_f}\, \frac{\left\langle q(l_1) g^d_n(r)\right| H_{gq\bar q} \left|q^{\beta'}_{\lambda'}(l')\right\rangle}{\varepsilon_{l'} - \varepsilon_{l_1} - \varepsilon_r} + (q \leftrightarrow r) \\
+ \int d^3s\, \frac{\left\langle g^c_k(q) g^d_n(r)\right| H_{ggg} \left|g(s)\right\rangle}{D_f}\, \frac{\left\langle q^\beta_\lambda(l) g(s)\right| H_{gq\bar q} \left|q^{\beta'}_{\lambda'}(l')\right\rangle}{\varepsilon_{l'} - \varepsilon_l - \varepsilon_s} \\
+ \frac{\left\langle q\, gg\right| \left(H_{ggq\bar q} + H_{q-\mathrm{inst}}\right) \left|q\right\rangle}{D_f} \Bigg\} \\
+ \left|(\Psi^{\beta'}_{\lambda'}(l'))_{ggg\,\rho}\right\rangle ,
\end{aligned} \tag{4.70}$$

where the factor $\frac{1}{2}$ compensates the double counting of the identical gluons, the instantaneous matrix elements are (D.12) and (D.16), and the ρ dependent component (c) factorizes by (3.4) into the product of (4.62) and the Weizsäcker–Williams amplitude. The diagrams (a)–(e) are related by crossing to the diagrams (a)–(g) of the $q\bar q g$ component of the one gluon state.

4.5.4 Quark and a quark–antiquark pair outgoing

At order g^2 the quark can also emit a gluon which splits into a pair, or create a pair through the instantaneous $J_q \cdot J_q$ interaction:



$$\begin{aligned}
\left|(\Psi^{\beta'}_{\lambda'}(l'))_{qq\bar q}\right\rangle = \frac{1}{2}\int d^3l\, d^3l_2\, d^3m\ \left|q^\beta_\lambda(l) q^{\beta_2}_{\lambda_2}(l_2) \bar q^\alpha_{\lambda_m}(m)\right\rangle \\
\times \Bigg\{ \int d^3k\, \frac{\left\langle q(l_2)\bar q(m)\right| H_{gq\bar q} \left|g(k)\right\rangle \left\langle q(l) g(k)\right| H_{gq\bar q} \left|q(l')\right\rangle}{D_f \left(\varepsilon_{l'} - \varepsilon_l - \varepsilon_k\right)}
\end{aligned}$$

$$+\frac{\langle q(l)q(l_2)\bar{q}(m)|\,H_{q-\mathrm{inst}}\,|q(l')\rangle}{D_f}\Bigg\}$$
$$-\big((\beta,\lambda,l)\leftrightarrow(\beta_2,\lambda_2,l_2)\big), \tag{4.71}$$

with $D_f=\varepsilon_{l'}-\varepsilon_l-\varepsilon_{l_2}-\varepsilon_m$ and the matrix element (D.19). The antisymmetrization accounts for the two identical quarks in the final state. The ρ dependent pair creation off the valence charge with a spectator quark is the disconnected $\mathfrak{A}_1$ term.

4.5.5 Single-antiquark incoming state

The antiquark sector follows from the quark sector by the replacements dictated by the antiquark vertex (D.6): the colour matrix is replaced by $t^b_{\beta\beta'}\to -t^b_{\alpha'\alpha}$ (incoming colour index first), and the spinor bilinear of the quark line by that of the antiquark line, in which the incoming helicity stands on the left, $\Phi^j_{\lambda\lambda'}(p;l,l')\to\Phi^j_{\lambda'\lambda}(p;m',m)$. For an incoming antiquark $\Big|\bar{q}^{\alpha'}_{\lambda'}(m')\Big\rangle$ and outgoing $\bar{q}^{\alpha}_{\lambda}(m)$ one finds

$$\Big|(\Psi^{\alpha'}_{\lambda'}(m'))_{\bar{q}g}\Big\rangle=-\frac{g}{8\pi^{3/2}}\int d^3m\,d^3p\;|\bar{q}g\rangle\;\frac{t^b_{\alpha'\alpha}\,\delta^{(3)}(m'-m-p)\,\Phi^j_{\lambda'\lambda}(p;m',m)}{\sqrt{p^+}\,\big(\varepsilon_{m'}-\varepsilon_m-\varepsilon_p\big)}, \tag{4.72}$$

$$\begin{aligned}\Big|(\Psi^{\alpha'}_{\lambda'}(m'))_{\bar{q}\,\rho}\Big\rangle=-\frac{g^2}{32\pi^3}\int d^3m\;|\bar{q}^{\alpha}_{\lambda}(m)\rangle\;\frac{t^b_{\alpha'\alpha}\rho^b(m'-m)}{(p^+)^2}\,\chi^\dagger_{\lambda'}\Bigg\{\\ \theta(m'^+-m^+)\left[\frac{2}{\bar{\Delta}-\varepsilon_p}-\frac{p^+(\boldsymbol{\sigma}\cdot\boldsymbol{m}')(\boldsymbol{\sigma}\cdot\boldsymbol{m})}{m^+m'^+(\bar{\Delta}-\varepsilon_p)\bar{\Delta}}\right]\\ -\theta(m^+-m'^+)\left[\frac{2}{\varepsilon_p}-\frac{p^+(\boldsymbol{\sigma}\cdot\boldsymbol{m}')(\boldsymbol{\sigma}\cdot\boldsymbol{m})}{m^+m'^+\varepsilon_p\bar{\Delta}}\right]\Bigg\}\chi_\lambda,\end{aligned} \tag{4.73}$$

with $\bar{\Delta}=\varepsilon_{m'}-\varepsilon_m$ and $p=\pm(m'-m)$. We have verified (4.73) explicitly: the cancellation of the instantaneous term proceeds as for the quark, with the instantaneous matrix element (D.9) carrying the antiquark charge.

4.5.6 Summary of results

The dressed quark state through $\mathcal{O}(g^2)$ is

$$\begin{aligned}\Big|\Psi^{\beta'}_{\lambda'}(l')\Big\rangle=N\Big|q^{\beta'}_{\lambda'}(l')\Big\rangle+\Big|(\Psi^{\beta'}_{\lambda'}(l'))_{qg}\Big\rangle+\Big|q^{\beta'}_{\lambda'}(l')\Big\rangle\otimes\big|\Psi^{\mathrm{LO}}_{g\,\rho}\big\rangle\\ +\Big|(\Psi^{\beta'}_{\lambda'}(l'))_{q\,\rho}\Big\rangle+\Big|(\Psi^{\beta'}_{\lambda'}(l'))_{qgg}\Big\rangle+\Big|(\Psi^{\beta'}_{\lambda'}(l'))_{qq\bar{q}}\Big\rangle+\ldots,\end{aligned} \tag{4.74}$$

where $\big|\Psi^{\mathrm{LO}}_{g\,\rho}\big\rangle$ is the leading order Weizsäcker–Williams component of the vacuum and the dots denote the remaining disconnected components (the order g^2 vacuum components with the quark as a spectator). As in the gluon case, the normalization is fixed by $\langle\Psi|\Psi\rangle=1$. The ρ dependent overlaps reproduce those of the vacuum, and the only new contribution is the norm of the qg component. With $z=p^+/l'^+$ and $\boldsymbol{\kappa}=\boldsymbol{p}-z\boldsymbol{l}'$ of (2.7), for fixed λ',

$$\sum_{\lambda,j}\big|\Phi^j_{\lambda\lambda'}(p;l,l')\big|^2=\frac{2\,\boldsymbol{\kappa}^2\big[1+(1-z)^2\big]}{(l'^+)^2z^2(1-z)^2}, \tag{4.75}$$

and

$$\big\langle(\Psi^{\beta''}_{\lambda''}(l''))_{qg}\big|(\Psi^{\beta'}_{\lambda'}(l'))_{qg}\big\rangle = \delta^{\beta'\beta''}\delta_{\lambda'\lambda''}\delta^{(3)}(l'-l'')\,\frac{g^2C_F}{8\pi^3}\int_{\Lambda/l'^+}^1 dz\;\frac{1+(1-z)^2}{z}\int\frac{d^2\kappa}{\boldsymbol{\kappa}^2}$$
$$= \delta\delta\delta\;\frac{g^2C_F}{8\pi^3}\Big[2\log\frac{l'^+}{\Lambda}-\frac{3}{2}\Big]\int\frac{d^2\kappa}{\boldsymbol{\kappa}^2}. \quad (4.76)$$

The longitudinal integral contains the soft gluon logarithm, and the transverse integral is scaleless and vanishes in dimensional regularization. The normalization of the quark state is therefore the same N as for all the other sectors, as it must be since Ω is a single operator.

4.6 Calculation of the quark–antiquark incoming state

For an incoming pair $\Big|q^{\beta'}_{\lambda'_1}(l')\bar q^{\alpha'}_{\lambda'_2}(m')\Big\rangle$ the possible final configurations are the vacuum, one gluon, two gluons, a pair with one gluon and a pair. The first three test the unitarity relations of the coefficients that annihilate pairs; the last fixes $\mathfrak{D}_3$.

4.6.1 Vacuum, one and two gluons outgoing

At order g the pair annihilates into a gluon:

$$\big|(\Psi(l',m'))_g\big\rangle = \int d^3k\;|g^a_i(k)\rangle\;\frac{\langle g^a_i(k)|\,H_{gq\bar q}\,|q\bar q\rangle}{\varepsilon_{l'}+\varepsilon_{m'}-\varepsilon_k}$$
$$= -\frac{g}{8\pi^{3/2}}\int d^3k\;|g^a_i(k)\rangle\;\frac{t^a_{\alpha'\beta'}\,\delta^{(3)}(k-l'-m')\,\big[\Phi^i_{\lambda'_1\lambda'_2}(k;l',m')\big]^*}{\sqrt{k^+}\,\big(\varepsilon_k-\varepsilon_{l'}-\varepsilon_{m'}\big)}. \quad (4.77)$$

Comparing with (4.32) and (4.26) one reads off $\mathfrak{A}_4=-\mathfrak{A}_3^\dagger$, which is the first relation of (4.8); the minus sign is the sign of the energy denominator. At order g^2 the pair annihilates into the gluon which is absorbed by the valence, or annihilates instantaneously on ρ. By the reduction formula (3.3), the first is $(-A^\dagger)\mathfrak{A}_4\,\varepsilon_k/(\varepsilon_{l'}+\varepsilon_{m'})$ and the second is the Hermitian conjugate of (4.30) with the opposite sign of the energy denominator. Their sum reproduces (4.9) exactly: the instantaneous terms are anti-Hermitian and cancel between $\mathfrak{A}_1$ and $\mathfrak{A}_2^\dagger$, while $-\mathfrak{A}_1^\dagger+A^\dagger\mathfrak{A}_3^\dagger = A^\dagger\mathfrak{A}_3^\dagger\,\varepsilon_k/(\varepsilon_{l'}+\varepsilon_{m'})+\dots$. Similarly, the transition $q\bar q\to gg$ computed in LCPT (the time reversed diagrams of section 4.3) reproduces the relation (4.10), including the terms with $\mathfrak{F}$ and $\mathfrak{G}$ which arise from the t- and u-channel exchanges.

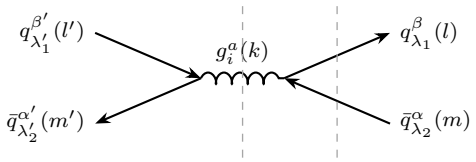
4.6.2 Quark–antiquark outgoing

The overall energy denominator is

$$D_f \equiv \varepsilon_{l'}+\varepsilon_{m'}-\varepsilon_l-\varepsilon_m. \quad (4.78)$$

Five connected diagrams contribute at order g^2.

(a) s-channel: the pair annihilates into a gluon which splits again.

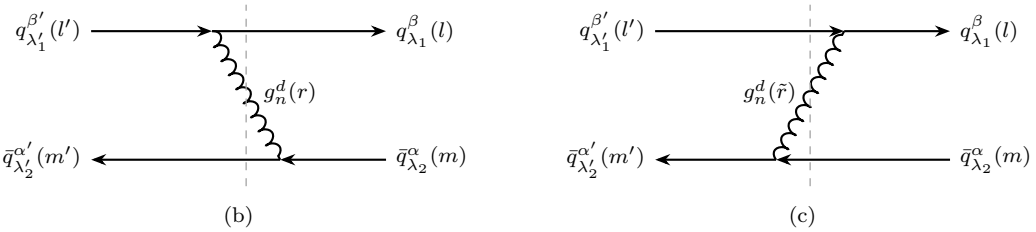


$$\big|(\Psi(l',m'))^1_{q\bar q}\big\rangle = \int d^3k\, d^3l\, d^3m\ |q\bar q\rangle\ \frac{\langle q\bar q|\, H_{gq\bar q}\, |g^a_i(k)\rangle}{D_f}\ \frac{\langle g^a_i(k)|\, H_{gq\bar q}\, \left|q^{\beta'}_{\lambda_1'}(l')\bar q^{\alpha'}_{\lambda_2'}(m')\right\rangle}{\varepsilon_{l'}+\varepsilon_{m'}-\varepsilon_k}. \tag{4.79}$$

With $(t^a_{\beta'\alpha'})^* = t^a_{\alpha'\beta'}$ and $k = l+m = l'+m'$,

$$\big|(\Psi(l',m'))^1_{q\bar q}\big\rangle = \frac{g^2}{64\pi^3}\int d^3l\, d^3m\ |q\bar q\rangle \times t^a_{\beta\alpha}t^a_{\alpha'\beta'}\,\delta^{(3)}(l+m-l'-m')\,\frac{\Phi^i_{\lambda_1\lambda_2}(k;l,m)\left[\Phi^i_{\lambda_1'\lambda_2'}(k;l',m')\right]^*}{k^+\left(\varepsilon_{l'}+\varepsilon_{m'}-\varepsilon_k\right)D_f}. \tag{4.80}$$

(b), (c) t-channel gluon exchange.



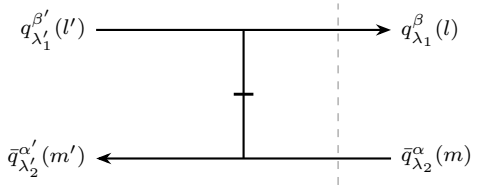
In (b) the quark emits the gluon $r = l'-l$ (so that $l'^+ > l^+$) which the antiquark absorbs; in (c) the antiquark emits $\tilde r = m'-m$ (so that $m'^+ > m^+$) which the quark absorbs. Using (D.5), (D.6) and their conjugates,

$$\big|(\Psi(l',m'))^2_{q\bar q}\big\rangle = -\frac{g^2}{64\pi^3}\int d^3l\, d^3m\ |q\bar q\rangle\ t^d_{\beta\beta'}t^d_{\alpha'\alpha}\,\delta^{(3)}(l+m-l'-m') \times \frac{\Phi^n_{\lambda_1\lambda_1'}(r;l,l')\,\Phi^n_{\lambda_2'\lambda_2}(r;m',m)}{r^+\left(\varepsilon_{l'}-\varepsilon_l-\varepsilon_r\right)D_f}\,\theta(r^+), \tag{4.81}$$

$$\big|(\Psi(l',m'))^3_{q\bar q}\big\rangle = -\frac{g^2}{64\pi^3}\int d^3l\, d^3m\ |q\bar q\rangle\ t^d_{\beta\beta'}t^d_{\alpha'\alpha}\,\delta^{(3)}(l+m-l'-m') \times \frac{\Phi^n_{\lambda_1\lambda_1'}(\tilde r;l,l')\,\Phi^n_{\lambda_2'\lambda_2}(\tilde r;m',m)}{\tilde r^+\left(\varepsilon_{m'}-\varepsilon_m-\varepsilon_{\tilde r}\right)D_f}\,\theta(\tilde r^+). \tag{4.82}$$

The two orderings populate the complementary regions $l'^+ \gtrless l^+$.

(d) Instantaneous gluon exchange between the quark and the antiquark.

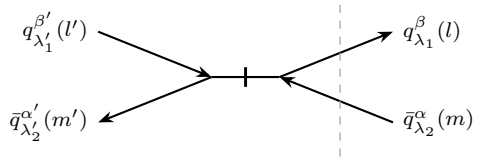


This is the $J_q\cdot J_q$ term of (2.25); the exchanged line carries $P^+ = l'^+ - l^+$ and the antiquark charge supplies a minus sign:

$$\big|(\Psi(l',m'))^4_{q\bar q}\big\rangle = -\frac{g^2}{(2\pi)^3}\int d^3l\, d^3m\ |q\bar q\rangle\ \frac{t^a_{\beta\beta'}t^a_{\alpha'\alpha}\,\delta_{\lambda_1\lambda_1'}\delta_{\lambda_2\lambda_2'}\,\delta^{(3)}(l+m-l'-m')}{(l'^+-l^+)^2\,D_f}. \tag{4.83}$$

The diagrams (b), (c) and (d) together build the full t-channel one-gluon exchange between the quark and the antiquark in light-cone gauge.

(e) Instantaneous s-channel gluon exchange.



The $J_q \cdot J_q$ interaction also annihilates the incoming pair and creates the outgoing one instantaneously; the exchanged line carries $P^+ = k^+ = l^+ + m^+$:

$$\big|(\Psi(l',m'))^5_{q\bar{q}}\big\rangle = \frac{g^2}{(2\pi)^3}\int d^3l\, d^3m\ |q\bar{q}\rangle\ \frac{t^a_{\beta\alpha}t^a_{\alpha'\beta'}\,\delta_{\lambda_1\lambda_2}\delta_{\lambda'_1\lambda'_2}\,\delta^{(3)}(l+m-l'-m')}{(k^+)^2\ D_f}. \qquad (4.84)$$

It is the instantaneous part of the s-channel exchange (a), in the same way as (d) is that of (b), (c).

Disconnected contributions. The exchange of a gluon between the quark (or antiquark) and the valence charge, with the other member of the pair a spectator, is the coefficient $\mathfrak{P}$ ($\bar{\mathfrak{P}}$) of (4.26) computed in section 4.5; it is not part of $\mathfrak{D}_3$.

4.6.3 Summary of results

The connected $q\bar{q} \to q\bar{q}$ amplitude at order g^2 is the sum of (4.80)–(4.84); its overall denominator D_f vanishes on the energy shell and is understood with the prescription of section 3.2. The remaining components of the dressed pair state are products of the single quark and single antiquark components and of the order g annihilation amplitude (4.77).

4.7 Calculation of the quark–gluon incoming state

Finally, for an incoming state $\left|q^{\beta'}_{\lambda'}(l')g^c_k(q)\right\rangle$ the connected order g^2 transition into $\left|q^{\beta}_{\lambda}(l)g^d_n(r)\right\rangle$ is soft quark–gluon (Compton) scattering. The light-cone time ordered contributions are: the s-channel (the quark absorbs the gluon and re-emits one); the u-channel in its two orderings (the quark first emits the gluon r and then absorbs q, or the gluon q first splits into a pair whose antiquark annihilates with the incoming quark into the gluon r); the t-channel in its two orderings (the gluon q splits into two gluons one of which is absorbed by the quark, or the quark emits a gluon which merges with q); the instantaneous quark exchange; and the instantaneous $J_q \cdot J_g$ gluon exchange. In addition there are the ρ dependent products in which the incoming gluon is absorbed by the valence while the quark emits the outgoing gluon, or the valence emits the outgoing gluon while the quark absorbs the incoming one; by (3.4) these factorize. With the reduction formula (3.3), each contribution is the product of two of the first order amplitudes already computed, weighted by the corresponding ratio of energy denominators, or an instantaneous matrix element divided by $D_f = \varepsilon_{l'} + \varepsilon_q - \varepsilon_l - \varepsilon_r$. The resulting coefficient $\mathfrak{Q}_2$ is given in section 5; like $\mathfrak{D}_3$, its ρ independent part has an on-shell pole.

5 Coefficients of the evolution operator up to $\mathcal{O}(g^2)$ through LCPT

To determine the coefficients of the evolution operator we compare its action on the elementary Fock states, Eqs. (4.21)–(4.27), with the light-cone wave functions computed in section 4, using the rule (4.20). Since these states probe different sectors of Fock space, they constrain different structures of Ω, and the coefficients are fixed one by one. Wherever possible we express the second order coefficients through the first order ones by the reduction formula (3.3), which in terms of coefficients reads: for a transition $i \to f$ through an intermediate state m generated by the first order structures with coefficients X ($i \to m$) and Y ($m \to f$),

$$K_{f\leftarrow i} \supset Y\,X\,\frac{E_m - E_f}{E_i - E_f}, \qquad \text{and} \qquad K_{f\leftarrow i} \supset \frac{(2\pi)^{3(n+M)/2}\,H^{(2)}_{fi}}{E_i - E_f} \tag{5.1}$$

for the instantaneous terms, the internal momenta being integrated with $\int_p$. This is the form in which the coefficients D_1, D_2 of Paper I were written. In the terms that involve the three-gluon coefficients C and $C_2 = -C^\dagger$, the combinatorial factors associated with identical gluons are those of Paper I, whose conventions for these coefficients we adopt throughout.

5.1 Order g

Matching (4.32), (4.62) and (4.72) with (4.22) and (4.25) ($K = (2\pi)^{9/2}\psi$) gives

$$\mathfrak{A}^{b\alpha\beta}_{3\,j\lambda_2\lambda_1}(p,m,l) = \frac{(2\pi)^3 g\,t^b_{\beta\alpha}}{2\sqrt{2}\sqrt{p^+}}\;\frac{\delta^{(3)}(p-l-m)\;\Phi^j_{\lambda_1\lambda_2}(p;l,m)}{\varepsilon_p - \varepsilon_l - \varepsilon_m}, \tag{5.2}$$

$$\mathfrak{F}^{\beta'\beta b}_{\lambda'\lambda j}(l',l,p) = \frac{(2\pi)^3 g\,t^b_{\beta\beta'}}{2\sqrt{2}\sqrt{p^+}}\;\frac{\delta^{(3)}(l'-l-p)\;\Phi^j_{\lambda\lambda'}(p;l,l')}{\varepsilon_{l'} - \varepsilon_l - \varepsilon_p}, \tag{5.3}$$

$$\mathfrak{G}^{\alpha'\alpha b}_{\lambda'\lambda j}(m',m,p) = -\frac{(2\pi)^3 g\,t^b_{\alpha'\alpha}}{2\sqrt{2}\sqrt{p^+}}\;\frac{\delta^{(3)}(m'-m-p)\;\Phi^j_{\lambda'\lambda}(p;m',m)}{\varepsilon_{m'} - \varepsilon_m - \varepsilon_p}. \tag{5.4}$$

Each of them is the vertex (2.22)–(2.24) divided by the energy released in the splitting, the quark analogue of the three-gluon coefficient $C = (2\pi)^{9/2}\,\langle gg|\,H_{ggg}\,|g\rangle\,/(E_i - E_f)$ of Paper I. Their conjugates $\mathfrak{A}_4$, $\mathfrak{F}_2$, $\mathfrak{G}_2$ follow from (4.8); for $\mathfrak{A}_4$ this was verified directly in (4.77).

5.2 Pair creation off the valence charge: $\mathfrak{A}_1$

Comparing (4.21) with (4.31) ($K = (2\pi)^3\psi$),

$$\boxed{\mathfrak{A}^{\alpha\beta}_{1\,\lambda_2\lambda_1}(m,l) = -g^2\,t^a_{\beta\alpha}\,\rho^a(-k)\left[\frac{\delta_{\lambda_1\lambda_2}}{k^+\boldsymbol{k}^2} + \frac{\chi^\dagger_{\lambda_1}\big(\boldsymbol{l}\cdot\boldsymbol{m} + i\,(\boldsymbol{l}\times\boldsymbol{m})\,\sigma^3\big)\chi_{\lambda_2}}{2\,l^+m^+\boldsymbol{k}^2\,(\varepsilon_l+\varepsilon_m)}\right], \qquad k = l+m,} \tag{5.5}$$

with $\boldsymbol{l}\times\boldsymbol{m} \equiv \epsilon^{ij}l^i m^j$. In the form (5.1), $\mathfrak{A}_1 = \int_k \mathfrak{A}_3(k,m,l)A(k)\,(\varepsilon_l+\varepsilon_m-\varepsilon_k)/(\varepsilon_l+\varepsilon_m) - (2\pi)^3\,\langle q\bar{q}|\,H_{q-\mathrm{inst}}|0\rangle/(\varepsilon_l+\varepsilon_m)$, with the Weizsäcker–Williams coefficient of Paper I

$$A^b_j(p) = -\frac{\sqrt{2}\,g\,p^j\rho^b(-p)}{\sqrt{p^+}\,\boldsymbol{p}^2}. \tag{5.6}$$

Check. With (5.5), $\mathfrak{A}_1+\mathfrak{A}_2^\dagger$ obtained from the direct LCPT computation of section 4.6 equals $\int_p \mathfrak{A}_3(p,m,l)A(p)$, which is the relation (4.9). Using $4(k^+)^2\varepsilon_k = 2k^+\boldsymbol{k}^2$, this amounts to the identity $\frac{1}{2k^+\boldsymbol{k}^2 E} + \frac{1}{4(k^+)^2E(E-\varepsilon_k)} = \frac{1}{2k^+\boldsymbol{k}^2(E-\varepsilon_k)}$ with $E=\varepsilon_l+\varepsilon_m$; the instantaneous parts cancel.

5.3 Pair creation off a soft gluon: $\mathfrak{B}_3$

Comparing (4.22) with the connected wave functions of section 4.2 ($K=(2\pi)^6\psi$),

$$\boxed{\mathfrak{B}_{3\,\lambda_2\lambda_1 in}^{\alpha\beta ad}(m,l,k,r) = A_n^d(r)\,\mathfrak{A}_{3\,i\lambda_2\lambda_1}^{a\alpha\beta}(k,m,l) + (2\pi)^6\Big[\psi^{(1)}+\psi^{(2)}+\psi^{(3)}+\psi^{(6)}+\psi^{(7)}\Big]_{gq\bar q}},\tag{5.7}$$

where $\psi^{(n)}_{gq\bar q}$ denote the coefficient functions of the $q\bar qg$ components (4.38), (4.39), (4.41), (4.45) and (4.46), i.e. the expressions multiplying $|q\bar q\,g\rangle$ under the integral; their prefactors become $(2\pi)^3g^2/8$ for (a), (b), (c) and $(2\pi)^3g^2/2$ for (f), (g). The first term is the factorized contribution (4.44). Equivalently, by (5.1),

$$\begin{aligned}\mathfrak{B}_3(m,l,k,r) = A(r)\mathfrak{A}_3(k,m,l) + \frac{1}{D_f}\Big[&\int_p \mathfrak{F}(p,l,r)\mathfrak{A}_3(k,m,p)\,(\varepsilon_p-\varepsilon_l-\varepsilon_r)\\ &+\int_{m'}\mathfrak{G}(m',m,r)\mathfrak{A}_3(k,m',l)\,(\varepsilon_{m'}-\varepsilon_m-\varepsilon_r)\\ &+\int_{k'}\mathfrak{A}_3(k',m,l)\,C(k,k',r)\,(\varepsilon_{k'}-\varepsilon_l-\varepsilon_m)\\ &+(2\pi)^6\,\langle q\bar q\,g|\,H_{ggq\bar q}+H_{q-\mathrm{inst}}\,|g\rangle\Big],\end{aligned}\tag{5.8}$$

with $D_f=\varepsilon_k-\varepsilon_l-\varepsilon_m-\varepsilon_r$. In each product the weight cancels the energy denominator of the coefficient that acts second, so that (5.8) is (vertex)×(first order coefficient)$/D_f$, as in (4.38)–(4.41).

5.4 Gluon pair annihilation into quarks: $\mathfrak{B}_1$

The two-gluon state probes the symmetric combination $\mathfrak{B}_1(p,k)+\mathfrak{B}_1(k,p)$, Eq. (4.23). Assigning to $\mathfrak{B}_{1\,ji\lambda_2\lambda_1}^{ba\alpha\beta}(p,k,m,l)$ the contributions in which the gluon $g_i^a(k)$ is absorbed (by the valence charge, by the quark or by the antiquark) while $g_j^b(p)$ splits, and half of the contributions that are symmetric by themselves,

$$\boxed{\begin{aligned}\mathfrak{B}_{1\,ji\lambda_2\lambda_1}^{ba\alpha\beta}(p,k,m,l) = &-A_i^{\dagger a}(k)\,\mathfrak{A}_{3\,j\lambda_2\lambda_1}^{b\alpha\beta}(p,m,l)\\ &+(2\pi)^6\Big[\tfrac12\psi^{(1)}+\psi^{(4)}+\psi^{(5)}+\tfrac12\psi^{(6)}+\tfrac12\psi^{(7)}\Big]_{q\bar q},\end{aligned}}\tag{5.9}$$

where $\psi^{(1)},\psi^{(6)},\psi^{(7)}$ are the coefficient functions of (4.50), (4.56), (4.57), and $\psi^{(4)},\psi^{(5)}$ those of (4.54), (4.55) with $(a,i,k)\leftrightarrow(b,j,p)$ in the displayed term. The first term is the factorized contribution (4.53). In the form (5.1) the t- and u-channel terms read

$$\begin{aligned}\mathfrak{B}_1(p,k,m,l) \supset \frac{1}{D_f}\Big[&\int_{l_1}\mathfrak{F}_2(k,l_1,l)\,\mathfrak{A}_3(p,m,l_1)\,(\varepsilon_{l_1}+\varepsilon_k-\varepsilon_l)\\ &+\int_{m_1}\mathfrak{G}_2(k,m_1,m)\,\mathfrak{A}_3(p,m_1,l)\,(\varepsilon_{m_1}+\varepsilon_k-\varepsilon_m)\Big],\end{aligned}\tag{5.10}$$

which makes manifest that they produce the $\mathfrak{F}$ and $\mathfrak{G}$ terms of the unitarity relation (4.10). Since $D_f = \varepsilon_k + \varepsilon_p - \varepsilon_l - \varepsilon_m$ can vanish, the ρ independent part of $\mathfrak{B}_1$ has an on-shell pole whose residue is the $gg \to q\bar{q}$ amplitude.

5.5 Factorized coefficients: $\mathfrak{C}_3$, $\mathfrak{C}_4$, $\mathfrak{D}_1$

For these sectors the transition proceeds through two independent subprocesses, the two time orderings combine by (3.4), and the coefficients factorize. With $K = (2\pi)^9\psi$ and $(2\pi)^9/64\pi^3 = (2\pi)^6/8$, Eqs. (4.58), (4.61) and (4.59) give

$$\mathfrak{C}^{bcde\alpha\beta}_{4\,jkon\lambda_2\lambda_1}(p,q,r,s,m,l) = \mathfrak{A}^{b\alpha\beta}_{3\,j\lambda_2\lambda_1}(p,m,l)\; C^{edc}_{nok}(q,r,s), \tag{5.11}$$

$$\mathfrak{C}^{bcde\alpha\beta}_{3\,jkon\lambda_2\lambda_1}(p,q,r,s,m,l) = \mathfrak{A}^{b\alpha\beta}_{3\,j\lambda_2\lambda_1}(p,m,l)\; C^{edc}_{2\,nok}(q,r,s), \tag{5.12}$$

$$\mathfrak{D}^{cb\alpha'\beta'\alpha\beta}_{1\,kj\lambda_2'\lambda_1'\lambda_2\lambda_1}(q,p,m',l',m,l) = \mathfrak{A}^{b\alpha\beta}_{3\,j\lambda_2\lambda_1}(p,m,l)\; \mathfrak{A}^{c\alpha'\beta'}_{3\,k\lambda_2'\lambda_1'}(q,m',l'), \tag{5.13}$$

where C and $C_2 = -C^\dagger$ are the $g \to gg$ and $gg \to g$ coefficients of Paper I (Eqs. (4.8) and (5.5) there). Explicitly, e.g.,

$$\mathfrak{C}_4 = \frac{i(2\pi)^6 g^2}{8}\, t^b_{\beta\alpha} f^{cde}\, \frac{\delta^{(3)}(p-l-m)\delta^{(3)}(q-r-s)}{\sqrt{p^+q^+r^+s^+}}\, \frac{\Phi^j_{\lambda_1\lambda_2}(p;l,m)\,\mathcal{V}^{kon}(q;r,s)}{\big(\varepsilon_p - \varepsilon_l - \varepsilon_m\big)\big(\varepsilon_q - \varepsilon_r - \varepsilon_s\big)}. \tag{5.14}$$

The factorizations hold with no leftover powers of 2π, which is a stringent check of the prefactors of $\mathfrak{A}_3$ and of the rule (4.20). The unitarity partners $\mathfrak{C}_1$, $\mathfrak{C}_2$, $\mathfrak{D}_2$ follow from (4.12)–(4.14); since $\mathfrak{A}_3$ contains no ρ, two $\mathfrak{A}_3$'s commute and (4.14) gives $\mathfrak{D}_2(l,m,l',m',q,p) = \mathfrak{A}_3^\dagger(q,m,l)\mathfrak{A}_3^\dagger(p,m',l')$, the crossed assignment, which is also what a direct LCPT computation of $q\bar{q}\,q\bar{q} \to gg$ gives.

5.6 Quark–antiquark scattering: $\mathfrak{D}_3$

Comparing (4.26) with (4.80)–(4.84) ($K = (2\pi)^6\psi$; with the ordering $d\,b$ of (4.4) no extra sign arises),

$$\begin{aligned}
\mathfrak{D}^{\beta'\alpha'\alpha\beta}_{3\,\lambda_1'\lambda_2'\lambda_2\lambda_1}(l',m',m,l) = \frac{(2\pi)^3 g^2}{8}\,\frac{\delta^{(3)}(l+m-l'-m')}{D_f}\Bigg\{ & \\
t^a_{\beta\alpha}t^a_{\alpha'\beta'}\,\frac{\Phi^i_{\lambda_1\lambda_2}(k;l,m)\,\big[\Phi^i_{\lambda_1'\lambda_2'}(k;l',m')\big]^*}{k^+\big(\varepsilon_{l'}+\varepsilon_{m'}-\varepsilon_k\big)} & \\
- t^d_{\beta\beta'}t^d_{\alpha'\alpha}\Bigg[\frac{\Phi^n_{\lambda_1\lambda_1'}(r;l,l')\,\Phi^n_{\lambda_2'\lambda_2}(r;m',m)\,\theta(r^+)}{r^+\big(\varepsilon_{l'}-\varepsilon_l-\varepsilon_r\big)} & \\
+\frac{\Phi^n_{\lambda_1\lambda_1'}(\tilde{r};l,l')\,\Phi^n_{\lambda_2'\lambda_2}(\tilde{r};m',m)\,\theta(\tilde{r}^+)}{\tilde{r}^+\big(\varepsilon_{m'}-\varepsilon_m-\varepsilon_{\tilde{r}}\big)}\Bigg] & \\
-\frac{8\,t^a_{\beta\beta'}t^a_{\alpha'\alpha}\,\delta_{\lambda_1\lambda_1'}\delta_{\lambda_2\lambda_2'}}{(l'^+-l^+)^2}+\frac{8\,t^a_{\beta\alpha}t^a_{\alpha'\beta'}\,\delta_{\lambda_1\lambda_2}\delta_{\lambda_1'\lambda_2'}}{(k^+)^2}\Bigg\}, &
\end{aligned} \tag{5.15}$$

with $k = l + m$, $r = l' - l$, $\tilde{r} = m' - m$ and $D_f = \varepsilon_{l'} + \varepsilon_{m'} - \varepsilon_l - \varepsilon_m$. In the form (5.1),

$$\begin{aligned}
\mathfrak{D}_3(l',m',m,l) = \frac{1}{D_f}\Big[& \int_k \mathfrak{A}_3(k,m,l)\mathfrak{A}_4(l',m',k)\,(\varepsilon_k - \varepsilon_l - \varepsilon_m) \\
& + \int_r \mathfrak{G}_2(r,m',m)\mathfrak{F}(l',l,r)\,(\varepsilon_{m'} + \varepsilon_r - \varepsilon_m)
\end{aligned}$$

$$+\int_{\tilde r}\mathfrak{F}_2(\tilde r,l',l)\mathfrak{G}(m',m,\tilde r)\,(\varepsilon_{l'}+\varepsilon_{\tilde r}-\varepsilon_l)$$
$$+(2\pi)^6\,\langle q\bar q|\,H_{q-\rm inst}\,|q\bar q\rangle\,\Big]. \tag{5.16}$$

Check. Write $D_a=\varepsilon_k-\varepsilon_l-\varepsilon_m$ and $D'_a=\varepsilon_k-\varepsilon_{l'}-\varepsilon_{m'}$, so that $D_a-D'_a=D_f$. The s-channel term of $\mathfrak{D}_3(l',m',m,l)+\mathfrak{D}_3^\dagger(l,m,m',l')$ is

$$\frac{(2\pi)^3g^2}{8}\,\frac{t^at^a\,\Phi\Phi^*\delta^{(3)}}{k^+}\Big[-\frac{1}{D'_aD_f}+\frac{1}{D_aD_f}\Big]$$
$$=-\frac{(2\pi)^3g^2}{8}\,\frac{t^at^a\,\Phi\Phi^*\delta^{(3)}}{k^+D_aD'_a}=-\int_k\mathfrak{A}_3(k,m,l)\,\mathfrak{A}_3^\dagger(k,m',l'), \tag{5.17}$$

which is the first term of (4.17). In the same way the t-channel terms (b) and (c) reproduce the $\mathfrak{F}$, $\mathfrak{G}$ terms of (4.17), while the instantaneous term is anti-Hermitian and drops out. Note that the t-channel part of $\mathfrak{D}_3$ is not anti-Hermitian by itself; it satisfies unitarity only together with the coefficients $\mathfrak{F}$ and $\mathfrak{G}$.

5.7 Quark rescattering on the valence charge: $\mathfrak{P}$ and $\bar{\mathfrak{P}}$

Comparing (4.25) with (4.69) ($K=(2\pi)^3\psi$, $(2\pi)^3/32\pi^3=1/4$),

$$\boxed{\mathfrak{P}^{\beta'\beta}_{\lambda'\lambda}(l',l)=\frac{g^2}{4}\,\frac{t^b_{\beta\beta'}\,\rho^b(l'-l)}{(p^+)^2}\,\chi^\dagger_\lambda\Big\{\theta(l'^+-l^+)\,\Big[\frac{2}{\Delta-\varepsilon_p}-\frac{p^+\,(\boldsymbol{\sigma}\cdot\boldsymbol{l})(\boldsymbol{\sigma}\cdot\boldsymbol{l}')}{l^+l'^+(\Delta-\varepsilon_p)\,\Delta}\Big]$$
$$-\theta(l^+-l'^+)\,\Big[\frac{2}{\varepsilon_p}-\frac{p^+\,(\boldsymbol{\sigma}\cdot\boldsymbol{l})(\boldsymbol{\sigma}\cdot\boldsymbol{l}')}{l^+l'^+\,\varepsilon_p\,\Delta}\Big]\Big\}\chi_{\lambda'},} \tag{5.18}$$

with $\Delta=\varepsilon_{l'}-\varepsilon_l$ and $p=\pm(l'-l)$ in the two regions. In the form (5.1),

$$\mathfrak{P}(l',l)=-A^\dagger(p)\,\mathfrak{F}(l',l,p)\,\frac{\varepsilon_p}{\Delta}\Big|_{p=l'-l}$$
$$-\mathfrak{F}^\dagger(l,l',p)\,A(p)\,\frac{\varepsilon_{l'}+\varepsilon_p-\varepsilon_l}{\Delta}\Big|_{p=l-l'}+\frac{(2\pi)^3\,\langle q(l)|\,H_{q-\rm inst}\,|q(l')\rangle}{\Delta}. \tag{5.19}$$

The antiquark coefficient $\bar{\mathfrak{P}}^{\alpha'\alpha}_{\lambda'\lambda}(m',m)$ is $(2\pi)^3$ times the coefficient function of (4.73).

Check. Let $p=l'-l$ with $l'^+>l^+$. Then $\mathfrak{P}(l',l)$ is given by the first line of (5.18), and $\mathfrak{P}(l,l')$ (incoming l, outgoing l') by the second line with $l\leftrightarrow l'$. Using $\rho^b(-p)^\dagger=\rho^b(p)$ and $\big(t^b_{\beta'\beta}\big)^*=t^b_{\beta\beta'}$, and denoting $S=\chi^\dagger_\lambda(\boldsymbol{\sigma}\cdot\boldsymbol{l})(\boldsymbol{\sigma}\cdot\boldsymbol{l}')\chi_{\lambda'}/(l^+l'^+)$,

$$\mathfrak{P}(l',l)+\mathfrak{P}^\dagger(l,l')=\frac{g^2}{4}\frac{t^b\rho^b(p)}{(p^+)^2}\Big\{2\delta_{\lambda\lambda'}\Big[\frac{1}{\Delta-\varepsilon_p}-\frac{1}{\varepsilon_p}\Big]-p^+S\Big[\frac{1}{(\Delta-\varepsilon_p)\Delta}+\frac{1}{\varepsilon_p\Delta}\Big]\Big\}$$
$$=\frac{g^2}{4}\frac{t^b\rho^b(p)}{(p^+)^2}\,\frac{4\varepsilon_p-2\Delta-p^+S}{(\Delta-\varepsilon_p)\,\varepsilon_p}$$
$$=\frac{g^2}{4}\frac{t^b\rho^b(p)\;p^j\Phi^j_{\lambda\lambda'}(p;l,l')}{(p^+)^2(\Delta-\varepsilon_p)\,\varepsilon_p}=-\int_{p'}A^\dagger(p')\,\mathfrak{F}(l',l,p'), \tag{5.20}$$

where we used (4.68), (5.6) and (5.3). This is the relation (4.18) (the self-energy term is momentum diagonal and scaleless). The pole at $\Delta=0$ cancels in the sum, as it must: $\mathfrak{P}+\mathfrak{P}^\dagger$ is a product of first order coefficients.

5.8 Coefficients with three outgoing particles: $\mathfrak{Q}_1$, $\mathfrak{Q}_2$, $\mathfrak{K}_1$

Comparing (4.25), (4.27) with (4.70), (4.71) and with the quark–gluon amplitude of section 4.7, and using (5.1), we obtain

$$\begin{aligned}\mathfrak{Q}_1(l',l,q,r) = \frac{1}{2}\Bigg\{ & A(q)\mathfrak{F}(l',l,r) + A(r)\mathfrak{F}(l',l,q) \\ & + \frac{1}{D_f}\Big[\int_{l_1} \mathfrak{F}(l_1,l,q)\mathfrak{F}(l',l_1,r)\,(\varepsilon_{l_1}-\varepsilon_l-\varepsilon_q) + (q\leftrightarrow r) \\ & + \int_s C(s,q,r)\,\mathfrak{F}(l',l,s)\,(\varepsilon_s-\varepsilon_q-\varepsilon_r) \\ & + (2\pi)^6\,\langle q\,gg|\,H_{ggq\bar q}+H_{q-\mathrm{inst}}\,|q\rangle\,\Big]\Bigg\}, \end{aligned} \tag{5.21}$$

with $D_f = \varepsilon_{l'}-\varepsilon_l-\varepsilon_q-\varepsilon_r$;

$$\begin{aligned}\mathfrak{K}_1(l',l,l_2,m) = \frac{1}{2}\Bigg\{ & \frac{1}{D_f}\Big[\int_k \mathfrak{A}_3(k,m,l_2)\,\mathfrak{F}(l',l,k)\,(\varepsilon_k-\varepsilon_{l_2}-\varepsilon_m) \\ & + (2\pi)^6\,\langle q\,q\bar q|\,H_{q-\mathrm{inst}}\,|q\rangle\,\Big] - \big(l\leftrightarrow l_2\big)\Bigg\}, \end{aligned} \tag{5.22}$$

with $D_f = \varepsilon_{l'}-\varepsilon_l-\varepsilon_{l_2}-\varepsilon_m$; and

$$\begin{aligned}\mathfrak{Q}_2(l',q,l,r) = & -A^\dagger(q)\mathfrak{F}(l',l,r) + A(r)\mathfrak{F}_2(q,l',l) \\ & + \frac{1}{D_f}\Big[\int_{l_1} \mathfrak{F}(l_1,l,r)\mathfrak{F}_2(q,l',l_1)\,(\varepsilon_{l_1}-\varepsilon_l-\varepsilon_r) \\ & + \int_{l_1} \mathfrak{F}_2(q,l_1,l)\mathfrak{F}(l',l_1,r)\,(\varepsilon_{l_1}+\varepsilon_q-\varepsilon_l) \\ & - \int_{m_1} \mathfrak{A}_4(l',m_1,r)\mathfrak{A}_3(q,m_1,l)\,(\varepsilon_{l'}+\varepsilon_{m_1}-\varepsilon_r) \\ & + \int_s \mathfrak{F}_2(s,l',l)\,C(q,s,r)\,(\varepsilon_s+\varepsilon_{l'}-\varepsilon_l) \\ & + \int_s C_2(s,q,r)\,\mathfrak{F}(l',l,s)\,(\varepsilon_s+\varepsilon_q-\varepsilon_r) \\ & + (2\pi)^6\,\langle q\,g|\,H_{ggq\bar q}+H_{q-\mathrm{inst}}\,|q\,g\rangle\,\Big], \end{aligned} \tag{5.23}$$

with $D_f = \varepsilon_{l'}+\varepsilon_q-\varepsilon_l-\varepsilon_r$. The six terms in the bracket of (5.23) are the s-channel, the two orderings of the u-channel (the second, with a soft pair in the intermediate state, carries the Fermi minus sign) and the two orderings of the t-channel, followed by the instantaneous terms; the first line contains the factorized ρ dependent products. The instantaneous matrix elements are given in appendix D. The antiquark coefficients $\bar{\mathfrak{Q}}_{1,2}$, $\bar{\mathfrak{K}}_1$ follow by $\mathfrak{F}\to\mathfrak{G}$, $\mathfrak{A}_3\to\mathfrak{A}_3$ with the roles of quark and antiquark exchanged.

5.9 Coefficients fixed by unitarity

The remaining coefficients follow from the constraint equations of section 4: $\mathfrak{A}_4$, $\mathfrak{F}_2$, $\mathfrak{G}_2$ from (4.8); $\mathfrak{A}_2$ from (4.9); $\mathfrak{B}_2$, $\mathfrak{B}_4$ from (4.10), (4.11); $\mathfrak{C}_1$, $\mathfrak{C}_2$, $\mathfrak{D}_2$ from (4.12)–(4.14); $\mathfrak{Q}_3$

and $\mathfrak{K}_2$ from (4.15), (4.16). Finally, every structure of $\Omega^{(2)}_\times$, Eq. (4.6), is the normal ordered product of two first order structures X, Y acting on disjoint particles; by (3.4), and since at most one of the two coefficients contains ρ,

$$K_{:\hat{X}\hat{Y}:} = X\,Y, \tag{5.24}$$

e.g. the coefficient of $a^\dagger b^\dagger d^\dagger b^\dagger b\,a$ is $\mathfrak{A}_3\,\mathfrak{F}$ and that of $a^\dagger a^\dagger b^\dagger b\,a^\dagger a$-type structures is $\mathfrak{F}\,C$. These coefficients satisfy the unitarity relations identically.

The overall energy denominator $E_i - E_f$ can vanish only in the coefficients that preserve the number of soft quarks, antiquarks and gluons, namely $\mathfrak{P}$, $\bar{\mathfrak{P}}$, the ρ independent parts of $\mathfrak{D}_3$ and $\mathfrak{Q}_2$, and in the $2 \to 2$ coefficients $\mathfrak{B}_1$, $\mathfrak{B}_2$. These are understood with the prescription of section 3.2, and their residues are on-shell scattering amplitudes. In all other coefficients the overall denominator either never vanishes (away from collinear configurations of measure zero) or cancels between the two time orderings, which then combine into a product of first order coefficients, as in $\mathfrak{C}_3$, $\mathfrak{C}_4$, $\mathfrak{D}_1$, the factorized parts of $\mathfrak{B}_1$, $\mathfrak{B}_3$, $\mathfrak{Q}_1$, $\mathfrak{Q}_2$, and (5.24).

5.10 The generator

Having determined all the coefficients, we reconstruct the evolution operator. As in Paper I we write $\Omega = e^{iG}$ with $G = G_{\rm YM} + G_q$ Hermitian and known through $\mathcal{O}(g^2)$. Since $\Omega^{(2)} = iG^{(2)} + \frac{1}{2}(iG^{(1)})^2$ and the uncontracted part of $\frac{1}{2}(iG^{(1)})^2$ is $\Omega^{(2)}_\times$ together with $\mathfrak{C}_{1,\dots,4}$, $\mathfrak{D}_{1,2}$ and the factorized parts of the other coefficients, the connected part of G_q is

$$\begin{aligned}
G_q = &- i\int_{p,l,m}\Big[\mathfrak{A}_3\,b^\dagger d^\dagger a - \mathfrak{A}_3^\dagger\,a^\dagger db\Big] - i\int_{p,l,l'}\Big[\mathfrak{F}\,a^\dagger b^\dagger b - \mathfrak{F}^\dagger\,b^\dagger b\,a\Big] - i\int_{p,m,m'}\Big[\mathfrak{G}\,a^\dagger d^\dagger d - \mathfrak{G}^\dagger\,d^\dagger d\,a\Big]\\
&- \frac{i}{2}\int_{l,m}\Big[\mathfrak{A}_1 - \mathfrak{A}_2^\dagger\Big]b^\dagger d^\dagger - \frac{i}{2}\int_{l,m}\Big[\mathfrak{A}_2 - \mathfrak{A}_1^\dagger\Big]db - \frac{i}{2}\int_{p,q,l,m}\Big[\mathfrak{B}_1 - \mathfrak{B}_2^\dagger\Big]b^\dagger d^\dagger aa\\
&- \frac{i}{2}\int_{p,q,l,m}\Big[\mathfrak{B}_2 - \mathfrak{B}_1^\dagger\Big]a^\dagger a^\dagger db - \frac{i}{2}\int_{p,q,l,m}\Big[\mathfrak{B}_3 - \mathfrak{B}_4^\dagger\Big]a^\dagger a\,b^\dagger d^\dagger\\
&- \frac{i}{2}\int_{p,q,l,m}\Big[\mathfrak{B}_4 - \mathfrak{B}_3^\dagger\Big]db\,a^\dagger a\\
&- \frac{i}{2}\int_{l,m,l',m'}\Big[\mathfrak{D}_3 - \mathfrak{D}_3^\dagger\Big]b^\dagger d^\dagger db - \frac{i}{2}\int_{l,l'}\Big[\mathfrak{P} - \mathfrak{P}^\dagger\Big]b^\dagger b - \frac{i}{2}\int_{m,m'}\Big[\bar{\mathfrak{P}} - \bar{\mathfrak{P}}^\dagger\Big]d^\dagger d\\
&- \frac{i}{2}\int\Big[\mathfrak{Q}_1 - \mathfrak{Q}_3^\dagger\Big]a^\dagger a^\dagger b^\dagger b - \frac{i}{2}\int\Big[\mathfrak{Q}_3 - \mathfrak{Q}_1^\dagger\Big]b^\dagger b\,aa - \frac{i}{2}\int\Big[\mathfrak{Q}_2 - \mathfrak{Q}_2^\dagger\Big]a^\dagger b^\dagger b\,a\\
&- \frac{i}{2}\int\Big[\mathfrak{K}_1 - \mathfrak{K}_2^\dagger\Big]b^\dagger b^\dagger d^\dagger b - \frac{i}{2}\int\Big[\mathfrak{K}_2 - \mathfrak{K}_1^\dagger\Big]b^\dagger d\,bb\\
&+ \big(q \to \bar{q}\big),
\end{aligned} \tag{5.25}$$

where the arguments and labels are those of (4.3)–(4.5), $(q \to \bar{q})$ denotes the antiquark analogues of the $\mathfrak{Q}$ and $\mathfrak{K}$ terms, and $\mathfrak{F}$, $\mathfrak{G}$, $\mathfrak{A}_3$ are of order g while all the other coefficients are of order g^2. G_q is Hermitian by construction. By the unitarity relations, each combination $[K - K'^\dagger]/2$ in (5.25) equals K minus half of the corresponding product of first order coefficients; for the particle number preserving coefficients it is their anti-Hermitian part, which contains the instantaneous terms and the principal value parts of the on-shell poles.

6 Diagonalization of the QCD Hamiltonian at $\mathcal{O}(g)$ and $\mathcal{O}(g^2)$

The evolution operator provides a systematic framework for diagonalizing the soft QCD Hamiltonian of a fast-moving hadron. We use the unitary similarity transformation

$$H \to \Omega^\dagger H\, \Omega, \tag{6.1}$$

with Ω constructed perturbatively in g with the coefficients fixed in section 5. The role of Ω is to cancel the off-diagonal matrix elements of H that connect different Fock sectors, order by order in the coupling. In Paper I it was shown that in pure Yang–Mills theory the coherent operator (3.7) alone is not sufficient: the three-gluon interaction generates additional structures at the same order. With quarks, the quark–gluon vertex generates three further structures at order g, and all of them must be included for a complete diagonalization.

Splitting the Hamiltonian by order in the coupling,

$$\begin{aligned} &H = H_0 + H^{(1)} + H^{(2)}, \qquad H^{(1)} = H_g + H_{ggg} + H_{gq\bar{q}}, \\ &H^{(2)} = H_{gggg} + H_{gg-\mathrm{inst}} + H_{gggg-\mathrm{inst}} + H_{ggq\bar{q}} + H_{q-\mathrm{inst}}, \end{aligned} \tag{6.2}$$

and using $\Omega^{(1)\dagger} = -\Omega^{(1)}$ and $\Omega^{(2)} = iG^{(2)} + \frac{1}{2}\big(\Omega^{(1)}\big)^2$, one finds

$$\Omega^\dagger H\Omega\Big|_{g^1} = H^{(1)} + \big[H_0, \Omega^{(1)}\big], \tag{6.3}$$

$$\Omega^\dagger H\Omega\Big|_{g^2} = H^{(2)} + \tfrac{1}{2}\big[H^{(1)}, \Omega^{(1)}\big] + \big[H_0, iG^{(2)}\big], \tag{6.4}$$

where in (6.4) we used $\big[[H_0, \Omega^{(1)}], \Omega^{(1)}\big] = -[H^{(1)}, \Omega^{(1)}]$, which follows once (6.3) vanishes. For a normal ordered structure $\mathcal{O}$ whose creation operators carry total energy E_{out} and whose annihilation operators carry E_{in},

$$\big[H_0, \mathcal{O}\big] = \big(E_{\mathrm{out}} - E_{\mathrm{in}}\big)\, \mathcal{O}. \tag{6.5}$$

6.1 Order g

Substituting the first order part of the ansatz into (6.3) and using (6.5),

$$\begin{aligned} \Omega^\dagger H\Omega\Big|_{g^1} = H_g + H_{ggg} + H_{gq\bar{q}} &+ \int_p \Big[A^b_j(p)\, \varepsilon_p\, a^{\dagger b}_j(p) + A^{\dagger b}_j(p)\, \varepsilon_p\, a^b_j(p)\Big] \\ &+ \int_{p,q,r} \big(\varepsilon_q + \varepsilon_r - \varepsilon_p\big)\Big[C(p,q,r)\, a^\dagger(r)a^\dagger(q)a(p) \\ &\qquad + C^\dagger(r,q,p)\, a^\dagger(r)a(q)a(p)\Big] \\ &+ \int_{p,l,m} \big(\varepsilon_l + \varepsilon_m - \varepsilon_p\big)\Big[\mathfrak{A}_3(p,m,l)\, b^\dagger(l)d^\dagger(m)a(p) \\ &\qquad + \mathfrak{A}^\dagger_3(p,m,l)\, a^\dagger(p)d(m)b(l)\Big] \\ &+ \int_{p,l,l'} \big(\varepsilon_l + \varepsilon_p - \varepsilon_{l'}\big)\Big[\mathfrak{F}(l',l,p)\, a^\dagger(p)b^\dagger(l)b(l') \\ &\qquad + \mathfrak{F}^\dagger(l',l,p)\, b^\dagger(l')b(l)a(p)\Big] \\ &+ \int_{p,m,m'} \big(\varepsilon_m + \varepsilon_p - \varepsilon_{m'}\big)\Big[\mathfrak{G}(m',m,p)\, a^\dagger(p)d^\dagger(m)d(m') \end{aligned}$$

$$+\,\mathfrak{G}^\dagger(m',m,p)\,d^\dagger(m')d(m)a(p)\Big], \tag{6.6}$$

where colour, polarization and helicity labels are suppressed. The Hamiltonian $H^{(1)}$ is purely off-diagonal — every term of H_g, H_{ggg} and $H_{gq\bar{q}}$ changes the Fock sector — and the commutator with H_0 multiplies each structure by its energy imbalance. Hence (6.3) can be solved structure by structure: writing $H_{gq\bar{q}}$ in the form (2.21), the vanishing of (6.6) requires

$$\mathfrak{A}_3 = \frac{V_3}{\varepsilon_p - \varepsilon_l - \varepsilon_m}, \qquad \mathfrak{F} = \frac{V_F}{\varepsilon_{l'} - \varepsilon_l - \varepsilon_p}, \qquad \mathfrak{G} = \frac{V_G}{\varepsilon_{m'} - \varepsilon_m - \varepsilon_p}, \tag{6.7}$$

together with $A = -\mathfrak{a}/\varepsilon_p$ and $C = \mathfrak{c}/(\varepsilon_p - \varepsilon_q - \varepsilon_r)$ of Paper I, where $\mathfrak{a}$ and $\mathfrak{c}$ are the vertices of H_g and H_{ggg}. These are precisely the coefficients (5.2)–(5.4) obtained by matching to the LCWF: at this order the two characterizations of Ω — it generates the dressed states and it diagonalizes H — agree term by term. Upon substituting the explicit values, all off-diagonal contributions cancel identically and

$$\Omega^\dagger\big(H_0 + H_g + H_{ggg} + H_{gq\bar{q}}\big)\Omega = H_0 + \mathcal{O}(g^2). \tag{6.8}$$

The transformed Hamiltonian is thus fully diagonal at order g. We stress that the coefficients $\mathfrak{F}$ and $\mathfrak{G}$ are indispensable: without them the $a^\dagger b^\dagger b$ and $a^\dagger d^\dagger d$ pieces of $H_{gq\bar{q}}$ survive in (6.6) and the Hamiltonian is not diagonal at order g.

6.2 Order g^2

At order g^2 the transformed Hamiltonian is (6.4). Denote $\mathcal{K} \equiv H^{(2)} + \frac{1}{2}[H^{(1)}, \Omega^{(1)}]$. By (6.5), $iG^{(2)}$ can be chosen to cancel any normal ordered piece of $\mathcal{K}$ with $E_{\rm out} \neq E_{\rm in}$, its coefficient being $-\mathcal{K}/(E_{\rm out} - E_{\rm in})$; as shown in section 3.2, the resulting $\Omega^{(2)}$ is exactly the LCPT coefficient μ_2, i.e. the coefficients of section 5. What remains is the part of $\mathcal{K}$ with $E_{\rm out} = E_{\rm in}$. With the $i\epsilon$ prescription the off-diagonal elements between states of different energy cancel, and what is left for the particle number preserving structures is multiplied by $i\epsilon/(E_i - E_f + i\epsilon)$, which is nonvanishing only on the energy shell; since the shell is a set of measure zero, this leftover does not contribute to matrix elements between normalizable states, exactly as in Paper I. This applies in particular to the ρ dependent terms that rotate the colour of a soft quark in the valence background, i.e. to the instantaneous $\rho \cdot J_q$ interaction and to the products $H_g \times \mathfrak{F}$, $H_g \times \mathfrak{G}$: they transfer momentum between the soft quark and the valence charge and are removed by $iG^{(2)}$, the remainder being the on-shell part discussed above. Substituting all the coefficients we obtain

$$\begin{aligned}
\Omega^\dagger H \Omega = H_0 &- \int_p \varepsilon_p\, A_j^{\dagger b}(p) A_j^b(p) \\
&- \int_{p,q,r} \big(\varepsilon_q + \varepsilon_r - \varepsilon_p\big)\, C^\dagger(r,q,p) C(p,q,r)\, a^\dagger(p) a(p) \\
&- \int_{p,l,m} \big(\varepsilon_l + \varepsilon_m - \varepsilon_p\big)\, \mathfrak{A}_3^\dagger(p,m,l) \mathfrak{A}_3(p,m,l)\, a^\dagger(p) a(p) \\
&- \int_{l',l,p} \big(\varepsilon_l + \varepsilon_p - \varepsilon_{l'}\big)\, \mathfrak{F}^\dagger(l',l,p) \mathfrak{F}(l',l,p)\, b^\dagger(l') b(l') \\
&- \int_{m',m,p} \big(\varepsilon_m + \varepsilon_p - \varepsilon_{m'}\big)\, \mathfrak{G}^\dagger(m',m,p) \mathfrak{G}(m',m,p)\, d^\dagger(m') d(m')
\end{aligned}$$

$$+\,\mathcal{O}(g^3), \tag{6.9}$$

where in each loop term the internal momenta are fixed by the δ functions contained in the coefficients and the colour and spin sums are implied. The first correction is the coherent background field energy of the Weizsäcker–Williams field. The remaining terms are the second order light-cone energy shifts, $-\sum|\text{vertex}|^2/(E_{\text{out}}-E_{\text{in}})$, of a single soft gluon (from the gluon and the quark loops) and of a single soft quark and antiquark. They are evaluated with the variables of (2.6) and (2.7): for the quark loop, using (4.33), every power of $\boldsymbol{\kappa}$ cancels between the numerator and the energy denominator and

$$\int_{l,m}\big(\varepsilon_l+\varepsilon_m-\varepsilon_p\big)\,\mathfrak{A}_3^\dagger\mathfrak{A}_3 \;\propto\; g^2N_fT_F\int_0^1\frac{dz}{z(1-z)}\big[z^2+(1-z)^2\big]\int d^2\kappa, \tag{6.10}$$

and likewise for the gluon loop and for the quark self-energy, with the splitting functions P_{gg} and P_{gq} in place of $z^2+(1-z)^2$. The transverse integrals carry no scale — the soft partons are massless and there is no external invariant — and vanish in dimensional regularization, as do the integrals $\int d^2\kappa/\boldsymbol{\kappa}^2$ in the normalizations (4.34) and (4.76); the same holds for the self-energy (seagull) contributions of the instantaneous interactions. Hence

$$\begin{aligned}\int_{p,q,r}\big(\varepsilon_q+\varepsilon_r-\varepsilon_p\big)C^\dagger C&=\int_{l,m}\big(\varepsilon_l+\varepsilon_m-\varepsilon_p\big)\mathfrak{A}_3^\dagger\mathfrak{A}_3\\&=\int_{l,p}\big(\varepsilon_l+\varepsilon_p-\varepsilon_{l'}\big)\mathfrak{F}^\dagger\mathfrak{F}=\int_{m,p}\big(\varepsilon_m+\varepsilon_p-\varepsilon_{m'}\big)\mathfrak{G}^\dagger\mathfrak{G}=0.\end{aligned} \tag{6.11}$$

This is the statement that a free massless soft gluon or quark receives no mass correction, as gauge invariance and chiral symmetry require. The remaining coherent term is

$$\int_p\varepsilon_p\,A_j^{\dagger b}(p)A_j^b(p)=\int_p\frac{g^2\,\rho^b(p)\,\rho^b(-p)}{(p^+)^2}, \tag{6.12}$$

so that the final diagonalized Hamiltonian through $\mathcal{O}(g^2)$ is

$$\boxed{\Omega^\dagger H\Omega=H_0-\int_p\frac{g^2\,\rho^a(p)\,\rho^a(-p)}{(p^+)^2}+\mathcal{O}(g^3),\qquad\rho^a=\rho_g^a+\rho_q^a\,.} \tag{6.13}$$

The only surviving correction is the ρ^2 background field energy, demonstrating explicitly that the perturbatively constructed evolution operator diagonalizes the soft QCD Hamiltonian through $\mathcal{O}(g^2)$ within the CGC framework. Three remarks are in order.

The quark sector adds no new diagonal structure. Every quark induced term at $\mathcal{O}(g^2)$ is either removed by $iG^{(2)}$, because it changes the Fock sector or transfers momentum to the valence charge, or is proportional to a scaleless integral. The form of the background field energy, and the fact that it is the only survivor, are unchanged from pure Yang–Mills theory; the sole quark modification of (6.13) is that ρ now contains the valence quark charge density (2.17). The pair creation off the vacuum is of order g^2, so its contribution to the ground state energy is of order g^4.

Consistency with the coefficients. The order g condition reproduced A, C, $\mathfrak{A}_3$ and forced $\mathfrak{F}$, $\mathfrak{G}$; at order g^2 the construction requires $iG^{(2)}$ to cancel the off-diagonal part of $\mathcal{K}$, which is the same information carried by the second order coefficients obtained by LCPT matching. The two routes are the same calculation read in opposite directions, and the unitarity relations of section 4 are what verifies that they agree.

Why N_f is absent. The N_f dependent part of the running coupling originates in the quark loop of the gluon self-energy. It does not appear in (6.13) because this is the energy of the soft sector in a fixed valence background with massless on-shell soft partons; the corresponding ultraviolet pole is carried by the scaleless integral in (4.35), where it combines with the gluon loop into $11N_c-4T_FN_f$. As in Paper I, the running coupling becomes explicit in physical observables once the $\mathcal{O}(g^3)$ coefficients are included.

7 Concluding remarks

We have constructed the soft light-cone wave function and the corresponding unitary evolution operator Ω of a fast-moving hadron explicitly through $\mathcal{O}(g^2)$ in QCD with N_f massless quark flavours, within the CGC framework, extending the pure Yang–Mills construction of Paper I. Starting from the light-cone QCD Hamiltonian in the eikonal and Born–Oppenheimer approximations, we showed that the valence sector couples to the soft modes only through its total colour charge density $\rho = \rho_g + \rho_q$, which obeys the same non-Abelian algebra as in the pure gauge theory, while the soft quark degrees of freedom generate new operator structures in Ω. The evolution operator was organized as a fully normal ordered series in the soft gluon, quark and antiquark creation and annihilation operators, with coefficients that are non-commuting functionals of ρ.

The coefficients were determined by the same two complementary procedures as in Paper I. The unitarity condition $\Omega^\dagger\Omega = 1$, imposed order by order in the coupling, provided the relations among the coefficients, and the coefficients of real transitions were fixed by matching the action of Ω on the vacuum, one, two and three-gluon states, a single quark and antiquark, a quark–antiquark pair and a quark–gluon state to the LCWFs computed in LCPT. Three features of the construction are specific to the quark sector. First, a complete ansatz must contain, besides the structures that create and annihilate soft pairs, the coefficients $\mathfrak{F}$ and $\mathfrak{G}$ for gluon emission by soft quarks and antiquarks, which are of order g, and the coefficients $\mathfrak{P}$, $\bar{\mathfrak{P}}$ for the eikonal rescattering of a soft quark on the valence charge, which are of order g^2; without them Ω is neither unitary nor able to diagonalize the Hamiltonian. They enter the unitarity relations of the pair coefficients through the t- and u-channel quark and gluon exchanges. Second, in the coefficients $\mathfrak{A}_1$ and $\mathfrak{P}$ the instantaneous interaction with the valence charge cancels exactly the helicity conserving part of the corresponding one gluon exchange, as required by unitarity. Third, every second order coefficient is a sum of products of first order coefficients weighted by ratios of energy denominators plus instantaneous terms, Eq. (3.3); for independent subprocesses the overall energy denominator cancels and the coefficients factorize exactly, while on-shell poles survive only in the particle number preserving coefficients and in the $2 \to 2$ scattering coefficients, where they are defined by the Møller $i\epsilon$ prescription.

As an application we used Ω to diagonalize the soft QCD Hamiltonian. At order g the Hamiltonian is diagonalized exactly to its free part once the coherent operator and the three-gluon coefficient of Paper I are supplemented by $\mathfrak{A}_3$, $\mathfrak{F}$ and $\mathfrak{G}$. At order g^2 all quark induced off-diagonal terms are removed; the quark loop contribution to the gluon self-energy and the quark self-energy are scaleless and vanish in dimensional regularization, the ultraviolet pole of the gluon normalization being proportional to $11N_c-4T_FN_f$; and the only diagonal correction is the coherent background field energy $-\int_p g^2\rho^a(p)\rho^a(-p)/(p^+)^2$, now with ρ including the valence quark charge.

The construction presented here provides the full QCD evolution operator needed for the computation of physical observables at next-to-leading order with quarks. In particular,

together with the order g^3 coefficients it is the input for the single inclusive gluon spectrum at NLO at mid-rapidity in full QCD [44], for the inclusive production of soft quarks and quark–antiquark pairs [28, 29], and for the quark contributions to multi-particle correlations [23, 24]. We leave these applications for future work.

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A From the QCD Lagrangian to the light-cone QCD Hamiltonian

The QCD Lagrangian with N_f massless flavours is

$$\mathcal{L}_{\rm QCD} = -\frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu} + \bar\psi\, i\gamma^\mu D_\mu\psi, \qquad D_\mu = \partial_\mu - igA^a_\mu t^a. \tag{A.1}$$

The Yang–Mills part is treated as in appendix A of Paper I; the only change is that the colour current entering Gauss's law is now $J^{a\nu} = \bar\psi\gamma^\nu t^a\psi$ plus the gluon current. In light-cone gauge $A^+ = 0$ and with $\psi_\pm = \Lambda_\pm\psi$, the Dirac equation splits into a dynamical equation for ψ_+ and a constraint for ψ_-,

$$i\partial^+\psi_- = \frac{1}{2}\gamma^0\big(i\gamma^i D_i\big)\psi_+ \quad\Rightarrow\quad \psi_- = \frac{1}{2i\partial^+}\,\gamma^0\gamma^i\big(i\partial_i + gA_i\big)\psi_+, \tag{A.2}$$

while the + component of the Yang–Mills equation determines A^-,

$$\partial^+ A^{a-} = -\frac{1}{\partial^+}\Big(D_i^{ab}\partial^+ A_i^b - gJ^{a+}\Big), \qquad J^{a+} = \bar\psi\gamma^+ t^a\psi = \sqrt{2}\,\psi_+^\dagger t^a\psi_+. \tag{A.3}$$

Eliminating ψ_- and A^- through (A.2) and (A.3) from the canonical Hamiltonian $P^- = \int dx^- d^2x\,[\Pi\,\partial^-\phi - \mathcal{L}]$, one obtains [36, 51, 52] the light-cone Hamiltonian (2.9)–(2.10), in which the fields are free fields at $x^+ = 0$. The term $\bar\psi\gamma^i A_i\,(\gamma^+/2i\partial^+)\,\gamma^j A_j\psi$ comes from the gA_i part of ψ_- and describes the instantaneous fermion propagation; the current–current terms come from the gJ^{a+} part of A^- and combine with the gluonic current of Paper I into the single operator (2.12). In the two-component representation $\psi_+ \leftrightarrow \xi$ of Refs. [51, 36], the quark–gluon vertex becomes the bilinear (2.5) between the two-component spinors, the factors $2k^i/k^+$, $(\boldsymbol{\sigma}\cdot\boldsymbol{P}_L)\sigma^i/P_L^+$ and $\sigma^i(\boldsymbol{\sigma}\cdot\boldsymbol{P}_R)/P_R^+$ arising from $\partial_i A_i/\partial^+$ and from the two $\boldsymbol{\sigma}\cdot\boldsymbol{\partial}/\partial^+$ operators of (A.2) acting on the outgoing and incoming fermion lines.

B State definitions and normalization

As in Paper I, $|0\rangle$ denotes the vacuum of the free soft Hamiltonian, $|0\rangle \equiv |0\rangle_{\rm soft}\otimes|v\rangle$, with $E_0 = 0$. The single particle states are

$$|g_i^a(k)\rangle = \frac{a_i^{\dagger a}(k)}{(2\pi)^{3/2}}|0\rangle, \qquad \Big|q_\lambda^\beta(l)\Big\rangle = \frac{b_\lambda^{\dagger\beta}(l)}{(2\pi)^{3/2}}|0\rangle, \qquad |\bar q_\lambda^\alpha(m)\rangle = \frac{d_\lambda^{\dagger\alpha}(m)}{(2\pi)^{3/2}}|0\rangle, \tag{B.1}$$

normalized as $\langle q^{\beta'}_{\lambda'}(l')|q^{\beta}_{\lambda}(l)\rangle = \delta^{\beta\beta'}\delta_{\lambda\lambda'}\delta^{(3)}(l-l')$, with energies ε_k, ε_l, ε_m. Multiparticle states are defined by the product of creation operators, each with a factor $(2\pi)^{-3/2}$, in the order

$$\left| q^{\beta}_{\lambda_1}(l)\bar{q}^{\alpha}_{\lambda_2}(m)\, g\cdots \right\rangle = \frac{b^{\dagger\beta}_{\lambda_1}(l)\, d^{\dagger\alpha}_{\lambda_2}(m)\, a^{\dagger}\cdots}{(2\pi)^{3n/2}}|0\rangle, \tag{B.2}$$

quark operators to the left of antiquark operators; gluon operators commute with both. The energy of a multiparticle state is the sum of the single particle energies. With the ordering (B.2) and the ordering $d\,b$ of the pair annihilation operators in (4.4),

$$d^{\alpha}_{\lambda_2}(m)\, b^{\beta}_{\lambda_1}(l)\; b^{\dagger\beta'}_{\lambda_1'}(l')\, d^{\dagger\alpha'}_{\lambda_2'}(m')|0\rangle = (2\pi)^6\delta^{\beta\beta'}\delta^{\alpha\alpha'}\delta_{\lambda_1\lambda_1'}\delta_{\lambda_2\lambda_2'}\delta^{(3)}(l-l')\delta^{(3)}(m-m')|0\rangle, \tag{B.3}$$

with a positive sign, so that all the matrix elements of Ω between states with one soft pair are read off without additional signs. In states with two identical fermions the antisymmetry of the creation operators is accounted for by the explicit antisymmetrization of the corresponding coefficients, Eqs. (4.16) and (5.22).

The matching rule (4.20) follows from these definitions. A normal ordered term with coefficient K, M creation and n annihilation operators, acting on an n-particle state contracts each annihilation operator with an incoming particle, which produces a factor $(2\pi)^3\delta^{(3)}$ against the measure $1/(2\pi)^3$ and the state normalization $(2\pi)^{-3/2}$, i.e. a net $(2\pi)^{-3/2}$ per incoming particle; each of the M creation operators comes with the measure $1/(2\pi)^3$ and builds a normalized outgoing particle with $(2\pi)^{3/2}$, i.e. again $(2\pi)^{-3/2}$. Hence $\Omega\,|n\rangle \supset (2\pi)^{-3(n+M)/2}\int d^3(M)\, K\,|M\rangle$.

The notation for the wave functions follows Paper I: the subscript lists the outgoing particles and the powers of ρ, the argument the incoming particles, and all outgoing and intermediate momenta are integrated over.

C The CGC Hamiltonian with quarks: eikonal approximation

The soft and valence modes of the quark field are introduced as in (2.14),

$$\xi = \Xi + \xi, \qquad \Xi^{\beta}(x) = 2^{-1/4}\int_{\vee}^{\infty}\frac{dk^+}{2\pi}\int\frac{d^2k}{(2\pi)^2}\sum_{\lambda}\left[\bar{b}^{\beta}_{\lambda}(k)\chi_{\lambda}e^{-ik\cdot x} + \bar{d}^{\dagger\beta}_{\lambda}(k)\chi_{\lambda}e^{ik\cdot x}\right], \tag{C.1}$$

with the soft field ξ restricted to $\Lambda < k^+ < \vee$, and similarly for the gluon field (appendix C of Paper I). Inserting the decomposition into (2.10) generates soft–soft, soft–valence and valence–valence terms. The eikonal approximation neglects all contributions suppressed by powers of $k^+_{\rm soft}/k^+_{\rm valence}$ and all terms in which inverse longitudinal derivatives act on valence fields. Concretely:

- the vertex $\bar{\psi}\gamma^i t^a\psi A^a_i$ with a valence fermion and a soft gluon changes the valence momentum by a soft amount; its transverse structure (2.5) contains $1/P^+_{L,R}$ of the valence fermion and the only unsuppressed part is the $2k^i/k^+$ term, which is the coupling of the soft gluon to the plus current of the valence quark. Together with the corresponding term of the gluonic sector it builds H_g with the total charge density (2.15);
- vertices with a soft fermion and a valence gluon, or with one soft and one valence fermion, cannot conserve the plus momentum within the soft and valence windows and are dropped;

- the instantaneous fermion exchange with valence fields carries $1/P^+$ of a valence line and is suppressed;
- in the instantaneous gluon exchange (2.12), the valence current reduces to its colour charge, $J^{a+}_{\rm val} \to \rho^a$, and the cross terms with the soft currents give the $\rho \cdot J_g$ term of Paper I and the $\rho \cdot J_q$ term of (2.25).

The valence quark charge density (2.17) is the Fourier transform of $J^{a+}_{\rm val,q} = \sqrt{2}\,\Xi^\dagger t^a \Xi$, normal ordered, restricted to its colour rotating ($b^\dagger b$ and $d^\dagger d$) part; the terms that create or annihilate valence pairs change the valence plus momentum by a valence amount and do not contribute. Using (2.4) for the valence modes one verifies (2.18) for ρ_q; since ρ_g and ρ_q commute, the algebra holds for their sum.

D Matrix elements

We list the matrix elements used in the calculation, between the states of appendix B; colour indices in the fundamental representation are $\alpha, \beta, \ldots$, spin labels λ, and Φ is defined in (2.5).

• Gluon emission and absorption by the valence charge (Paper I), with spectators X:

$$\langle g^a_i(k)|\, H_g|0\rangle = \frac{g\,k^i \rho^a(-k)}{4\pi^{3/2}|k^+|^{3/2}}, \tag{D.1}$$

$$\big\langle X\, g^b_j(p)\big|\, H_g\,|X\rangle = \frac{g\,p^j \rho^b(-p)}{4\pi^{3/2}|p^+|^{3/2}}, \qquad \langle X|\, H_g\,\big|X\, g^b_j(p)\big\rangle = \frac{g\,p^j \rho^b(p)}{4\pi^{3/2}|p^+|^{3/2}}. \tag{D.2}$$

• Three-gluon vertex (Paper I):

$$\big\langle g^c_l(q) g^b_j(p)\big|\, H_{ggg}\,|g^a_i(k)\rangle = \frac{igf^{abc}\delta^{(3)}(k-p-q)}{8\pi^{3/2}\sqrt{k^+p^+q^+}}\mathcal{V}^{ijl}(k;p,q). \tag{D.3}$$

• Quark–gluon vertices:

$$\Big\langle q^\beta_{\lambda_1}(l)\bar q^\alpha_{\lambda_2}(m)\Big|\, H_{gq\bar q}\,|g^a_i(k)\rangle = \tfrac{g\,t^a_{\beta\alpha}}{8\pi^{3/2}\sqrt{k^+}}\,\delta^{(3)}(k-l-m)\,\Phi^i_{\lambda_1\lambda_2}(k;l,m), \tag{D.4}$$

$$\Big\langle q^\beta_\lambda(l) g^b_j(p)\Big|\, H_{gq\bar q}\,\Big|q^{\beta'}_{\lambda'}(l')\Big\rangle = \tfrac{g\,t^b_{\beta\beta'}}{8\pi^{3/2}\sqrt{p^+}}\,\delta^{(3)}(l'-l-p)\,\Phi^j_{\lambda\lambda'}(p;l,l'), \tag{D.5}$$

$$\big\langle \bar q^\alpha_\lambda(m) g^b_j(p)\big|\, H_{gq\bar q}\,\Big|\bar q^{\alpha'}_{\lambda'}(m')\Big\rangle = -\tfrac{g\,t^b_{\alpha'\alpha}}{8\pi^{3/2}\sqrt{p^+}}\,\delta^{(3)}(m'-m-p)\,\Phi^j_{\lambda'\lambda}(p;m',m). \tag{D.6}$$

The absorption vertices are the complex conjugates, with $\big[\Phi^i_{\lambda\lambda'}(k;P_L,P_R)\big]^* = \Phi^i_{\lambda'\lambda}(k;P_R,P_L)$.

• Instantaneous interaction of soft quarks with the valence charge ($\rho \cdot J_q$):

$$\Big\langle q^\beta_{\lambda_1}(l)\bar q^\alpha_{\lambda_2}(m)\Big|\, H_{q-\rm inst}|0\rangle = \frac{g^2\,t^a_{\beta\alpha}\,\rho^a(-l-m)\,\delta_{\lambda_1\lambda_2}}{8\pi^3(l^+ + m^+)^2}, \tag{D.7}$$

$$\Big\langle q^\beta_\lambda(l)\Big|\, H_{q-\rm inst}\,\Big|q^{\beta'}_{\lambda'}(l')\Big\rangle = \frac{g^2\,t^a_{\beta\beta'}\,\rho^a(l'-l)\,\delta_{\lambda\lambda'}}{8\pi^3(l'^+ - l^+)^2}, \tag{D.8}$$

$$\langle \bar q^\alpha_\lambda(m)|\, H_{q-\rm inst}\,\Big|\bar q^{\alpha'}_{\lambda'}(m')\Big\rangle = -\frac{g^2\,t^a_{\alpha'\alpha}\,\rho^a(m'-m)\,\delta_{\lambda\lambda'}}{8\pi^3(m'^+ - m^+)^2}. \tag{D.9}$$

• Instantaneous quark exchange ($H_{ggq\bar q}$); the instantaneous fermion line carries $1/(2P^+)$, P^+ being the plus momentum flowing along the fermion number arrow:

$$\Big\langle q^\beta_{\lambda_1}(l)\bar q^\alpha_{\lambda_2}(m) g^d_n(r)\Big|\, H_{ggq\bar q}\,|g^a_i(k)\rangle = \frac{g^2\,\delta^{(3)}(k-l-m-r)}{16\pi^3\sqrt{k^+r^+}}\,\chi^\dagger_{\lambda_1}\Big[\frac{(t^dt^a)_{\beta\alpha}\sigma^n\sigma^i}{l^+ + r^+} - \frac{(t^at^d)_{\beta\alpha}\sigma^i\sigma^n}{m^+ + r^+}\Big]\chi_{\lambda_2}, \tag{D.10}$$

$$\Big\langle q^{\beta}_{\lambda_1}(l)\bar q^{\alpha}_{\lambda_2}(m)\Big|\, H_{gg q\bar q}\,\big|g^a_i(k)g^b_j(p)\big\rangle = \frac{g^2\,\delta^{(3)}(k+p-l-m)}{16\pi^3\sqrt{k^+p^+}}\,\chi^\dagger_{\lambda_1}\Big[\frac{(t^at^b)_{\beta\alpha}\sigma^i\sigma^j}{l^+-k^+}+\frac{(t^bt^a)_{\beta\alpha}\sigma^j\sigma^i}{l^+-p^+}\Big]\chi_{\lambda_2}, \tag{D.11}$$

$$\Big\langle q^{\beta}_{\lambda}(l)g^c_k(q)g^d_n(r)\Big|\, H_{gg q\bar q}\,\Big|q^{\beta'}_{\lambda'}(l')\Big\rangle = \frac{g^2\,\delta^{(3)}(l'-l-q-r)}{16\pi^3\sqrt{q^+r^+}}\,\chi^\dagger_{\lambda}\Big[\frac{(t^ct^d)_{\beta\beta'}\sigma^k\sigma^n}{l^++q^+}+\frac{(t^dt^c)_{\beta\beta'}\sigma^n\sigma^k}{l^++r^+}\Big]\chi_{\lambda'}, \tag{D.12}$$

$$\Big\langle q^{\beta}_{\lambda}(l)g^d_n(r)\Big|\, H_{gg q\bar q}\,\Big|q^{\beta'}_{\lambda'}(l')g^c_k(q)\Big\rangle = \frac{g^2\,\delta^{(3)}(l'+q-l-r)}{16\pi^3\sqrt{q^+r^+}}\,\chi^\dagger_{\lambda}\Big[\frac{(t^dt^c)_{\beta\beta'}\sigma^n\sigma^k}{l'^++q^+}+\frac{(t^ct^d)_{\beta\beta'}\sigma^k\sigma^n}{l'^+-r^+}\Big]\chi_{\lambda'}. \tag{D.13}$$

- Instantaneous gluon exchange between soft currents ($J_g\cdot J_q$ and $J_q\cdot J_q$):

$$\Big\langle q^{\beta}_{\lambda_1}(l)\bar q^{\alpha}_{\lambda_2}(m)g^d_n(r)\Big|\, H_{q-\mathrm{inst}}\,\big|g^a_i(k)\big\rangle = \frac{ig^2 f^{bad}\,t^b_{\beta\alpha}\,\delta_{\lambda_1\lambda_2}\delta_{in}\,(k^++r^+)\,\delta^{(3)}(k-l-m-r)}{2(2\pi)^3\sqrt{k^+r^+}\,(l^++m^+)^2}, \tag{D.14}$$

$$\Big\langle q^{\beta}_{\lambda_1}(l)\bar q^{\alpha}_{\lambda_2}(m)\Big|\, H_{q-\mathrm{inst}}\,\big|g^a_i(k)g^b_j(p)\big\rangle = -\frac{ig^2 f^{cab}\,t^c_{\beta\alpha}\,\delta_{\lambda_1\lambda_2}\delta_{ij}\,(p^+-k^+)\,\delta^{(3)}(k+p-l-m)}{2(2\pi)^3\sqrt{k^+p^+}\,(k^++p^+)^2}, \tag{D.15}$$

$$\Big\langle q^{\beta}_{\lambda}(l)g^c_k(q)g^d_n(r)\Big|\, H_{q-\mathrm{inst}}\,\Big|q^{\beta'}_{\lambda'}(l')\Big\rangle = \frac{ig^2 f^{acd}\,t^a_{\beta\beta'}\,\delta_{\lambda\lambda'}\delta_{kn}\,(r^+-q^+)\,\delta^{(3)}(l'-l-q-r)}{2(2\pi)^3\sqrt{q^+r^+}\,(q^++r^+)^2}, \tag{D.16}$$

$$\Big\langle q^{\beta}_{\lambda}(l)g^d_n(r)\Big|\, H_{q-\mathrm{inst}}\,\Big|q^{\beta'}_{\lambda'}(l')g^c_k(q)\Big\rangle = \frac{ig^2 f^{acd}\,t^a_{\beta\beta'}\,\delta_{\lambda\lambda'}\delta_{kn}\,(q^++r^+)\,\delta^{(3)}(l'+q-l-r)}{2(2\pi)^3\sqrt{q^+r^+}\,(q^+-r^+)^2}, \tag{D.17}$$

$$\begin{aligned}&\Big\langle q^{\beta}_{\lambda_1}(l)\bar q^{\alpha}_{\lambda_2}(m)\Big|\, H_{q-\mathrm{inst}}\,\Big|q^{\beta'}_{\lambda_1'}(l')\bar q^{\alpha'}_{\lambda_2'}(m')\Big\rangle\\ &\qquad=\frac{g^2\,\delta^{(3)}(l+m-l'-m')}{(2\pi)^3}\Big[\frac{t^a_{\beta\alpha}t^a_{\alpha'\beta'}\delta_{\lambda_1\lambda_2}\delta_{\lambda_1'\lambda_2'}}{(l^++m^+)^2}-\frac{t^a_{\beta\beta'}t^a_{\alpha'\alpha}\delta_{\lambda_1\lambda_1'}\delta_{\lambda_2\lambda_2'}}{(l'^+-l^+)^2}\Big],\end{aligned} \tag{D.18}$$

$$\begin{aligned}&\Big\langle q^{\beta}_{\lambda}(l)q^{\beta_2}_{\lambda_2}(l_2)\bar q^{\alpha}_{\lambda_m}(m)\Big|\, H_{q-\mathrm{inst}}\,\Big|q^{\beta'}_{\lambda'}(l')\Big\rangle\\ &\qquad=\frac{g^2\,\delta^{(3)}(l'-l-l_2-m)}{(2\pi)^3}\Big[\frac{t^a_{\beta\beta'}t^a_{\beta_2\alpha}\delta_{\lambda\lambda'}\delta_{\lambda_2\lambda_m}}{(l_2^++m^+)^2}-\big((\beta,\lambda,l)\leftrightarrow(\beta_2,\lambda_2,l_2)\big)\Big].\end{aligned} \tag{D.19}$$

The remaining four-point matrix elements of the pure gauge sector are given in appendix D of Paper I. Each instantaneous gluon line carries $1/(P^+)^2$ with P^+ the plus momentum it transfers; the relative minus signs reflect the antiquark charge and Fermi statistics.

E Perturbation theory with non-commutative matrix elements

The derivation of the wave functions through order g^2 follows the Rayleigh–Schrödinger expansion of appendix E of Paper I [54, 55], keeping the order of the operator valued matrix elements, the later interaction standing on the left. The presence of fermions only introduces the signs dictated by the anticommutators (2.4) and the state ordering (B.2). Here we prove the two identities used throughout.

Reduction formula. With the $i\epsilon$ prescription of section 3.2, $H_{mi}=(E_i-E_m+i\epsilon)\,\mu_1(m\leftarrow i)$ and $H_{fm}=(E_m-E_f+i\epsilon)\,\mu_1(f\leftarrow m)$. Substituting into the second order amplitude (3.2)

gives (3.3) identically. For two subprocesses X, Y acting on disjoint particles there are two intermediate states, with $E_m - E_f = \Delta_Y$ (X first) and Δ_X (Y first), and $E_i - E_f = \Delta_X + \Delta_Y$; this gives (3.4). If neither or only one of X, Y contains ρ, the two products coincide and the overall denominator cancels. In operator language, the same statement follows from $\Omega = e^{iG}$: writing $\Omega^{(2)} = iG^{(2)} + \frac{1}{2}(\Omega^{(1)})^2$ with $iG^{(2)}_{fi} = \mathcal{K}_{fi}/(E_i - E_f)$ (section 6) and $\mathcal{K} = H^{(2)} + \frac{1}{2}[H^{(1)}, \Omega^{(1)}]$, one finds for a single intermediate state, with $a = E_i - E_m$ and $b = E_m - E_f$,

$$\Omega^{(2)}_{fi} = H_{fm}H_{mi}\Big[\frac{1}{2ab} + \frac{1}{2(a+b)}\Big(\frac{1}{a} - \frac{1}{b}\Big)\Big] + \frac{H^{(2)}_{fi}}{a+b} = \frac{H_{fm}H_{mi}}{a\,(a+b)} + \frac{H^{(2)}_{fi}}{a+b}, \tag{E.1}$$

which is the LCPT amplitude μ_2. Thus the normal ordered coefficients of Ω at second order are the LCPT amplitudes, and the diagonalization of section 6 and the matching of section 5 are the same calculation.

Unitarity identity. For a given intermediate state m let $a = E_i - E_m + i\epsilon$ and $b = E_f - E_m - i\epsilon$, so that $E_i - E_f + 2i\epsilon = a - b$. Then

$$\begin{aligned}\mu_2(f \leftarrow i) + \big[\mu_2(i \leftarrow f)\big]^\dagger &= H_{fm}H_{mi}\Big[\frac{1}{(a-b)\,a} - \frac{1}{(a-b)\,b}\Big] \\ &= -\frac{H_{fm}H_{mi}}{a\,b} = -\big[\mu_1(m \leftarrow f)\big]^\dagger \mu_1(m \leftarrow i),\end{aligned} \tag{E.2}$$

which is (3.5); the instantaneous terms are anti-Hermitian, $H^{(2)}_{fi}/(E_i - E_f + i\epsilon) + \big[H^{(2)}_{if}/(E_f - E_i + i\epsilon)\big]^\dagger = 0$ up to terms proportional to ϵ, and drop out. The ordering $H_{fm}H_{mi}$ is preserved, so the identity holds for the non-commuting coefficients. Every unitarity relation of section 4 is a sum of such identities over the intermediate states.

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